

ANTANAS BURAÈAS



THE RESPONSE OF BALTIC ECONOMIES TO PEGGING OF NATIONAL CURRENCIES

Final Report for the Phare ACE Project P98-1065-R

*Monetary and Exchange Rate Strategies
Related to the Current European Union's Enlargement Processes*

Vilnius

April 15-16, 2002

THE RESPONSE OF BALTIC ECONOMIES TO PEGGING OF NATIONAL CURRENCIES

Antanas Buracas¹

Abstract: The report is devoted to the features of current monetary policy in Baltic economies under the currency board regime in the pre-accession to the EU and re-pegging of national currencies to euro. The identification of some interrelations between the changes in both euro and national currencies flows in the Baltic States, on the one side, and within some interconnected macroeconomic variables, on the other side, also the analysis of monetary transmission channels were attempted to develop using the EER in the econometric cointegrative modeling.

Keywords: *currency board, exchange rates, pegging to euro, modeling of monetary transmission in Baltics.*

JEL classification: E32, E52, P33.

INTRODUCTION

An attention is given to the identification of the currency anchoring influence onto the structural macroeconomic values using:

- distributed lag analysis of monetary transmission flows reacting to the previous pegging of the litas to USD and kroon to DEM and their re-pegging to euro;
- peculiarities of dynamical changes in nominal and real effective exchange rates (REER), their influence on the macroeconomic indices (in particular, sectoral gross value added, tradable and non-tradable sectors) depending from their differences when calculated to currencies of Euro, CEE, CIS zones.

The essence of orientation to euro of the Baltic States is ensuring that even outside the EMU area they would ease their monetary policies to stability, foster convergence before adopting the euro. As was stressed by the European Council (1997), the mechanism (ERM) will help to protect them in case of unwarranted pressure in the foreign exchange markets. Countries joining the ERM can expect overall credibility and, as a result, lower interest rates.

¹ *Antanas Buracas* - Prof. of Vytauti Magnus University, Lukesciu 15, Vilnius, LT 2043. E-mail: anbura@lrs.lt. Internet page: <http://members.xoom.com/buracas>. My deep gratitude to Prof. Dr. *Lucjan Orlowski*, Visiting Professor in the Sacred Heart University, USA, Senior Researcher in the Center for Economic and Social Research (CASE), and Member of the Advisory Council, Poland - for methodological suggestions and comments within statistically unreliable environment; to Dr. *Virgilijus Sakalauskas* - Assoc. Prof. of Informatics, Ph.D. (Math. Sci.), Vilnius University (from 1976), e-mail: virgis@aula.vukhf.lt and Mgr. *Danas Zuokas* - Collaborator of Statistical Center, Vytauti Magnus University: Danas_Zuokas@fc.vdu.lt - for collaboration in the econometric calculations; Mgr. *Rasa Spokeviciute* – for the collecting, restructuring and editing Baltic States statistics.

The research was undertaken with support from the European Community's Phare ACE Program (P98-1065-R project "Monetary and Exchange Rate Strategies Related to the Current European Union's Enlargement processes"). The content and conclusions of the publication is the sole responsibility of the author and it in no way represents the views of the Commission or its services.

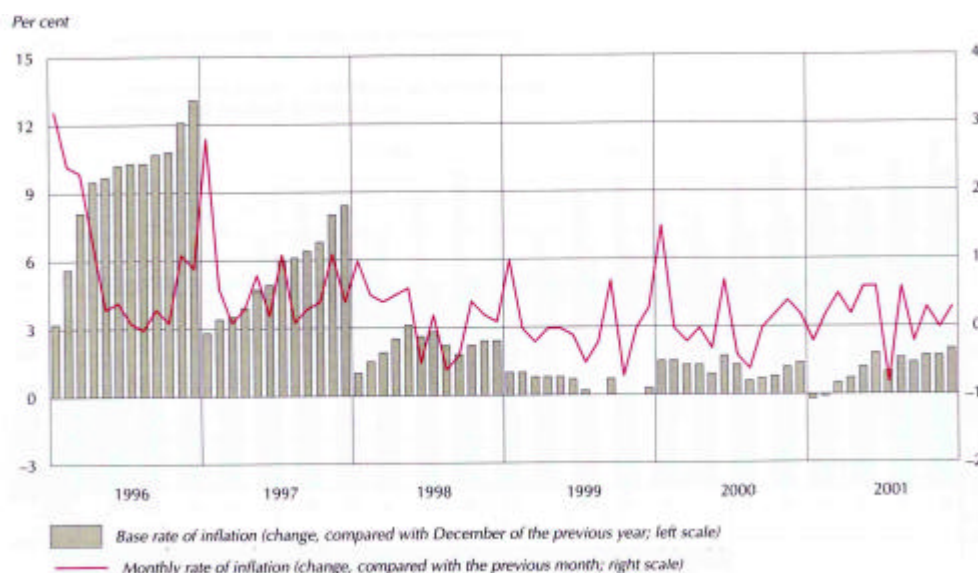
Other authors think that CBA's arrangements are facilitating nominal convergence, allowing a market test for exchange rate stability, helping to ensure that countries enter the euro zone at an appropriate exchange rate; and preparing central banks for operating within the euro zone (A. -M. Gulde a/o, 2000, p.3).

In the transitional period of the integration to the EU, the monetary policies both in Lithuania and Estonia are oriented to the fixed exchange rate anchoring. From Feb. 2, 2002, LTL is pegged to EUR (from the end of 1993 it was pegged to USD). EEK was pegged to EUR from 1999 and before since June 20, 1992, it was pegged to DEM. LVL was linked to the SDR basket of currencies since Mid-Feb. 1994 (at the rate 1 SDR = 0.7997 LVL) and expects to support this position until 2007. All these anchoring measures in monetary policies of the Baltic States helped to lower the annual price indices from three digits to single digit levels within consecutive 4-5 years.

At the same time, additionally, Lithuania seeks for the weak targeting of low inflation rates. This task for the monetary convergence of national monetary systems with euro was declared in the Memorandums of both central banks with the IMF for the nearest years. The fluctuation of the aggregated price levels around very slow (up to 1-2 yearly %) growth in the Baltics last 5-7 years determined the prevailing opinion that the DIT approach which is intensively discussed as a monetary policy measure in the CEE a/o countries is not very adequate monetary measure for the Baltic States. For ex., successful financial policies in Latvia have been reflected in the continuous fall of inflation from 958.7% in 1992 to 23.1% in 1995, and in 2000 - at 1.8%, in Lithuania – from 1170% in 1992 at 2% in 2001 (cf. charts L1, L2 and L3 below).

Chart L1

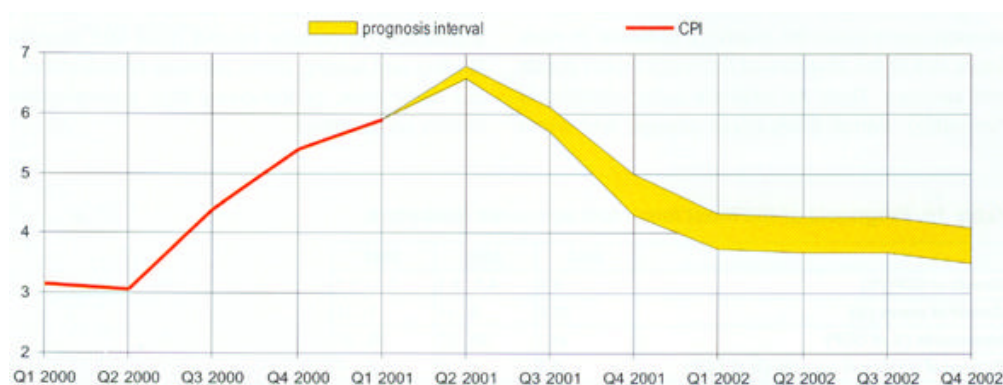
Fluctuations in the Inflation Rate in Lithuania



Source: Bank of Lithuania, 2001

Chart L2

Prediction of Quarterly CPI in Estonia, 200-2002



Source: Bank of Estonia, 2001.

Chart

CPI and Inflation in Latvia, 2000-2001



Source: Bank of Latvia, 2001.

1. PRE-ACCESSION MONETARY POLICY: LITHUANIA AND ESTONIA

1.1. The direct positive impact of integration of national economies to the EU under currency board regimes is different from the integration of countries with independent central banking. The transitional fluctuations of the small national monetary systems under currency board regime in the Baltic States are expected to be somewhat alleviated; the measurable effect of currency stabilization and risk reduction of open market operations are believed to be less tangible.

However, the main economic effect of the integration to the EU is expected to produce arise from: a) widening trade exchanges with the EU as a result of better adoption to its (the EU) *acquis* requirements; b) conciliation and adaptation of specific monetary instruments of EMU.

The achievement of price stability and interest rate convergence, the balance of risk appears to favor for small transition economies with fixed rather than flexible exchange rates in cases of fixed peg (Ghosh, Gulde and Wolf,

1999). There are widespread also opposite arguments that sophisticated float regime would bring even greater welfare at last in some of the Baltic States in the pre-accession period, in some cases attempted to confirm by econometric calculations (cf. esp. Lucio V. de Souza, E. Ledrut, 2001, p.20).

1.2. Starting from the monetary policy programs of BoL and BoE for 1997–1999, the Guidelines for the application of their monetary policy instruments and central banks Memorandums to the IMF, the *fixed exchange rate arrangement* was an acceptable way for Lithuania and Estonia to get ready for EU membership. One of the most important tasks in the medium-term monetary policy of the BoL was to *effect a smooth shift of the orientation of the litas from the US dollar towards the euro*. It was successfully done from 2nd Feb. this year. The Estonian Kroon was adequately re-pegged to the euro from Jan. 1999.

At the moment of re-pegging of LTL and EEK to euro, their previous exchange rates to anchor currency were fixed according to official exchange course meeting the requirements under the Treaty of the EU on common foreign exchange policy (Article 109). It helped to narrow the interval of market adaptation to the euro exchange rate and provide conditions for its participation in Exchange Rate Mechanism III after Lithuania and Estonia joins the EU.

The pegging helped to deal mostly successful with the monetary, financial and even some economic shocks in Baltic States in the early stage of transition to the market economy; in three years, 1992-94, inflation fell from triple to double digits. However, fixed exchange rate arrangements do not guarantee automatically their macroeconomic stabilization and effective control of inflationary forces and, in particular, at the final stage towards accession to the EU and EMU. So, first, Latvia has succeeded in the similar progress under an LVL exchange rate peg with SDR not supported by the CBA. And, second, Argentina's peso crisis under CBA arrangement in 2001-2002 show that coherent simplified cases of direct correlations between CBA and successful sustainable monetary policy arrangements have to be reviewed.

The policy measures that assisted the process of disinflation, in particular in Baltic States, included privatization and de-monopolization of state companies, trade liberalization, deregulation of financial systems and tight fiscal policies (L. Orlowsky, Oct. 2001, p. 3).

The medium-term tasks for the Lithuania and Estonia financial and monetary policy within last two-three years were geared to achieve a gradual reduction in the current account deficit, and to ensure that the government-borrowing requirements will be consistent with available financing. Fiscal policy was guided by the objectives of achieving a broadly balanced fiscal position in 2001-2002 and beyond, while safeguarding reasonably targeted social expenditure².

² Cf. Guidelines for the Application of the BoL's Monetary Policy Instruments, 1999; BoL's Memorandum to the IMF, 2000 (p.44, 72). The fiscal deficit target of 1.4 percent of GDP for 2001 was consistent with the authorities' macroeconomic objectives. The Monetary Policy Program of BoL for 1997–1999 announced exit strategy from the currency board but later it was postponed. As noticed Mr. Peter M. Keller: "Reliance on privatization proceeds from foreign investors and external borrowing to finance the fiscal expansion helped moderate the increase in interest rates".

The Lithuania's and Estonia's strategy of achieving an orderly current account adjustment through fiscal tightening and structural reforms in the context of the currency board arrangement brought significant positive results in 1999 - 2001. The economic growth resumed while inflation remained low, the current account deficit declined faster than expected, confidence in the currency board was maintained, and financial markets stabilized. As stated by Mr. Shigemitsu Sugisaki, Deputy Managing Director, representing the opinion of the Executive Board of the IMF, the growth of Lithuanian economy was export-led, as competitiveness was sustained through productivity growth and terms-of-trade gains, despite a further real appreciation of the litas. Structural reforms advanced significantly, including in the areas of energy sector restructuring, privatization, trade policy, and budget and treasury management. Estonia also reduced its current account deficit to an estimated 5-6 percent of GDP. However, domestic demand remained weak and unemployment in Lithuania continued to rise.

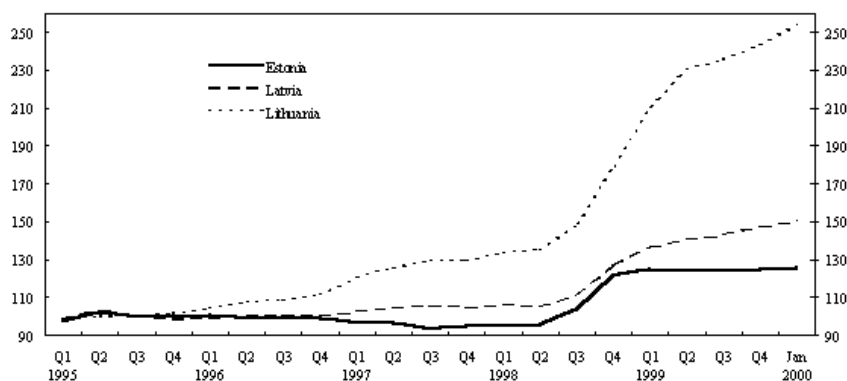
1.3. The Lithuania's central government and fiscal authorities plan to clear municipal arrears by mid-2001, also to complete the treasury modernization project by early 2001, and to set up a reserve stabilization fund to manage privatization receipts in a transparent manner within first half of 2001 are continued in 2002. The governments of Lithuania and Estonia are taking into account the spending pressures arising from EU and NATO accession, an aging population, structural unemployment, and energy sector reforms eroding the overall fiscal position.

One of important functional measures when striving for this was the consolidation of all state funds outside the government budget into single state budget in Lithuania. The aim was to reduce the national government financial deficit down to LTL 800 million, or 1.8 percent of GDP, in 2000, and to balance the general government budget by 2001 what was about achieved.

When the exchange rate is fixed, an independent interest rate or money supply policy cannot be made instrumental for achieving an acceptable inflation level. Now the Baltic States have one of the best price stability records for any economy in transition: Lithuania less than 1% of the CPI in 1999 and about 1.7% of the CPI in 2000. 40% of the CPI basket caused the sharp decrease in food prices due to the shutdown of the Russian market, appreciation of the litas and depressed consumer demand.

Chart 1.1.

Appreciation of Comparative Real Effective Exchange Rates: Baltic States



Calculated on the data of: the BoL, Bank of Latvia, and Bank of Estonia.

The increasing share of lending denominated in foreign exchanges usually reflects lower interest rates on resources denominated in those exchanges. The growth in shares of deposits denominated in foreign currencies in Lithuania (54,8 % in 2000, 43% in 1999 and 2001, up from 36% in 1998) witnesses about weak credibility and depth of the domestic banking system (Raimondas Kuodis, 2000)³.

The currency board regime in Lithuania does not seem to be explicitly procyclical: judging from the ratio of M2 to GDP which is about 20%, and that of domestic credit to GDP (15%) the link between the banking system and the real economy is relatively weak (Raimondas Kuodis, op.cit).

The traditional channels of monetary transmission are also rather narrow. The transmitting channels, which could be very sensitive to the increase in interest rates, such as housing mortgage or consumer credit, are being developed very slowly in all Baltic States.

Lithuania like Latvia and Estonia successfully aimed to ensure monetary and stability criteria applied by the Euro area partners concerning inflation convergence, the exchange rate mechanism, budget deficit and stock of state debt criteria.

Within the Guidelines for the Application of Monetary Policy Instruments the BoL and BoE have returned from the application of direct market instruments towards the application of indirect market instruments. Presently, the BoL uses limited way the following monetary policy instruments: reserve requirements; open market operations: repos (used to increase liquidity) and deposit auctions; collateralised overnight loans; lombard facility used for settlement purposes; also collateralised liquidity loans: lender of last resort loans in special cases. it also applies the over-night loans that are extended against collateral of Government securities. Since the banking system's demand for excess reserves is ever changing and is difficult to predict, the BoL and BoE does not announce their "targets" for interest rates or volumes prior to a market operation. This enables the BoL to get hold of information on market trends.

Open market operations facilitate transmission of external changes in interest rates to domestic markets taking account of the imperfect links between the latter and international capital markets. Finally, they meet the objective of promoting the development of the monetary and capital markets. As an additional safeguard for the stability of interbank settlements, the Bank of Lithuania introduced overnight lombard facility.

The principles of the BoL and BoE's monetary policy and the instruments for their implementation are similar to those of the European Central Bank (ECB). In particular, the reserve requirement is aimed at ensuring a higher and more sustainable liquidity of the banking system. The reserve requirement will be approximated with the procedure applicable in the EMU, including liabilities to non-residents.

The amount of direct Lithuania's trade in EUR has been growing very insignificantly until 2002: in January 1999 it accounted for 2,3 % of total currency exchange (from 0.2-0.5 in XEU in 1998), and in first half of 2000 - 4.1 –

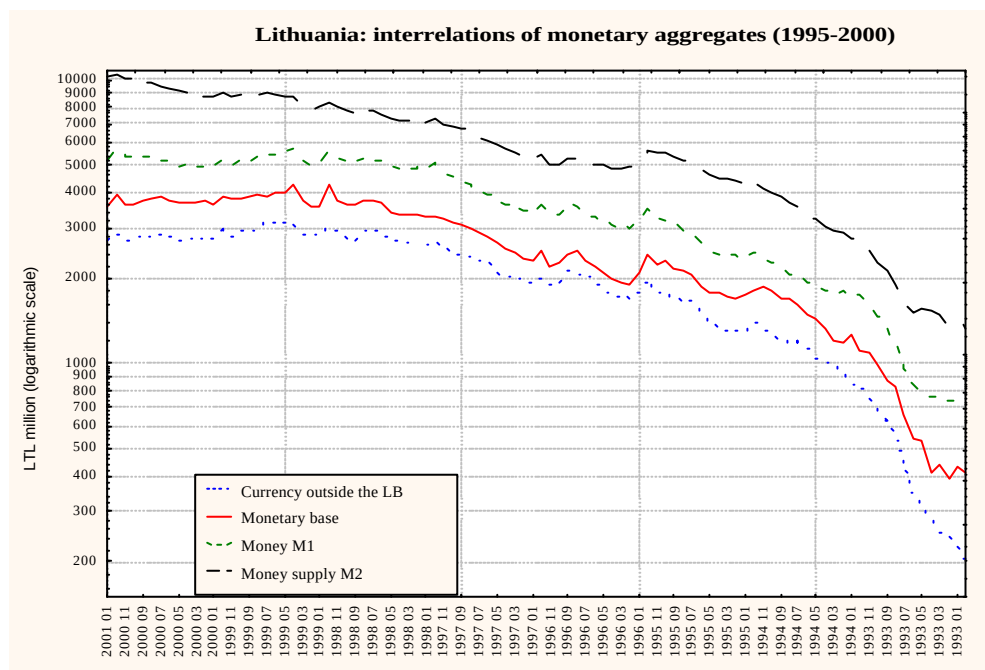
³*Currency Boards - Experience and Prospects*. Seminar organized by Eesti Pank, 5-6 May, Tallinn.

5.1 %, in Nov. 2001 – about 10 %⁴. Adequately, its trade with EU hardly exceeded 32 percent of all foreign trade before Russian finance crisis in 1997-8. However, last year and half the part of Lithuania's trade with the EU grew until about 50 percent (49.4 percent of export and 50.3 percent of import⁵).

The situation in Estonia was cardinally different: if in 1998 financial settlements in Estonian kroons and German marks accounted for 36 % of the special export then in June 1999 the settlements in Estonian kroons, Euro and Eurozone currencies amounted to 66 %. The share of exchange rate risk-free settlements in special import increased from 29 % in 1998 to 66 % by June 1999⁶. That resulted in the significant attention of Estonian statistician's too much more careful registration of any financial flows in Euro than in Lithuania.

1.4. The monetary aggregates of LTL in the period of 1995-2000 are shown in the chart 1.2 (and chart 1.2a in the annex). The divergence of their trends to 2000 (on left side) shows anyway the similarity of the monetary aggregates M0, M1 and M2 in Lithuania; their characters within observed period are rather identical logarithmically.

Chart 1.2.



Calculated on the data of the BoL.

The most difficult monetary variable to imitate within monetary transmission processes was the interest rates as independent factor specifically influencing such small open economies as are Baltic States: the interest rates themselves were substantially determined by the world economic conjuncture, often much more than internal financial processes (cf. charts 1.5 and 1.5a in the annex).

⁴ Currency Markets, BoL, Nr.7, 2000, p.8; Nr.12, 2001, p. 4.

⁵ Calculated on basis of: Quarterly Bulletin, BoL, 2001, N.3, p.82.

⁶ Cf. Teet Rajasalu a/o. Implications of the Introduction of the Euro for the Estonian Financial Markets // Eesti Pank. Bulletin, No 4, 2000, p. 26.

2. INFLUENCE OF EXCHANGE RATE REGIMES ON MACROECONOMIC CHANGES IN BALTICS

2.1. The effects of the exchange rate regimes in the Baltic countries are still subject to analysis and assessment of the currency board influence, in particular, on:

- Consecutive macroeconomic growth and its structural changes in these countries;
- Compatibility with ERM II in pre-accession period with account of sustainability indicators, such as sectoral price and wage, current account (esp. export performance) developments, interest rates, competitiveness, financial account flows;
- Stability to various shocks of small open economic systems, even after accession to the EU.

Since many of these variables adjust much more slowly than nominal exchange rate, an assessment under a currency board requires that this regime has been effectively in operation for a number of years, mentioned Christian Thimann, Head of Bilateral Division⁷.

After pegging the LTL a/o national currencies of the Baltic States to the euro, their exchange rates to the euro and one to other cross exchange rates became fixed. In this connection the exchange risk and cost of conversions between these currencies for both the corporate and financial sectors was eliminated. The re-designing of inter-bank funds transfer system by establishing an integrated payment system, which allows the settlements, defined by the ECB in real time expected to be completed by 2004.

The alternative simulations of monetary policy instrumentation in the Baltics using the traditional progressive and innovative econometrics of structural currency interrelations (like Harrod-Balassa-Samuelson, Mundell-Fleming or Dornbusch a/o.) are still problematic and questionable as a result of:

- absence of some important statistical ingredients identifying structural breaks in the monetary transmission channels and sectoral productivity parameters of the Baltic States (interrelations with the effective exchange rates);
- mostly adequate macrostatistics of these transition economies but their indicators of low reliability after intensive privatization limiting with rationality of econometric constructs⁸.

The changes in current account of Lithuania shows specific trend to the reciprocity between slope of its balance and credit / debit value dynamics but it is possibly determined by exogenous impact of the Russian crises, not by interdeterminacy of these aggregates by themselves what is not obvious.

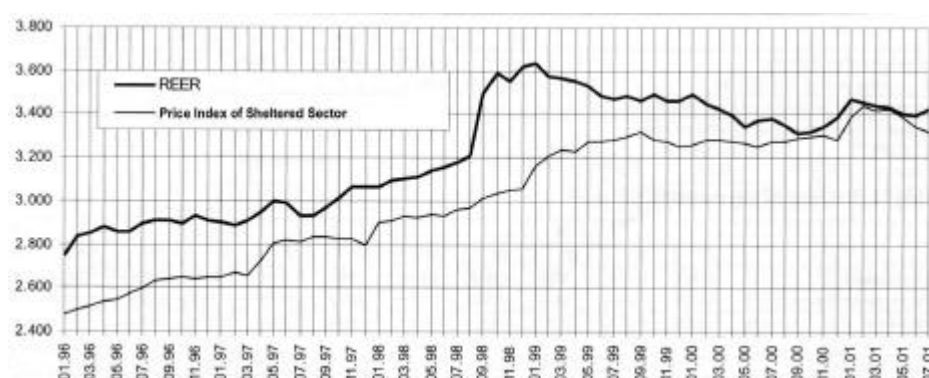
⁷ Preprint of his report *Monetary Policy Challenges in Transition and Towards Accession* // Vilnius, 8-9 Feb. 2001, *Economic Policy in the Framework of Accession to the EU and EMU*, p.25.

⁸ The participants of the international conference “Economic Policy in the Framework of Accession to the EU and EMU” hold under auspices of the EU delegation in Vilnius, 8-9 Feb., 2001, designated the successes when reforming monetary statistics in Baltic States however it is still insufficient and unreliable for more sophisticated monetary analytical purposes in-deep within the pre-accession period.

The shadow sector amounts up to 40-42 % of the GDP, according to official valuations of the Lithuanian Statistical Office (table 2.1a in annex)⁹. The influence of sheltered sector widespread onto about any economic sector including the price indices (CPI a/o.) and the real exchange rate indices what is clearly demonstrated by the BoE (cf. charts 2.1 and 2.1a in annex, Eesti Pank Bulletin, 2001, No 2 (59), pp. 69, 71).

Chart 2.1

REER and Prices in Sheltered Sector in Estonia, 1996-2001

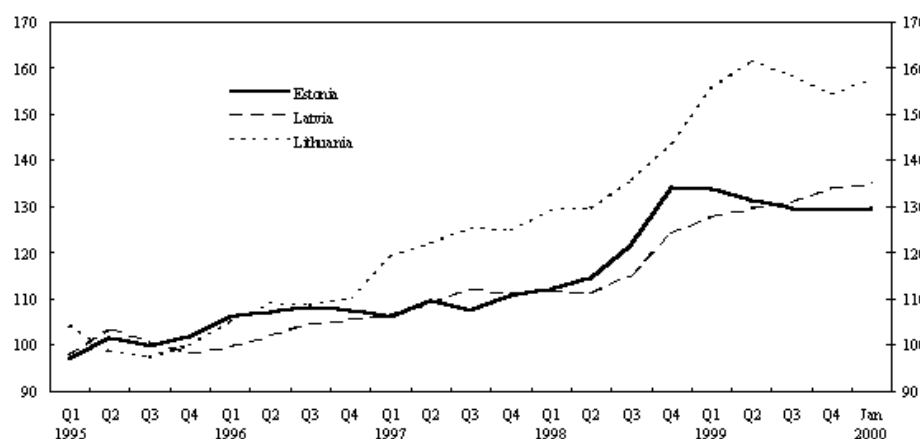


Source: Eesti Pank Bulletin, 2001, No 3 (60).

2.2. The interrelations between effective exchange rate (EER) dynamics and structural macroeconomic data are indirect but important analytical means for the identification and evaluation of the influence of pegged exchange rates' onto economic growth and financial flows and transitions in their base in such complicated analytical environment. They help in some degree as well for the determination of the effects (positive or negative) of pegging in the pre-accession to the EU period. The general character of comparative curves of the EERs in the Baltic States is presented in the charts 2.2 (nominal EERs) and 2.2a (real EERs).

Chart 2.2.

Comparative Nominal Effective Exchange Rates: Baltic States, 1995-2000 (1995=100)



Calculated on the data of: the BoL, Bank of Latvia, and Bank of Estonia.

The charts 2.2 and 2.2a shows that both real and nominal effective exchange rates of national currencies in all three Baltic States changed by very different way within period 1995-2000. But all of them clearly

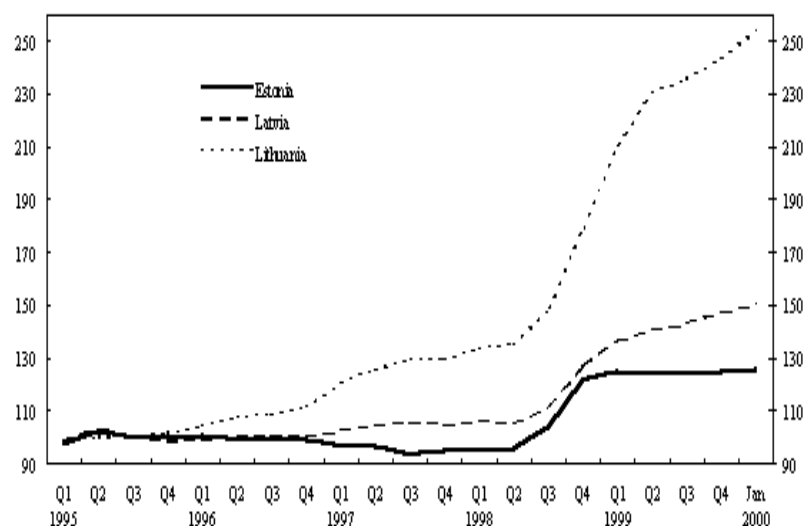
⁹ Also cf. Antanas Buracas (2000), p.37-38, table 4.1.

experienced the strong appreciation from the start of the Russian crisis in 1998. Besides, EEK and LTV reacted earlier and not so sharply as LTL; the similar break point in the dynamics of LTL was about 3 quarters later and 30 points sharper than EEK and LTV in the case of nominal effective exchange rates (1995=100). Trend of the real effective exchange rates (1995=100) was smother and about 120 points sharper for LTL than for EEK and about 100 points sharper for than for LTV.

Chart 2.2a.

Comparative Real Effective Exchange Rates: Baltic States, 1995-2000

1995=100

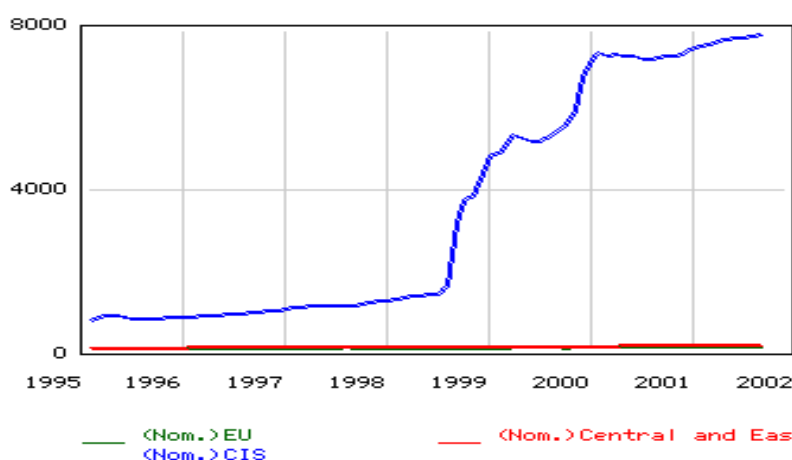


Calculated on the data of: the BoL, Bank of Latvia, and Bank of Estonia.

Nominal appreciation of Baltic national currencies mostly within last two years would enhance their NEER to react to asymmetric shocks and appreciate their REER as a result of faster productivity growth in the traded goods sector (comparing this with sector of non-traded goods, cf. A.M. Gulde a/o., 2000, p.18). Some specifical peculiarities of changes of the NEER for the LTL and EEK are presented on charts 2.3, 2.4 a/o.

Chart 2.3

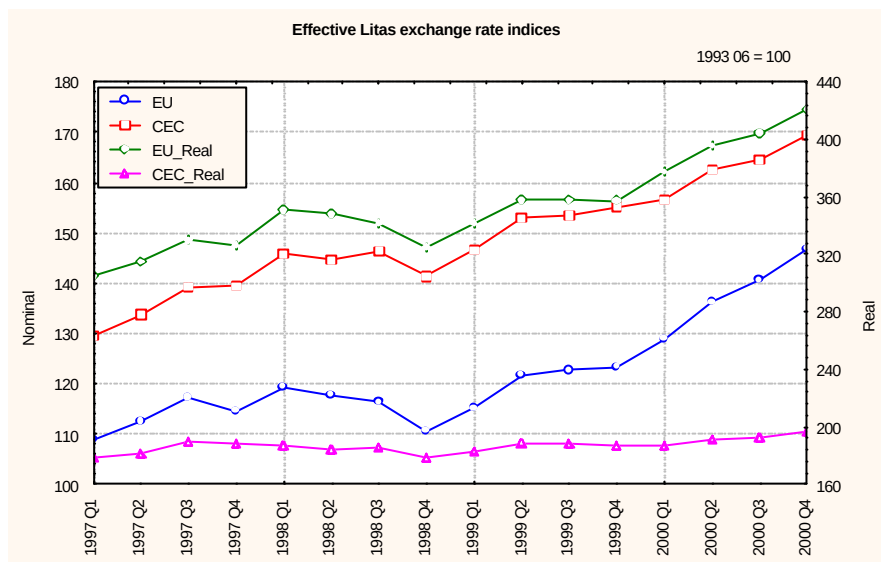
Quarterly Changes of the Nominal Effective Exchange Rates of the LTL



Calculated on the data of the BoL.

Naturally, that floating bands or corridors between real (REER) and nominal indices of LTL, EEK and LVL effective exchange rates are inadequate when taken against euro (before - ECU) and against currencies of CEE countries in transition (cf. chart 2.4). The slope of curves for the real and nominal effective exchange rate indices against the EU currencies is about similar on the chart 2.4 within observed period. But the slope of curves for the real and nominal effective exchange rate indices against CEEC's currencies is expressively different with widening the gap between their changes especially after Q4.1998, mostly as a result of RUB inflation.

Chart 2.4

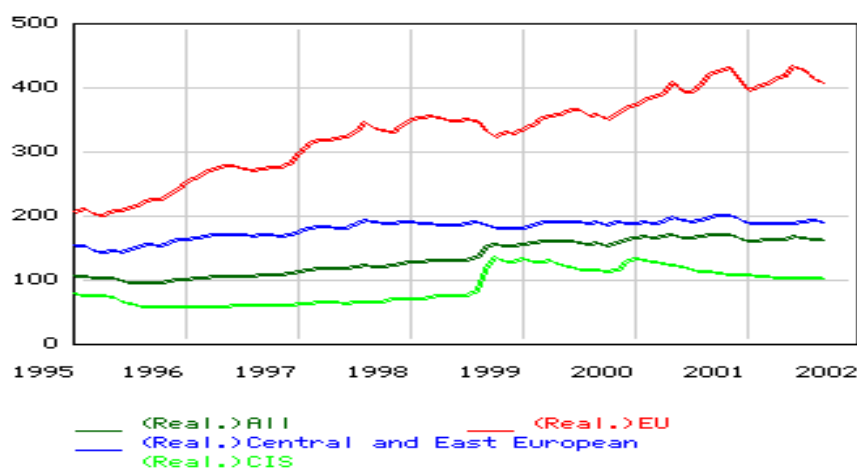


Calculated on the data of the BoL.

Under previous monetary situation when the LTL was about only currency in North Europe pegged to USD, Lithuanian economic system suffered from *additional "asymmetric shock"*. Its economy recorded substantial losses in current account balance in 1999-2001 as a result of low external competitiveness of Lithuania on the EMU markets where euro was loosing its weight against USD. Lithuania's trade exposure to the CIS was larger than Estonia's or Latvia's when the Russian crisis was echoed in the East and Central Europe, and even in more alienated world stock markets.

Chart 2.5.

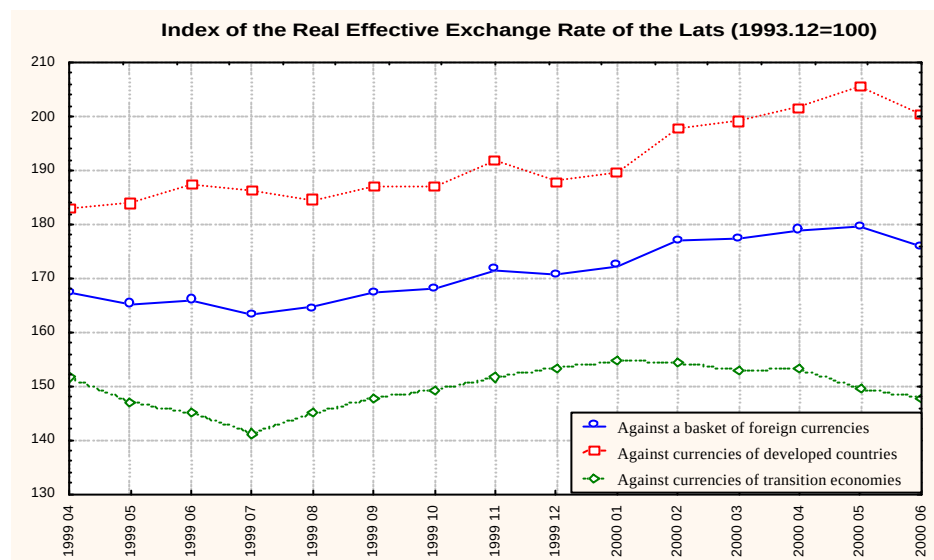
Quarterly Changes of the Real Effective Exchange Rates of the LTL: 1995-2002



Calculated on the data of the BoL.

At the same time, the depreciation of the EUR against USD about from the issuing the single EU currency made questionable for the Baltic States to use the *inflation-targeting monetary policies*. The significant appreciation of LTL and LVL respectively to EUR and RUB as a consequence of strong pegging to USD and SDR lowered not only inflation in Lithuania and Latvia below normal domestic policy targets but also their GDP and widened disbalance of their payment balances.

Chart 2.6



Calculated on the data of the Bank of Latvia.

In its September 1999 report, the IMF concluded that the Litas was still not overvalued. The sharp real exchange rate appreciation on a CPI basis between 1994 and 1999 reflected the wide interval of initial undervaluation of PPP of LTL. The nominal appreciation of both LTL and LVL to EUR resulted in a loss of competitiveness of both countries and the growth, depressive to Lithuania and Latvia, of the foreign trade balance deficit for about 3 years period (until spring 2000).

The situation was much more aggrieved by the fact that substantial part of the international loans from the European countries to both Baltic States matured at this period were contracted in USD and contracts with CIS countries, also in USD, were mostly defaulted.

2.3. The low inflation, low prices were predetermined in the Baltic States as a consequence of their blocked foreign trade with CIS countries within this period (until late 2000) and many traditional products (food, some other consumer items) left unexported. The low CPI was also determined by interconnected low purchasing power, weak internal consumption and low competitiveness in the internal markets against imported goods in the Baltics.

It is nontrivial to distinguish between negative effects resulting from pegging of LTL to USD especially in the view of its decreasing PPP, and that (negative effect) resulting due to inefficient economy. When for statistical purposes this distinction is made in terms of the GDP, the modeling solution could be found with parameters more adequately reflecting the actual functioning of the economic system in Lithuania.

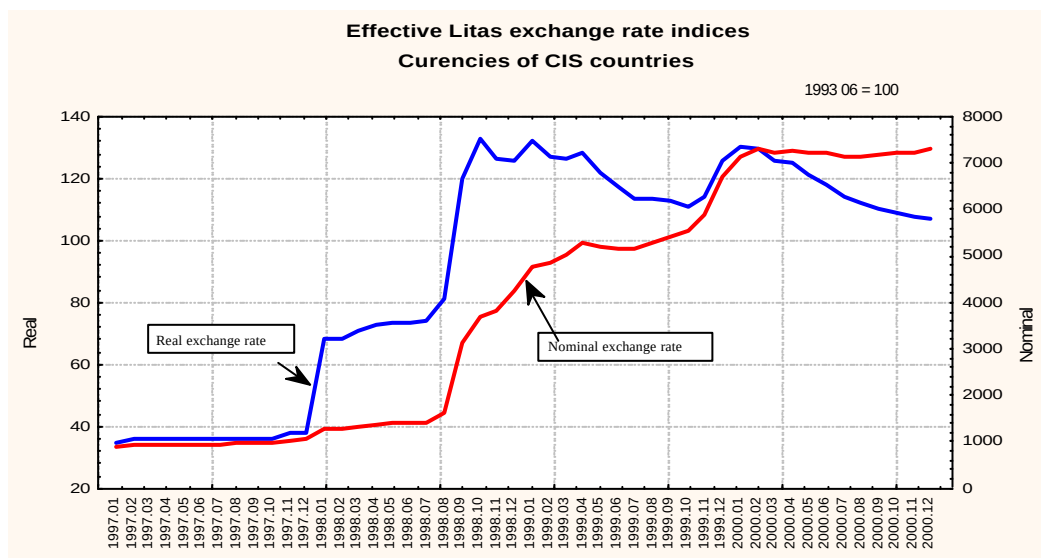
These interrelated facts of the national currency pegging under strong crisis in neighboring states, in particular, do not contradict to the argument, persuasive from common sense and practice: “The long-term objective of inflation rates closer to EU or euro area levels should help to anchor inflation expectations and outcomes over the medium term. The experience of emerging markets has shown that real convergence and low inflation do not stand in conflict, but rather that by focusing on low and stable inflation, monetary policy makes an important contribution to growth.” (Christian Thimann, p. 24).

Interesting lagging in the slopes of the LTL real and nominal effective exchange rate indices can be seen in their dynamics against CEEC’s currencies at the period of the last economic crisis in 1997.01-1998.08-09 (cf. charts 2.5 and esp. 2.6). It determined a huge, about doubling, gap from the beginning between the both levels when Russian crises’ influence was sheltered, especially in the financial area. It is important that this phenomenon was about repeated in the following period 1998.08-1999.09 of stronger crisis’ manifestation in Lithuanian economy.

When both gaps in the slopes of the LTL real and nominal effective exchange rate indices are like chain-interconnected, the regular lag between both of them is something about 8 months for the first one and 12 months for the second (cf. chart 2.6).

The stabilizing effect of resulting both: growth of the Lithuanian foreign trade with the EU partners and consecutive diminishment of the trade with partners in CIS countries went especially intensively within period 1997.01–1999.09. Lithuanian export to CIS part in total its export lowered from 46.4 percent to 17 percent in 2001. Such cardinal changes of international economic relations from one currency zone to another at unbelievably short period, much more deep than in the beginning of transitional period in Lithuania from central planning to the market economy in 1990-1994.

Chart 2.6



Calculated on the data: Bank of Lithuania, Eurostat.

The continuing *appreciation of anchor currencies* in the Baltic States in 1993-99 was resulting not from productivity gains but mostly from *undervaluation of the nominal exchange rates* of all three national currencies -

LTL, EEK and LVL – at the period of their introductory emission about in 1992-93, shortly after which their pegs were installed.

Litas' exchange rate flexibility depends to a certain extent on the loan markets. Then lowering of the reserve requirements by BoL from 10 percent to 8 (in the EU it equals to 2 percent) is possible to expect the additional money supply in the internal market and more competitive lower interest rates. In the following sections 3 and 4 of the report the regressive interconnections between loan factor and other monetary transmission channels and compound macroeconomic equations of the small open economies were verified.

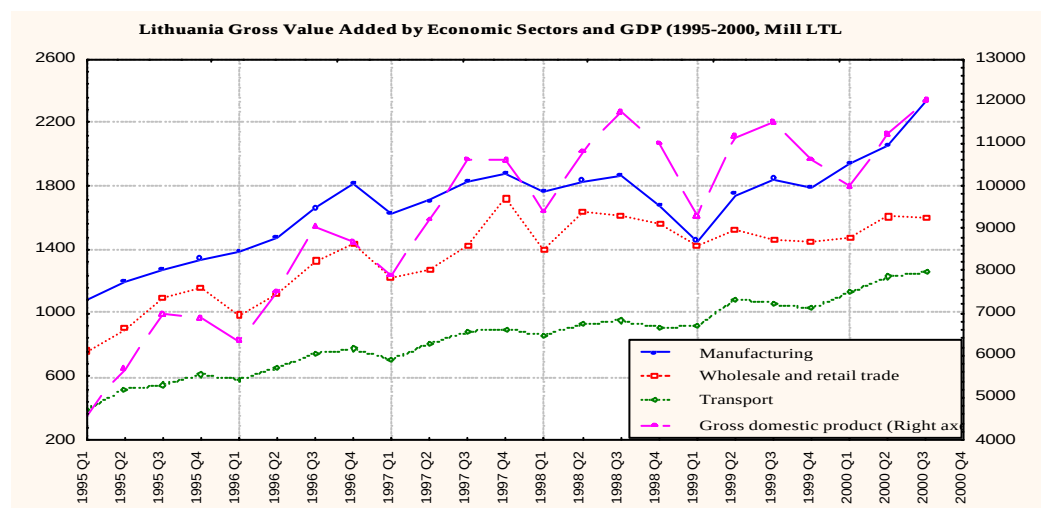
The understanding of the character of possible interrelations between the changes in the current account and the REER using graphical comparisons show rather complicated and not very clear interactions (cf. charts 2.7 and 2.7a in the annex). The same follows from more detailed econometric calculations when choosing the form of integrated macroeconomic interaction with monetary transmission mechanisms.

3. CORRELATIONS BETWEEN STRUCTURAL ECONOMIC CHANGES AND EER IN BALTIC STATES¹⁰

3.1. It is important to evaluate the structural monetary and economic changes in the national systems of the Baltic States connected with the fluctuations within effective exchange rates (EER) of their national currencies to EUR.

In this context we have to mark especially the significant autocorrelations between the relative exchange rates of LTL and dynamics of the gross value added sectoral indicators within manufacturing, wholesale and retail trade of the Republic of Lithuania. They are mostly interrelated similar way with the general slope of the changes within the GDP of Lithuania (cf. charts 3.1 and 3.2) in 1995-2001.

Chart 3.1.



Calculated on the data of the BoL, Lithuanian Department of Statistics.

¹⁰ Calculations and their analysis in Part 3 were done together with Dr. *Virgilijus Sakalauskas*.

It is possible to identify visually the waves of crisis in Russia pressing and intervening the period of EUR denomination to USD on the productivity of Lithuanian economic sectors following from interrelated changes within its foreign trade with the CIS and the EU area.

The interrelations and changes not so strong but similar in their character may be identified in Latvia. The exchange rates of the USD to LVL and EUR to LVL exchanged their positions as a result of unexpected appreciation of USD to EUR in post-crisis period of 1999-2000 (cf. chart 3.2).

Chart 3.2

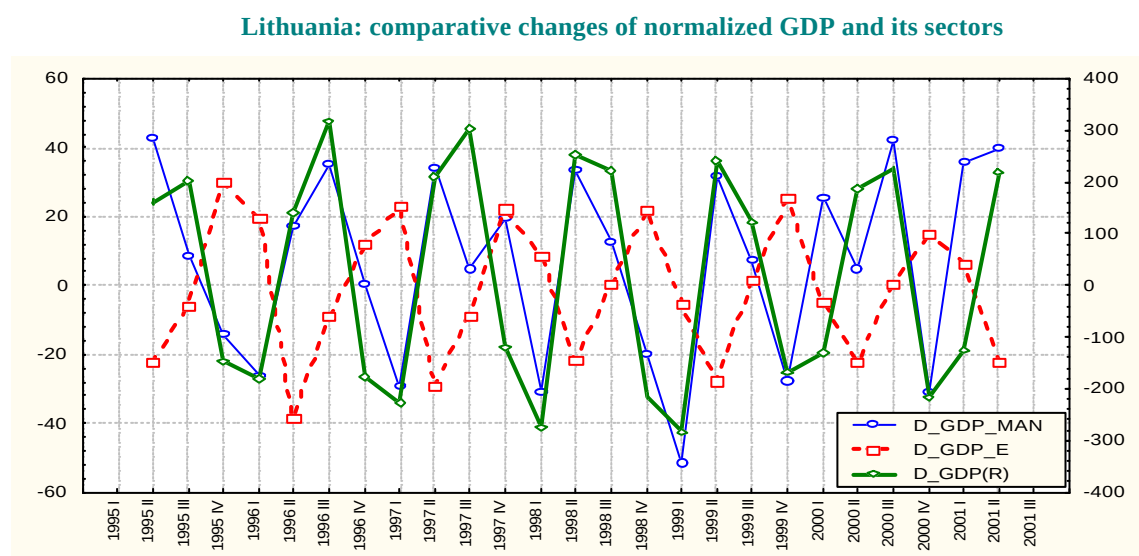


Source: data of the Bank of Latvia.

The interrelations between changes of the GDP and its components produced in the manufacturing (GDP_MAN) and services (GDP_S) are trivially direct and mostly of the same direction; those with energy sector (GDP_E) are changing in opposite direction what in fact possibly could be interpreted as changes with a lag (mostly t-1), cf. equation 3.1 and esp. chart 3.2:

$$\text{GDP} = -8,3219 + 2,9627 \text{ GDP_MAN} - 7,8494 \text{ GDP_E} + 0,3932 \text{ GDP}(-1) \quad (3.1)$$

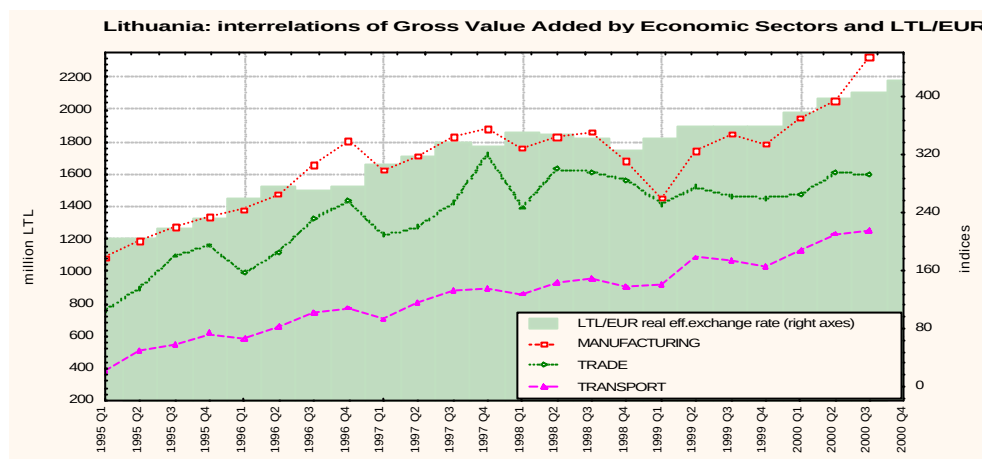
Chart 3.2



Calculated on the data of the BoL, Lithuanian Department of Statistics.

The regression equation between the GVA in manufacturing, trade and transport sectors of the Republic of Lithuania (LT_man_t) and the effective exchange rate of LTL to EUR (both direct and lagged) based on the quarterly data for 1995-2001 shows that regression coefficients are strong (cf. annex 2, equat. 3.1a). The graphical view (chart 3.3) confirms the complicated character of these interrelations but also their principal workability in necessary connection.

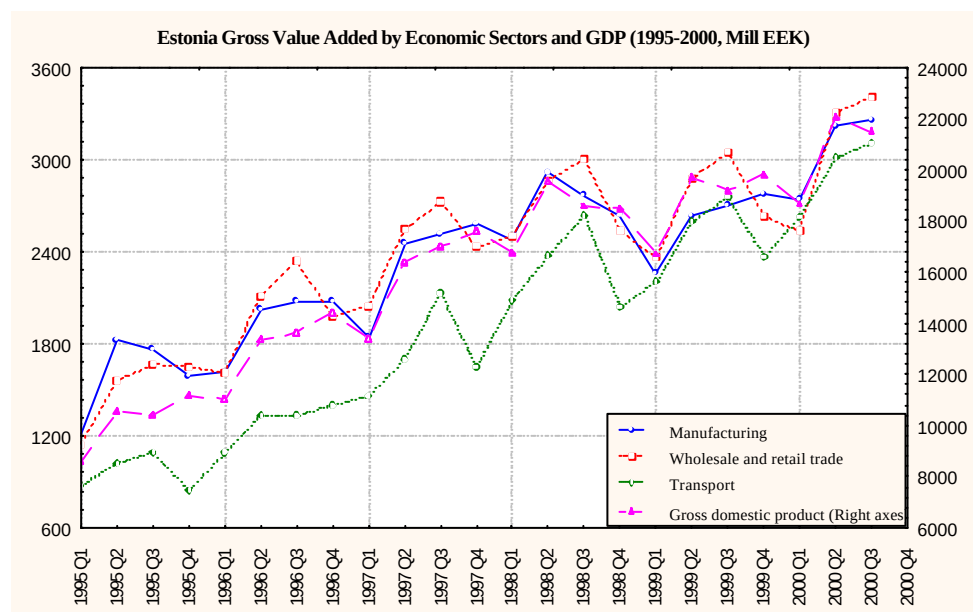
Chart 3.3.



Calculated on the data of the BoL, Lithuanian Department of Statistics, Eurostat.

The same significant autocorrelation between the exchange rates of EEK and dynamics of the gross value added sectoral indicators could be seen in the chart 3.3. The difference from chart for Lithuania is more condensed batch of the curves in the case of Estonia.

Chart 3.3a.



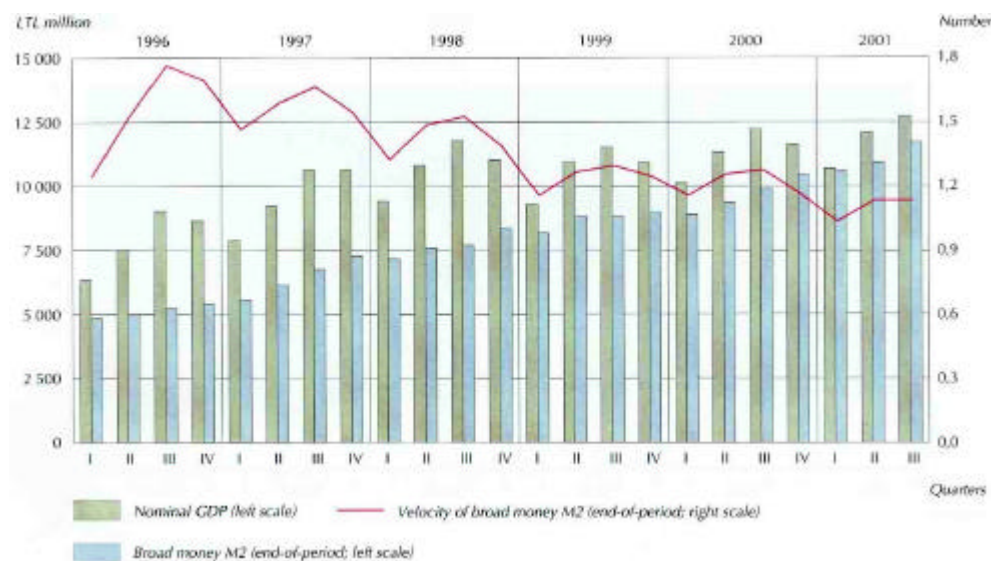
Calculated on the data of the BoE, Statistical Bureau of Estonia.

3.2. The interrelations between dynamics of the LTL exchange rates, on the one side, and the monetary aggregates M0, M1 and M2, on the other, in Lithuania are clearly expressed but not so strong as in the case of the gross value added by main economic sectors within observed period.

The slopes of REER of EEK and REER of LTL are not fixed like the similarities of the slopes of M1 and M2 for both for Lithuania and Estonia (See charts below, also 1.4 and 1.4a in Annex). The monetary aggregates of EEK also have negative values for the current time correlations with EEK/EUR.

Chart 3.4

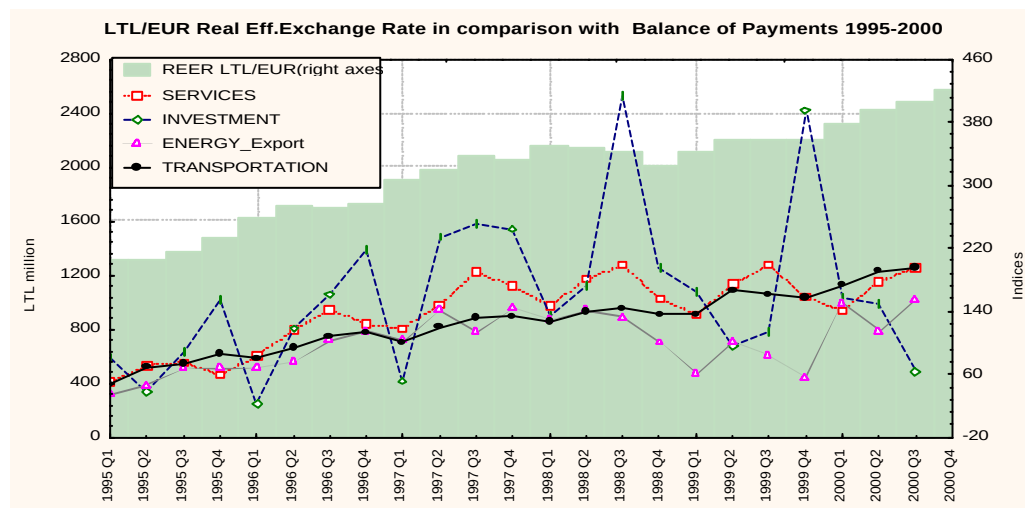
GDP, Broad Money and its Velocity in Lithuania, 1996-2001



Source: Bank of Lithuania, 2001.

3.3. The interrelations between changes in real effective exchange rates of the national currencies and sectors, or more exactly, groups of decisive items within balances of payments (BoP) are incontrovertible, however much more complicated than in case of direct interrelations with the changes within gross value added by economic sectors (incl. manufacturing, services, transportation, also investments, cf. chart 3.5, also chart 3.5a, regressions 3.5, 3.6, 3.7 in the annex).

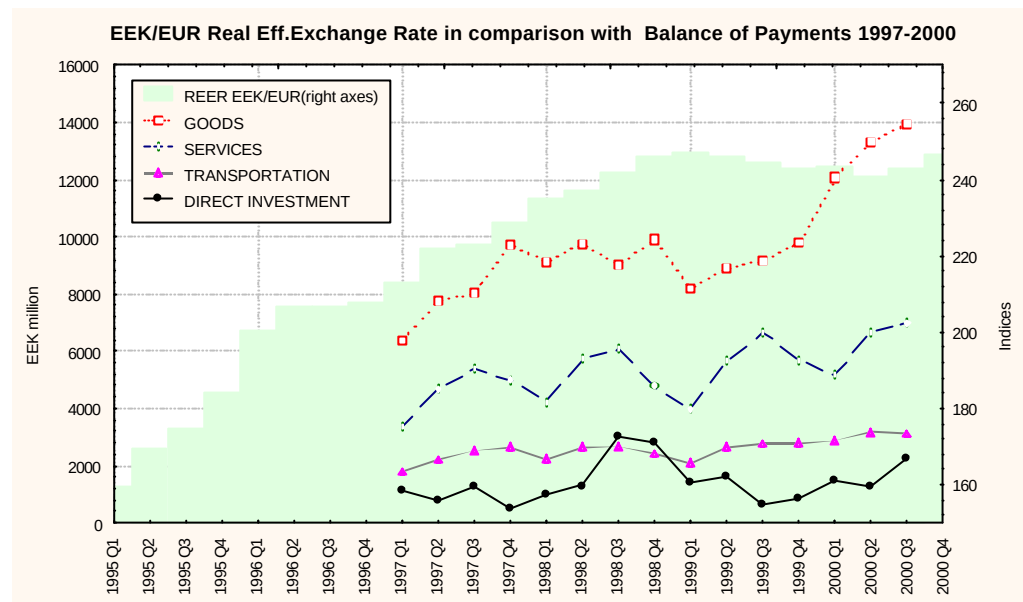
Chart 3.5.



Calculated on the data of the BoE, Statistical Bureau of Estonia, and Eurostat.

Graphically the obverse character of dynamics and interconnections of both variables is expressed especially impressively. In general, in 1993-1995 the real exchange rates of the Baltic States were negative but their current accounts were positive, and in 1996-2000 the situation become opposite

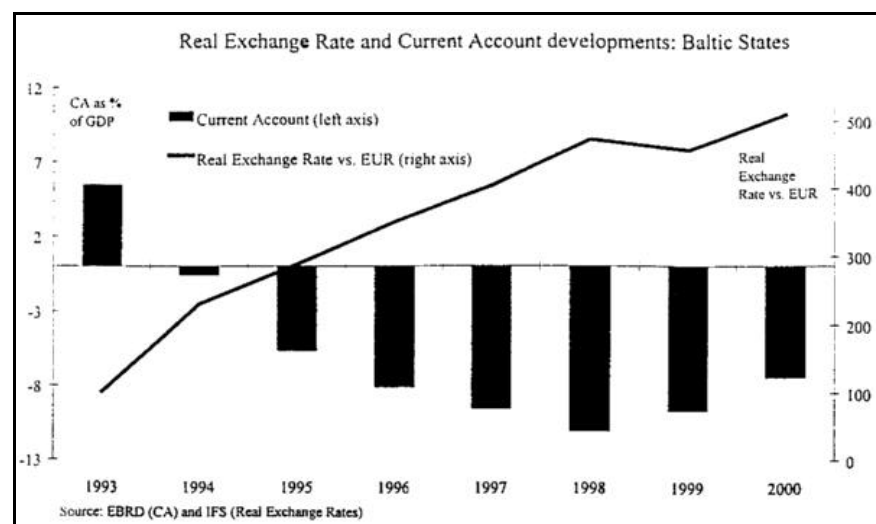
Chart 3.6.



Calculated on the data of the Bank of Estonia, Estonian Department of Statistics, Eurostat.

The changes and interrelations of the EER and current accounts balances of the BoP in Baltic States are presented in the chart 3.7 and, more detailed for Lithuania and Estonia, also in the charts 2.7 and 2.8 of Annex.

Chart 3.7.



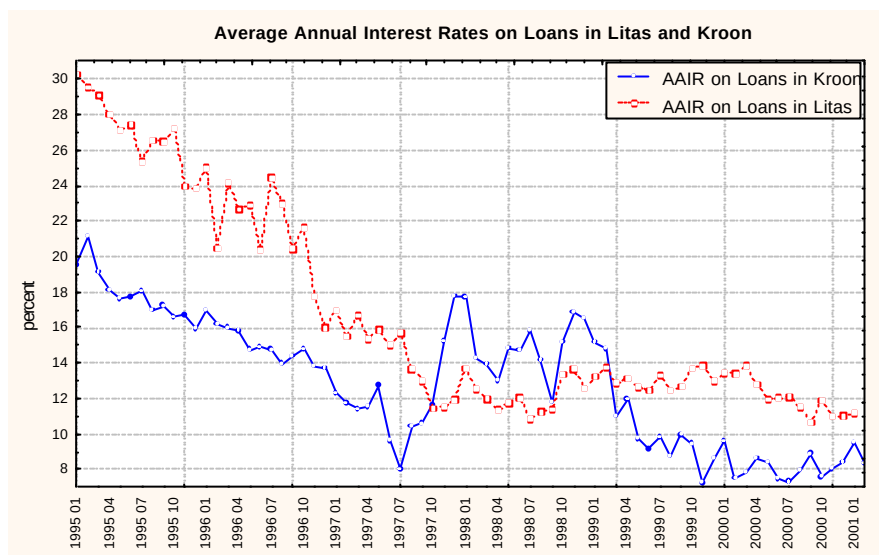
Calculated on the data of the BoL, Bank of Latvia, Bank of Estonia; Lithuanian Department of Statistics, Eurostat.

The changes in national capital account are even more complicated after the euro was issued as a single EMU currency due to more sophisticated motivation of investment processes. The shift to direct payments in euro naturally require to assess the risk of increasing the share of euro-denominated short-term external claims which

means related higher portfolio risks (as a result of smaller diversity of securities in the EMU zone and their increased volatility in comparison to global asset markets¹¹.

3.4. The fluctuations of the interest rates for the loans were lowering until the Russian crisis but now they are competitive with similar rates within the CEEC credit markets. The interesting fact is fixed in 07.2000 when interest rates for short-term kroon loans in Estonia were below of the interest rates for long-term kroon loans; similar situation was fixed also in Lithuania.

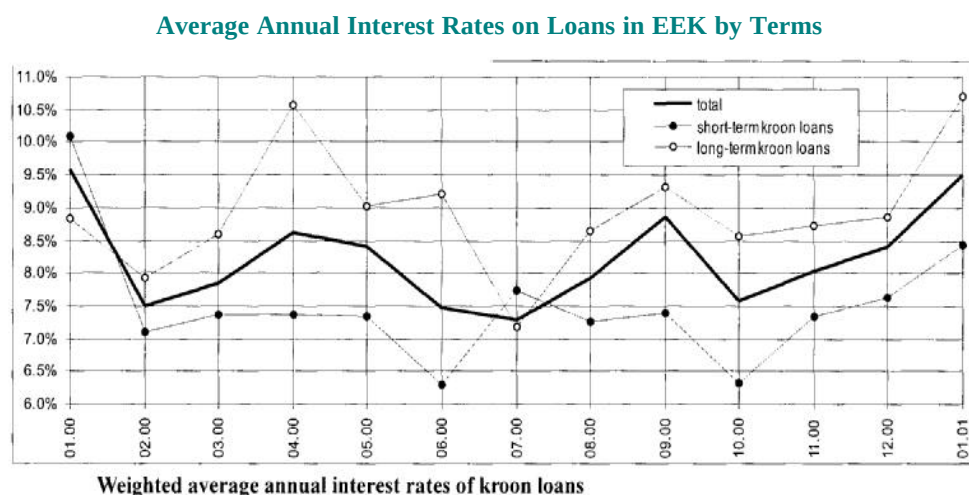
Chart 3.8



Calculated on the data of the BoL and Bank of Estonia.

The correlations between the annual interest rates for the loans both in areas of the national pegged currencies and those areas with key currencies are mostly inversive and not strong so they mostly used in specifications only in limited cases (cf. charts 3.8-3.9).

Chart 3.8a

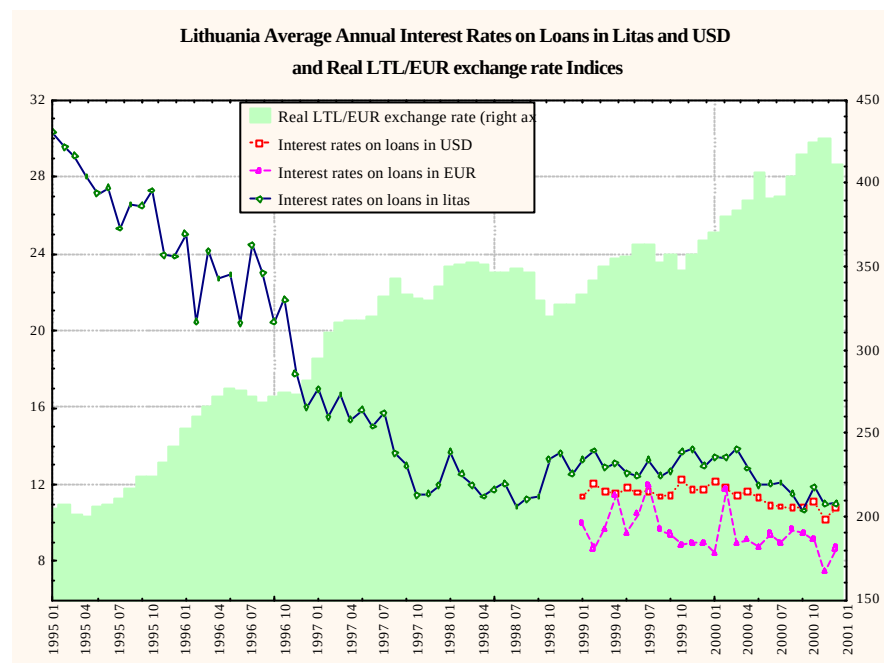


Source: Bank of Latvia.

¹¹ Teet Rajasalu a/o. op. cit., p. 7.

Their (AAIR) imitation is complicated by continuing process of privatization, changes in basic rates of international money markets, big investments from abroad, fluctuations of state debts; about all these factors strongly influence small open economies like the Baltic States.

Chart 3.9



Calculated on the data of the BoL, Eurostat.

3.5. The comparison of the results from regression analysis presented above for the prediction purposes is difficult as result of different absolute statistical values. One of possible ways is to compare MPE (mean percentage error), MAPE (mean absolute percentage error) or use the coefficient of determination of regression equations. We prefer the last one possibility. The results of comparisons are presented in the table 3.1¹².

Table 3.1.

The Coefficients of Square Determination for Regressions Between REER of LTL, Macroeconomic and Structural Activity Parameters

Factors	R-square
GDP	0.9971
M1	0.9943
Manufacturing	0.9935
Trade	0.9933
Transport	0.9917
M0	0.9913
M2	0.9902
Export of services	0.9892
Export of energy	0.9585

Note: for regression through the origin (no intercept), R-square represents the proportion of explained variability about the origin. This value cannot be compared to the R-square value when the intercept is included.

As follows from the table above, the R-square coefficient has its highest values for the correlations between changes of the GDP, in particular, manufacturing, trade and M1, on the one side, and changes in their relations to REER. Some differences between the REER to LTL and REER to EUR before re-pegging national currencies were some indirect indicator of different impact and effect of adequate anchor currency zones, i.e. USD and EUR (this logical sequence not effective in the case of EEK, so as its previous anchor DM as well known was main component of the EUR).

The comparable results were obtained when simulating basic interrelations for economic and monetary transmission flows in Estonia. Using the distributed lag analysis the regression equation coefficients and R-square values were calculated for the correlation intensity between the real effective EEK/EUR exchange rate and the essential sectoral indicators. The table 3.2 of comparative R-square values is presented below.

Table 3.2.

**The Coefficients of Square Determination for Regressions
Between REER of EEK, Macroeconomic and Structural Activity Parameters**

Factors (Estonia)	R-square
GDP	0.9963
Manufacturing	0.9946
Trade	0.9942
M1	0.9911
M0	0.9852
Transport	0.9848

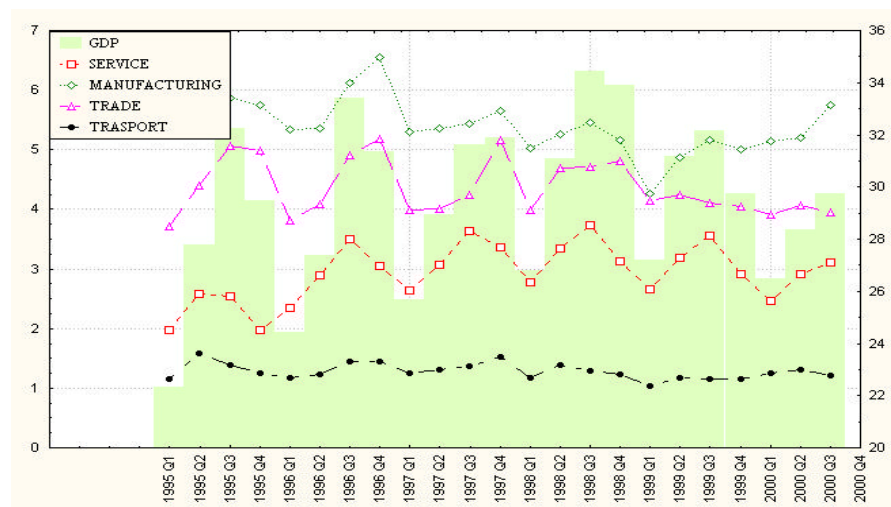
Note: cf. remarks below the table 3.1

The evaluations of the multicollinearity between REER and main financial and economic parameters of the Baltic States discussed before shows strong values of the multiple correlation (cf. $R(x, y)$ and r^2). The statistical hypothesis on independence of main variables from the REER also was tested (Student t-criteria). So as the probability of the 1st range errors is insignificant, the true is that their dependencies are substantial (cf. tables 3.4 and 3.5).

¹² More detailed calculations are lifted to appendix. Here we presented only the final comparative table of R-square values.

Chart 3.10.

**Multifactor Regression of the Lithuania's GDP Endogenous Response
To Main Variables Weighted by the REER of the LTL to EUR**



The parameters of multifactor regression of the Lithuania's GDP endogenous response to main variables weighted by the REER of the LTL to EUR are detailed in the chart 3.10.

Table 3.3

Regression Summary for Dependent Variable: GDP_EU						
MULTIPLE REGRESS. N=23	R= .94281661 R ² = .88890317 Adjusted R ² = .86421498 F(4,18)=36.005 p<.000000 Std.Error of estimate: 1.1511					
	BETA	St. Err. of BETA	B	St. Err. of B	t(18)	p-level
Intercept			3.55488	3.175744	1.11939	.277691
SERVI_EU	.639920	.084589	4.09394	.541164	7.56507	.000001
MANUF_EU	.058248	.122629	.39343	.828285	.47499	.640503
TRADE_EU	.668137	.105591	4.44814	.702972	6.32762	.000006
TRANS_EU	-.252168	.128286	-5.92691	3.015203	-1.96567	.064957

The multiple regressions for the Lithuania GDP with account of the coefficients of square determination for regressions between REER/ EUR, on the one side, macroeconomic and structural activity parameters, on the other, is as follows:

$$LT-GDP_t = .67 LT-Trad /EUR_t + .64 LT-Serv /EUR_t + .58 LT-Manuf /EUR_{t-1} - .25 LT-Trans/ EUR_{t-1}.$$

Table 3.4.

Numerical Values of the Variance for Correlations between Lithuanian Economy and REER LTL/EUR

Correlations [quarter.sta]											
Marked correlations are significant at p < .05000											
BASIC STATS Var. X & Var. Y	Mean	Std.Dv.	r(X,Y)	r ²	t	p	N	Constant dep: Y	Slope dep: Y	Constant dep: X	S de
REER LTL/EUR	313.	59.4									
Manufacturing	1676.	291.5	.8961	.8030	9.252	.0000	23	301.4	4.397	6.56	
REER LTL/EUR	313.	59.4									
TRADE	1356.	252.6	.8525	.7268	7.474	.0000	23	222.0	3.625	40.92	
REER LTL/EUR	313.	59.4									
TRANSPORT	848.	230.4	.9598	.9213	15.679	.0000	23	-316.6	3.723	102.97	
REER LTL/EUR	313.	59.4									
GDP	9254.	2106.8	.9058	.8205	9.797	.0000	23	-792.1	32.127	76.36	
REER LTL/EUR	313.	59.4									
MON_BASE	3000.	792.9	.9017	.8130	9.555	.0000	23	-763.1	12.035	110.03	
REER LTL/EUR	313.	59.4									
SUPPLYM2	4246.	1032.2	.9118	.8314	10.176	.0000	23	-708.6	15.844	89.91	
REER LTL/EUR	313.	59.4									
GOODS	-1095.	427.0	-.4430	.1962	-2.264	.0343	23	-99.0	-3.184	245.24	
REER LTL/EUR	313.	59.4									
ENERGY_EXPORT	705.	208.3	.6646	.4417	4.076	.0005	23	-24.1	2.330	179.16	
REER LTL/EUR	313.	59.4									
ENERGY_IMPORT	-1369.	243.7	-.6854	.4698	-4.314	.0003	23	-489.3	-2.812	84.05	
REER LTL/EUR	313.	59.4									
SERVICES	931.	269.0	.8838	.7810	8.655	.0000	23	-320.7	4.002	131.07	

Table 3.5.

Numerical Values of the Variance for Correlations between Estonian Economy and REER EEK/EUR

Correlations [quarter.sta]											
Marked correlations are significant at p < .05000 (Casewise deletion of missing data)											
BASIC STATS Var. X & Var. Y	Mean	Std.Dv.	r(X,Y)	r ²	t	p	N	Constant dep: Y	Slope dep: Y	Constant dep: X	S de
REER EEK/EUR	220.	27.2									
M1	14213.	3588.9	.8756	.7668	8.309	.0000	23	-113E2	115.65	125.94	
REER_EST	220.	27.2									
M2	23508.	8262.9	.9030	.8154	9.633	.0000	23	-370E2	274.58	150.37	
REER_EST	220.	27.2									
GDP	16016.	3858.9	.9334	.8712	11.920	.0000	23	-132E2	132.55	114.91	
REER_EST	220.	27.2									
Manufacturing	2344.	536.0	.8736	.7632	8.227	.0000	23	-1451.	17.23	116.39	
REER_EST	220.	27.2									
Monetary base M0	7666.	2324.0	.8637	.7459	7.852	.0000	23	-8597.	73.86	142.76	
REER_EST	220.	27.2									
TRANSPORT	1899.	708.6	.8848	.7828	8.699	.0000	23	-3180.	23.07	155.73	
REER_EST	220.	27.2									
TRADE	2384.	590.2	.8817	.7774	8.564	.0000	23	-1833.	19.15	123.41	

The numerical values of the correlations between EERs and economic indicators are presented for Lithuania in table 3 and for Estonia – in table 3.6.

Table 3.6.

Statistical Values of the Variance for REER LTL/EUR Correlations for Lithuania

Descriptive Statistics (quarter.sta)									
BASIC STATS	Valid N	Mean	Confid. -95.000%	Confid. +95.000%	Median	Lower Quartile	Upper Quartile	Std.Dev.	Skewn
REER LTL/EUR	24	317.2	291.0	343.5	334.6	272.7	357.6	62.17	-.42
Manufacturing	23	1676.3	1550.3	1802.4	1744.9	1455.1	1845.3	291.47	-.16
TRADE	23	1355.7	1246.5	1465.0	1426.2	1160.1	1564.1	252.61	-.80
TRANSPORT	23	847.6	747.9	947.2	882.7	656.1	1032.3	230.41	-.08
GDP	23	9254.0	8342.9	10165.1	9401.1	7520.1	11016.4	2106.83	-.67
MON. BASE	24	3030.7	2697.3	3364.1	3260.1	2317.0	3722.3	789.65	-.30
SUPPLYM2	24	4296.3	3857.4	4735.2	4835.6	3366.3	5193.8	1039.35	-.42
GOODS	23	-1094.7	-1279.3	-910.0	-998.6	-1501.0	-754.6	426.97	-.25
ENERGY EXPORT	23	704.6	614.5	794.7	720.0	517.0	894.0	208.30	-.15
ENERGY IMPORT	23	-1368.7	-1474.1	-1263.3	-1384.1	-1551.5	-1199.4	243.72	.31
SERVICES	23	930.7	814.4	1047.0	974.5	794.1	1155.3	268.99	-.58

Table 3.7.

Statistical Values of the Variance for REER EEK/EUR Correlations for Estonia

Descriptive Statistics (quarter.sta)									
BASIC STATS	Valid N	Mean	Confid. -95.000%	Confid. +95.000%	Median	Lower Quartile	Upper Quartile	Std.Dev.	Skewn
REER EEK/EUR	24	221.3	209.8	232.7	231.9	206.8	243.2	27.12	-.98
M1	24	14534.7	12910.0	16159.4	14872.3	11285.9	16905.6	3847.52	.11
M2	24	24236.3	20506.1	27966.5	26007.0	16237.6	29805.3	8833.80	.12
GDP	23	16015.5	14346.8	17684.2	16761.2	13372.1	19183.1	3858.88	-.36
Manufacturing	23	2343.6	2111.8	2575.4	2478.3	1843.6	2736.4	536.04	-.28
M0	24	7896.7	6825.2	8968.2	7776.9	5540.5	9498.8	2537.51	.36
TRANSPORT	23	1899.5	1593.1	2205.9	2037.8	1329.4	2593.0	708.57	.07
TRADE	23	2383.8	2128.5	2639.0	2499.8	1979.6	2855.8	590.20	-.28

4. MONETARY CONVERGENCE AND MACROECONOMIC CHANGES: ATTEMPTS TO SIMULATE SMALL OPEN BALTIC ECONOMIES¹³

4.1. The quantitative analysis of monetary transmission processes on the base of vector autoregressive models of interrelations of the main structural parameters of the national economies, on the one side, and the effective exchange rate dynamics, on the other, can be one of the starting points to build the econometric imitation of the open pre-accession monetary system and identify some aspects of pegging impact on its developments¹⁴.

After trials below to imitate the influence of the monetary transmission mechanisms under conditions of the CBA and pressing of the shadow economy in the Baltic States, even better is to recognize the wisdom of the Executive Board of the ECB summarizing the practical results of exploring new powerful system of the EMA monetary statistics and econometrics that "...any monetary policy action takes considerable time to affect the economy and...the precise impact is difficult to estimate. In other words, the impact of the monetary policy ...is subject to long, variable and uncertain lags". (*The Monetary Policy of the ECB*, p. 41)

Interesting empirical macroeconomic model of the structural monetary transmission based on Estonian statistics was presented by Rasmus Pikkani (2001) who also tested it for the adequacy applying it the dynamic forecasting. Getting the model together, as was mentioned by the author, requested to revise the specifications of the main variables and gap's parameters of endogenous impacts (ibid., p. 24-25). As seen from the publication, Pikkani also not tried to verify the relevance of his structurized econometric system for the evaluation of the impact of currency re-pegging to the monetary transmission channels in Estonia.

Very ingenious and original attempts to evaluate this impact, or, more precisely, the adequacy of pegged or floating exchange rate regimes for pre-EMU accession countries including the Baltic States were presented by *Lucio Vinhas de Souza* and Elisabeth Ledrut (2001). The essence of their approach is to find different independent variables within simplified system of economic and financial flows for every country who could differentiate the parallel behaviour of main processes anticipating the pegged and floating exchange rate regimes; later the authors are specifying the loss function measuring expected effect of pegging/ floating or the difference between alternative regimes. When trying to fix better this difference, the model authors introduced the surplus of interest rates over CPI (and its modification – difference between such surplus local values and their values in the euro area) as independent variable of the system. However, these variables have not over passed the probability tests in equations for Baltic a/o accession countries (ib., p.29-35).

With account of preliminary evaluation of the significance of main macroeconomic and monetary variables and their stochastic interrelations, the simplified system of econometric equations was experienced by the author for the

¹³ Econometric evaluations and forecasts in sections 4 - 6 were done mostly together with **Danas Zuokas**, collaborator of Statistical Center, Vytauti Magnus University. His E-mail - Danas_Zuokas@fc.vdu.lt. Dr. Virginijus Sakalauskas also participated at the preliminary stage of imitation studies.

¹⁴ Other opinions also may be found. As suggested by Centre for Central Banking Studies, BoL (L. Price, 1996), the econometrics based on VAR calculations is not [always] suitable for the transition economies, so the results of monetary convergence channels' modeling in the Baltics have to be evaluated carefully. However, there are not so much possibilities to choose between alternative econometric imitation means under other necessary restrictions.

monetary transmission simulation oriented to euroization effects in Lithuania (below, 3.2). It also requested repeatedly the additional specifications of the most important variables channeling endogenous impacts. The impact of processes anticipating the pegged and floating exchange rate regimes, with account of some tests mentioned in part 3 of this material, was tried to fix by specifying analogical parameters of monetary transmission channels, also the macroeconomic changes in national currency and in their normalized level by the EER to euro (the last one reflecting monetary market fluctuations through changing exchange rates between EUR to USD). As it is well known, the economic and finance parameters under the CBA are anchored to USD (in Lithuania), and differences between parallel specifications (based as mentioned before) are simplified index of priorities of the pegged regime over floating (or vice versa). The national exchange rate is an average of bilateral rates according to trading volumes with different trading countries, and the determination of nominal exchange rate with CPI movements relative to these countries would detail the pegging influence even more exactly, but requests incomparably much more additional statistics than used in this study.

The parameters of econometric equations were determined by widely used cointegration method, so setting in many cases two stage estimation and double equations for every dependent variable: first one – when delineating its long-run trend by the OLS and second – building the short run equation for determining the residuals of cointegration and estimation of the error correction.

4.2. The regressive system of econometric dependencies consist of the following equations with interacting monetary a/o macroeconomic variables imitating small open system but of limited interaction with foreign money markets:

$$\mathbf{D_LGDP = f (D_LGVA_MAN, D_LGVA_TRD, D_LGVA_SERV)} \quad 4.2.1$$

where L added to variables denotes Lithuania,

D – difference operator in logarithmic form expressing relative change in the variable against previous quarter,

GDP - gross domestic product ,

GVA_MAN – gross value added in manufacturing,

GVA_TR – gross value added in wholesale and retail trade,

GVA_SERV – gross value added in services.

The equation is expected to identify how much the gross domestic product is approximated not by its direct components but by gross value added within these sectors.

The money supply equation is approximated by following variables:

$$\mathbf{D_LM2 = f (D_LM2 (-1), D_LGDP, LNEER, D_LL, D2)} \quad 4.2.2$$

i.e. M2 is dependent from its previous value and previous GDP amounts, changes in loans (L), nominal effective exchange rate (NEER) and dummy variable (D2) for III quarter of 1997 (influence of Russian crisis);

$$\mathbf{LCPI = f (D_LY_pers, D_LGVA, LCPI (-1)),} \quad 4.2.3$$

i.e. consumer price index CPI is approximated by its previous amount and personal income Y_{pers} , total amount of gross value added LGVA; the specification of theoretically obvious relations with money supply don't show acceptable probability.

$$LEXP_G_LT = f(LGVA_MAN_LT, LREER_RUB, LREER_EU, D_LIMP_G) \quad 4.2.4$$

i.e. export of goods is dependent of gross value added in manufacturing, real EER towards ruble (LREER_RUB) area and eurozone (LREER_EU), also import of goods in previous period.

$$LEXP_S_LT = f(LGVA_SERV, LCPI, EXP_LS_{(-1)}) \quad 4.2.5$$

Export of services doesn't respond necessary way to simplified simulation as a dependent of GDP produced in service sector or gross value added in services, consumer prices and/or export of services in previous period; so it is imitated by nearest temporary dependencies having accountable impact on its changes.

$$LREER = f(LM2, D_LEXP_G, D) \quad 4.2.6$$

The real EER is rather sensitive to both integrated internal and foreign macroeconomic impact; but so as exogenous relations outside of small economy are not represented on this stage of modeling necessary way, it is also imitated by nearest temporary dependencies having measurable impact. The REER is supposed as dependent of changes in money supply and its changes and export of goods, without acceptable probability of its interrelations with inflation rate (INF) and loans, but including dummy variable to smooth shock of Russian crises of 1997.

$$LAAIR = f(LREER, LCA, LL_LT) \quad 4.2.7$$

Interest rate is undoubtedly very important instrumental variable dependent substantially of the REER, loans on the comparable currencies and capital account balance; some part of it is determined by the interest rates in EMU and other open monetary markets which impact is not included in this equation.

$$D_LIMP_G = f(LEXP_G, LCA) \quad 4.2.8$$

i.e. import of goods is imitated rather reliably as dependent from export of goods and current account balance.

4.3. The specifications of the parameters for this system of equations unsubstantially corrected have resulted with statistical tests presented in detail below in two consecutive steps. First-of-all, there were calculated logarithmic coefficients of the regressions and then with account of consistency of their tests formulated secondary regressions for the residual of cointegration. The evaluations of unknown coefficients were found by ordinary least squares method.

The estimated parameters of the *GDP* dependencies with parts of the GVA produced in the main economic sectors are as follows (cf. equation 4.3.1):

$$\text{LOG(LGDP_LT)} = 4.449908 + 0.298887 * \text{LOG(LGVA_MAN_LT)} + 0.523618 * \text{LOG(LGVA_TRD_LT)} \quad 4.3.1$$

	Coefficient	Std. Error	t-Statistic	Prob.
B(11)	4.449908	2.848644	1.913159	0.0709
B(12)	0.298887	0.213867	1.397534	0.1784
B(13)	0.523618	0.170931	3.063332	0.0064
R-squared	0.702703	Mean dependent var		22.64140
Adjusted R-squared	0.671409	S.D. dependent var		0.106811
S.E. of regression	0.061227	Akaike info criterion ¹⁵		-2.622337
Sum squared resid	0.071226	Schwarz criterion ¹⁶		-2.473559
Log likelihood	31.84571	Durbin-Watson stat		1.875801

The residual of cointegration of parameters determining the GDP of Lithuania is as follows:

$$\begin{aligned} \text{DLOG(LGDP_LT)} = & 0.434125 * \text{DLOG(LGVA_MAN_LT)} - 0.984223 * (\text{LOG(LGDP_LT}(-1)) - \%c11 - \\ & \%c12 * \text{LOG(LGVA_MAN_LT}(-1)) - \%c13 * \text{LOG(LGVA_TRD_LT}(-1))) + 0.864524 * \text{DLOG(LGVA_TRD_LT)} - \\ & 0.467244 * \text{DLOG(LGVA_SERV_LT)} \end{aligned} \quad 4.3.1a$$

	Coefficient	Std. Error	t-Statistic	Prob.
B(16)	-0.984223	0.257211	-3.826522	0.0014
B(17)	0.434125	0.200272	2.167673	0.0447
B(18)	0.864524	0.190461	3.539106	0.0003
B(19)	-0.467244	0.265791	-1.757935	0.0967
R-squared	0.843996	Mean dependent var		0.015166
Adjusted R-squared	0.816466	S.D. dependent var		0.129850
S.E. of regression	0.055629	Akaike info criterion		-2.770591
Sum squared resid	0.052607	Schwarz criterion		-2.571635
Log likelihood	33.09121	Durbin-Watson stat		1.435256

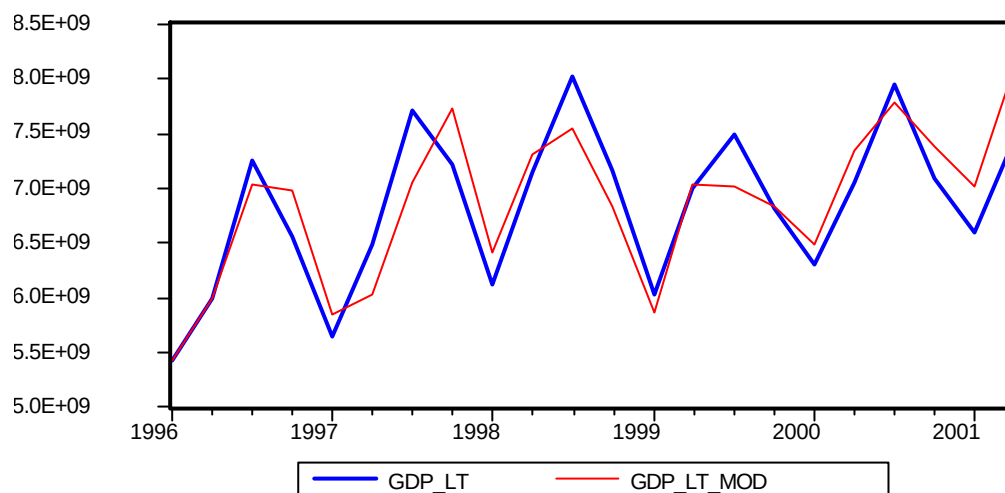
The graphical view shows significant seasonal character of changes with prevailing peaks of outputs in the middle of every year and significant lowering of its levels in 4th quarters. In all graphs below the observed changes are marked in thick blue line and simulated trend – in red (or sometimes - dash) thin line.

¹⁵ The AIC is used in model selection for non-nested alternatives; a model is better than another model if it has a smaller AIC value..

¹⁶ The SC is an alternative to the AIC that imposes a larger penalty for additional coefficients

Graph 4.3.1

The Observed and Imitated Changes in the GDP, 1996-2001



The probability tests for residual of cointegration normalized to nominal EER are characterized by better indices:

$$\begin{aligned} \text{DLOG(LGDP_LT / LNEER)} = & -1.038645 * (\text{LOG(LGDP_LT}(-1) / \text{LNEER}(-1)) - \%c11 * \text{LOG(LGVA_MAN_LT}(-1) / \\ & \text{LNEER}(-1)) - \%c12 * \text{LOG(LGVA_SERV_LT}(-1) / \text{LNEER}(-1))) + 0.382058 * \text{DLOG(LGVA_MAN_LT/LNEER)} + \\ & 0.622518 * \text{DLOG(LGVA_TR_LT / LNEER)} \end{aligned} \quad 4.3.1.b$$

	Coefficient	Std. Error	t-Statistic	Prob.
B(16)	-1.038645	0.241343	-3.303609	0.0004
B(17)	0.382058	0.162170	2.355913	0.0300
B(18)	0.622518	0.116923	4.324165	0.0000
R-squared	0.873386	Mean dependent var	-0.031222	
Adjusted R-squared	0.859318	S.D. dependent var	0.145633	
S.E. of regression	0.054624	Akaike info criterion	-2.845140	
Sum squared resid	0.053707	Schwarz criterion	-2.695923	
Log likelihood	32.87397	Durbin-Watson stat	1.702307	

The test of statistical parameters for the equation of *monetary supply* with some corrections is as follows:

$$\text{LOG(LM2_LT)} = 0.813832 * \text{LOG(LM2_LT}(-1)) + 0.163959 * \text{LOG(LGDP_LT)} + 0.086283 * \text{LOG(LNEER)} \quad 4.3.2$$

	Coefficient	Std. Error	t-Statistic	Prob.
B(21)	0.813832	0.074027	10.99377	0.0000
B(22)	0.163959	0.061588	2.662175	0.0159
B(23)	0.086283	0.049826	1.731687	0.1004
R-squared	0.984593	Mean dependent var	22.77021	

Adjusted R-squared	0.982881	S.D. dependent var	0.239539
S.E. of regression	0.031341	Akaike info criterion	-3.956212
Sum squared resid	0.017681	Schwarz criterion	-3.806994
Log likelihood	43.54022	Durbin-Watson stat	2.283238

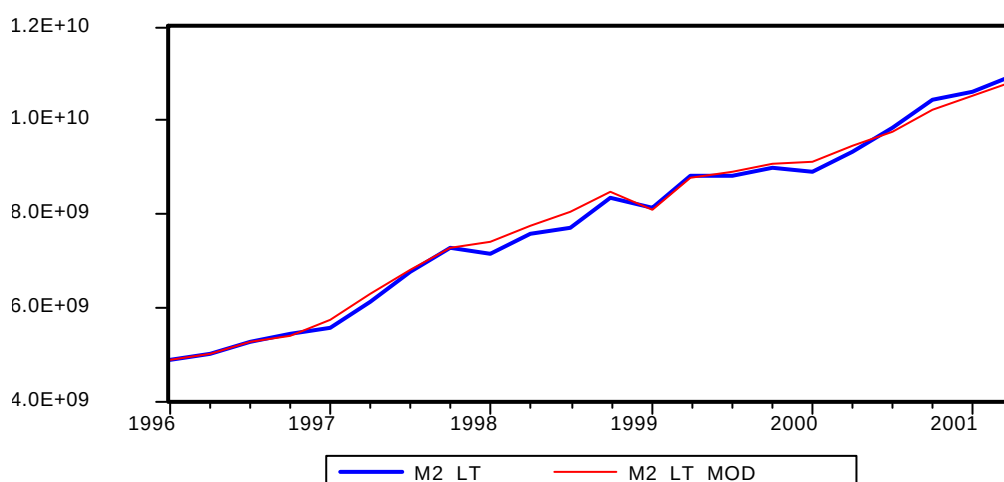
The results of cointegration for these independent variables of monetary supply were even better than in basic its regression form. That helped to include also other additional variables, as loans and interest rates, after testing and choosing best response of various independent variables influencing the money supply:

$$\text{DLOG(LM2_LT)} = -0.631874 * (\text{LOG(LM2_LT)}(-1)) - \%c21 * \text{LOG(LM2_LT)}(-2) - \%c22 * \text{LOG(LGDP_LT)}(-1) - \%c23 * \text{LOG(LNEER)}(-1)) + 0.023967 + 0.348624 * \text{DLOG(LL_LT)} - 0.034411 * \text{DLOG(LAAIR)} + 0.104962 * \text{D2} \quad 4.3.2a$$

	Coefficient	Std. Error	t-Statistic	Prob.
B(26)	-0.631874	0.189771	-3.329669	0.0050
B(27)	0.023967	0.005931	3.040655	0.0012
B(28)	0.348624	0.101285	3.441997	0.0040
B(29)	-0.034411	0.016464	-2.090027	0.0553
B(30)	0.104962	0.025777	3.071916	0.0011
R-squared	0.717052	Mean dependent var	0.039645	
Adjusted R-squared	0.636210	S.D. dependent var	0.037791	
S.E. of regression	0.022793	Akaike info criterion	-3.503761	
Sum squared resid	0.007274	Schwarz criterion	-3.255225	
Log likelihood	47.78573	Durbin-Watson stat	2.030223	

Graph 4.3.2

The Observed and Imitated Changes of Lithuania Money Supply, 1996-2001



The tests for the equation of *consumer prices* are as below:

$$\text{LOG(LCPI)} = 2.523910 - 0.085343 * \text{LOG(LY_PERS_LT)} - 0.088843 * \text{LOG(LGVA_LT)} \quad 4.3.3$$

	Coefficient	Std. Error	t-Statistic	Prob.
B(31)	2.523910	0.583897	3.322524	0.0004
B(32)	-0.085343	0.027332	-3.122491	0.0056
B(33)	-0.088843	0.029782	-2.983102	0.0076
R-squared	0.721390	Mean dependent var		0.013159
Adjusted R-squared	0.692062	S.D. dependent var		0.020589
S.E. of regression	0.011425	Akaike info criterion		-4.979897
Sum squared resid	0.002480	Schwarz criterion		-4.831118
Log likelihood	68.77887	Durbin-Watson stat		1.674762

After cointegration we have such equation parameters and statistical test measures:

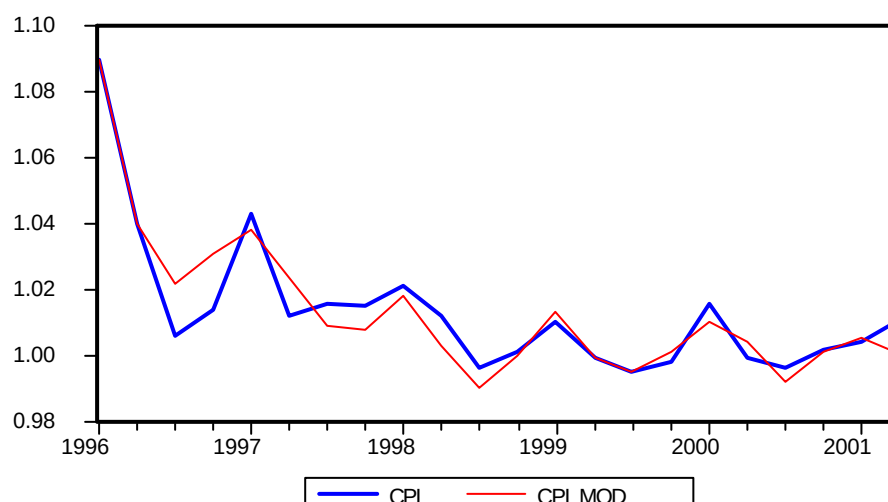
$$\text{DLOG(LCPI)} = -0.992657 * (\text{LOG(LCPI}(-1)) - \%c31 - \%c32 * \text{LOG(LY_PERS_LT}(-1)) - \%c33 * \text{LOG(LGVA_LT}(-1))) - 0.107360 * \text{DLOG(LGVA_LT)}$$

4.3.3a

	Coefficient	Std. Error	t-Statistic	Prob.
B(36)	-0.992657	0.165783	-4.987682	0.0000
B(37)	-0.107360	0.013866	-7.742440	0.0000
R-squared	0.803188	Mean dependent var		-0.003583
Adjusted R-squared	0.792829	S.D. dependent var		0.017527
S.E. of regression	0.007978	Akaike info criterion		-6.733951
Sum squared resid	0.001209	Schwarz criterion		-6.634473
Log likelihood	72.70649	Durbin-Watson stat		1.445349

Graph 4.3.3

The Observed and Imitated Changes of CPI in Lithuania, 1996-2001



The test of parameters of equation of *export of goods* is very acceptable and strongly explain it developments:

$$\text{LOG(LEXP_G_LT)} = 0.604122 * \text{LOG(GVA_MAN_LT)} - 0.315157 * \text{LOG(REER_RUB)} + 0.552507 * \text{LOG(REER_EU)} + 0.337151 * \text{LOG(IMP_G_LT)}$$

4.3.4

	Coefficient	Std. Error	t-Statistic	Prob.
B(41)	0.604122	0.131955	3.578228	0.0002
B(42)	-0.315157	0.049891	-6.316925	0.0000
B(43)	0.552507	0.122896	3.495715	0.0003
B(44)	0.337151	0.125038	2.696378	0.0148
R-squared	0.907514	Mean dependent var		22.06133
Adjusted R-squared	0.892100	S.D. dependent var		0.146972
S.E. of regression	0.048278	Akaike info criterion		-3.060732
Sum squared resid	0.041953	Schwarz criterion		-2.862361

The statistic tests after cointegration of these independent variables also confirm the acceptability of variables taken for imitation:

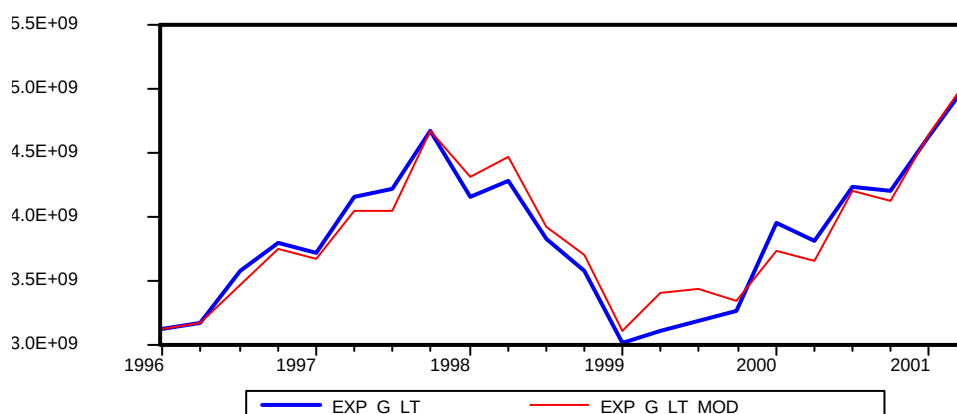
$$\begin{aligned} \text{DLOG}(\text{EXP_G_LT}) = & -0.737977 * (\text{LOG}(\text{EXP_G_LT}(-1)) - \%c41 * \text{LOG}(\text{GVA_MAN_LT}(-1)) - \%c42 * \text{LOG}(\text{REER_RUB}(-1)) - \\ & \%c43 * \text{LOG}(\text{REER_EU}(-1)) - \%c44 * \text{LOG}(\text{IMP_G_LT}(-1))) + 1.112998 * \text{DLOG}(\text{GVA_MAN_LT}) + \\ & 0.477378 * \text{DLOG}(\text{REER_EU}) - 0.185971 * \text{DLOG}(\text{REER_RUB}) - 0.063105 * \text{DLOG}(\text{GVA_AGR_LT}) \end{aligned} \quad 4.3.4a$$

	Coefficient	Std. Error	t-Statistic	Prob.
B(46)	-0.737977	0.174034	-3.240428	0.0006
B(47)	1.112998	0.110208	10.09910	0.0000
B(48)	0.477378	0.174255	2.739539	0.0145
B(49)	-0.185971	0.066408	-2.800446	0.0128
B(50)	-0.063105	0.015208	-3.149548	0.0008
R-squared	0.883695	Mean dependent var		0.022884
Adjusted R-squared	0.854618	S.D. dependent var		0.089098
S.E. of regression	0.033972	Akaike info criterion		-3.722297
Sum squared resid	0.018466	Schwarz criterion		-3.473601
Log likelihood	43.08412	Durbin-Watson stat		1.016662

The graphical view of comparative actual and imitated changes of export of goods in Lithuania is presented in the graph below.

Graph 4.3.4

The Observed and Imitated Changes of Export of Goods in Lithuania, 1996-2001



The statistical tests of equation parameters interpreting the *export of services* are still most unsuccessful:

$$\text{LOG(LEXP_SERV_LT)} = 0.962960 * \text{LOG(LGVA_SERV_LT)} - 3.315103 * \text{LOG(LCPI)}$$

4.3.5

	Coefficient	Std. Error	t-Statistic	Prob.
B(51)	0.962960	0.001267	760.0638	0.0000
B(52)	-3.315103	1.137797	-3.792507	0.0011
R-squared	0.665262	Mean dependent var		20.73055
Adjusted R-squared	0.648525	S.D. dependent var		0.181403
S.E. of regression	0.107545	Akaike info criterion		-1.535300
Sum squared resid	0.231320	Schwarz criterion		-1.436115
Log likelihood	18.88831	Durbin-Watson stat		1.710872

The statistic tests after cointegration of independent variables also confirm the problematic probability of variables taken for imitation of this process and dependence from other, not included additional factors:

$$\text{DLOG(EXP_S_LT)} = -0.851164 * (\text{LOG(EXP_S_LT}(-1)) - \%c51 * \text{LOG(GVA_SERV_LT}(-1)) - \%c52 * \text{LOG(CPI}(-1))) + 0.650079 * \text{DLOG(GVA_SERV_LT} - 4.367377 * \text{DLOG(CPI)} + 0.718865 * \text{DLOG(REER)}$$

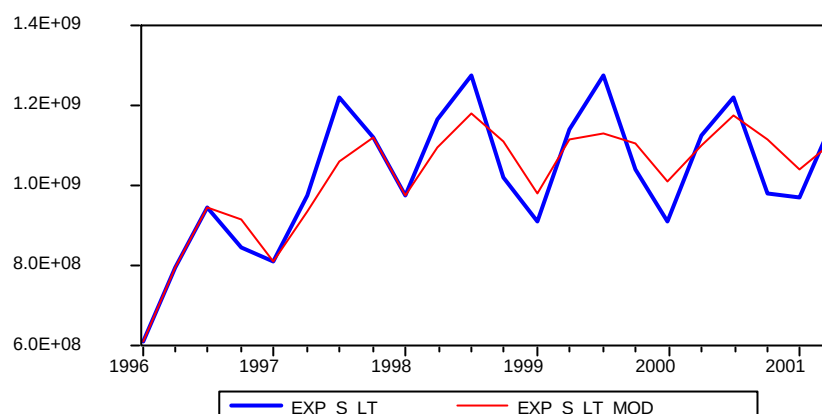
4.3.5a

	Coefficient	Std. Error	t-Statistic	Prob.
B(56)	-0.851164	0.240220	-3.543261	0.0025
B(57)	0.650079	0.340950	1.906670	0.0736
B(58)	-4.367377	1.539483	-3.486480	0.0028
B(59)	0.718865	0.558028	1.288224	0.2149
R-squared	0.673937	Mean dependent var		0.031120
Adjusted R-squared	0.616396	S.D. dependent var		0.169115
S.E. of regression	0.104743	Akaike info criterion		-1.504971
Sum squared resid	0.186509	Schwarz criterion		-1.306014
Log likelihood	19.80219	Durbin-Watson stat		1.731499

The graphical view of comparative calculations for the actual and imitated changes of export of services in Lithuania is presented in the graph 4.3.5.

Graph 4.3.5

The Observed and Imitated Changes of Export of Services in Lithuania, 1996-2001



So as there are not enough regressors in these equations, R-squared parameters are not enough high for reliable integration. However, we do not succeeded to find other statistically measured variables ameliorating this imitation; any attempt to take into account the EERs or monetary variables better connecting this equation with monetary transmission channels do not succeeded as a result of low probabilities of these interconnections.

The same problematic results we have after inclusion additionally the value of autoregressive variable of service export:

$$\text{DLOG(LEXP_S_LT)} = -0.769171 * (\text{LOG(LEXP_S_LT}(-1)) - \%c51 * \text{LOG(LGVA_SERV_LT}(-1)) - \%c52 * \text{LOG(LCPI}(-1))) - 4.670692 * \text{DLOG(LGVA_SERV_LT)} + 0.523897 * \text{DLOG(LCPI)} \quad 4.3.5b$$

	Coefficient	Std. Error	t-Statistic	Prob.
B(56)	-0.769171	0.235841	-3.261404	0.0043
B(57)	-4.670692	1.548992	-3.660892	0.0018
B(58)	0.523897	0.332506	1.575602	0.1325
R-squared	0.642107	Mean dependent var		0.031120
Adjusted R-squared	0.602341	S.D. dependent var		0.169115
S.E. of regression	0.106645	Akaike info criterion		-1.507066
Sum squared resid	0.204715	Schwarz criterion		-1.357848
Log likelihood	18.82419	Durbin-Watson stat		1.764730

If to take out the independent variable LGVA_SERV, the probabilities of interrelation become reasonable but again there are not enough statistically measured regressors in this equation, R-squared parameters are not enough high for its reliable integration.

Imitation of the *REER* is rather important for evaluations of monetary transmissions mechanisms and it looks acceptably successful after statistical test of its parameters:

$$\text{LOG(LREER)} = 0.736264 * \text{LOG(LM2_LT)} - 4.769283 - 0.274452 * \text{LOG(LEXP_G_LT)} \quad 4.3.6$$

	Coefficient	Std. Error	t-Statistic	Prob.
B(61)	-4.769283	1.224413	-3.711877	0.0002
B(62)	0.736264	0.033363	22.06823	0.0000
B(63)	-0.274452	0.057665	-3.759393	0.0001
R-squared	0.963298	Mean dependent var		3.925222
Adjusted R-squared	0.959434	S.D. dependent var		0.179897
S.E. of regression	0.036233	Akaike info criterion		-3.671565
Sum squared resid	0.024944	Schwarz criterion		-3.522786
Log likelihood	43.38721	Durbin-Watson stat		1.711116

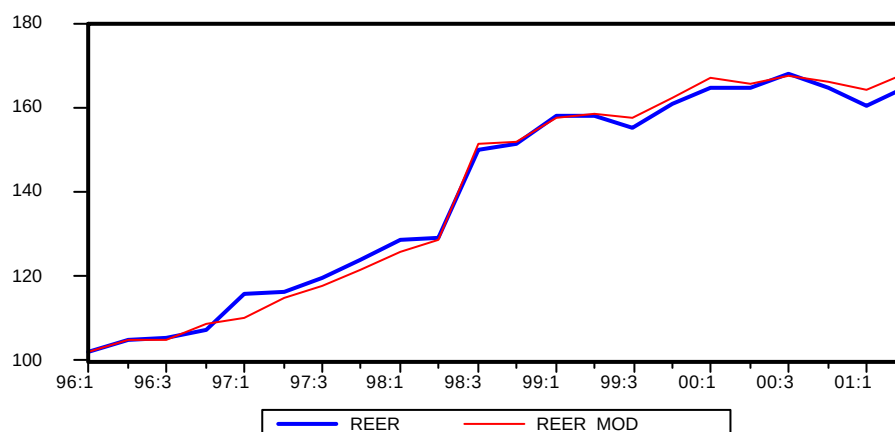
The REER cointegration equation parameters are not so good and it clearly influences worsening R-squared characteristics:

$$\text{DLOG(LREER)} = -0.221897 * (\text{LOG(LREER}(-1)) - \%c61 - \%c62 * \text{LOG(LM2_LT}(-1)) - \%c63 * \text{LOG(LEXP_G_LT}(-1))) + 0.019615 + 0.115060 * \text{D4} + -0.092054 * \text{DLOG(LEXP_G_LT)} \quad 4.3.6a$$

	Coefficient	Std. Error	t-Statistic	Prob.
B(66)	-0.221897	0.159992	-1.386932	0.1834
B(67)	0.019615	0.005671	3.458689	0.0030
B(68)	0.115060	0.027023	3.257859	0.0005
B(69)	-0.092054	0.065625	-1.402718	0.1787
R-squared	0.670058	Mean dependent var		0.022857
Adjusted R-squared	0.611833	S.D. dependent var		0.037905
S.E. of regression	0.023616	Akaike info criterion		-3.484166
Sum squared resid	0.009481	Schwarz criterion		-3.285209
Log likelihood	51.08374	Durbin-Watson stat		1.736496

Below are presented the charts demonstrating results of actual and modeled changes of the REER.
Graph 4.3.6

The Observed and Imitated Changes of the REER in Lithuania, 1996-2001



The trials to imitate factors determining *interest rate* by simplified way are complicated, as was mentioned, as a result of insufficient statistics about mechanisms influencing the internal monetary markets from abroad.

$$\text{LOG(LAAIR)} = 46.54398 - 2.941227 \cdot \text{LOG(LREER_EU)} - 1.028164 \cdot \text{LOG(LL_LT)} - 0.227847 \cdot \text{LOG(LCA_LT)}$$

4.3.7

	Coefficient	Std. Error	t-Statistic	Prob.
B(71)	46.54398	12.39785	3.754198	0.0016
B(72)	-2.941227	0.629087	-3.675392	0.0002
B(73)	-1.028164	0.467921	-2.197303	0.0421
B(74)	-0.227847	0.117291	-1.942585	0.0688
R-squared	0.709124	Mean dependent var		2.677087
Adjusted R-squared	0.657793	S.D. dependent var		0.371829
S.E. of regression	0.217514	Akaike info criterion		-0.043459
Sum squared resid	0.804313	Schwarz criterion		0.155497
Log likelihood	3.456324	Durbin-Watson stat		2.579291

The interest rate cointegration equation parameters are not the best but R-squared characteristics changes to better after inclusion of dummy variable D1 for 1997 IV.

$$\text{DLOG(AAIR)} = -1.225656 \cdot (\text{LOG(AAIR(-1))} - \%c71 - \%c72 \cdot \text{LOG(REER_EU(-1))} - \%c73 \cdot \text{LOG(L_LT_LT(-1))} - \%c74 \cdot \text{LOG(CA_LT(-1)))} - 0.683312 \cdot \text{D1} - 0.090571 \cdot \text{DLOG(CA_LT)}$$

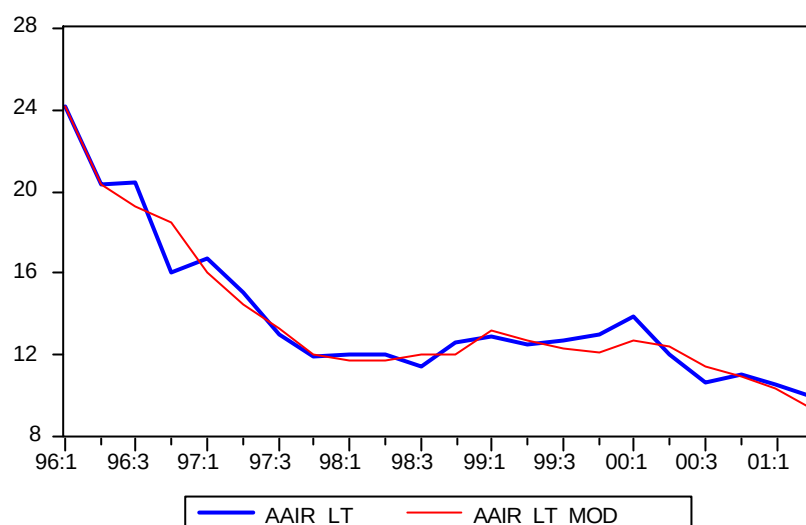
4.3.7a

	Coefficient	Std. Error	t-Statistic	Prob.
B(76)	-1.225656	0.183758	-6.669964	0.0000
B(77)	-0.683312	0.178030	-3.838193	0.0013
B(78)	-0.090571	0.065842	-1.375579	0.1868
R-squared	0.784631	Mean dependent var		-0.044874
Adjusted R-squared	0.759293	S.D. dependent var		0.332612
S.E. of regression	0.163186	Akaike info criterion		-0.650377
Sum squared resid	0.452702	Schwarz criterion		-0.501017
Log likelihood	9.503768	Durbin-Watson stat		1.079704

The graphical view of comparative actual and imitated changes of the interest rates for loans in Lithuania is presented in the graph 4.3.7.

Graph 4.3.7

The Observed and Imitated Changes of the Interest Rate in Lithuania, 1996-2001



Even very simplified imitation of *import of goods* may be characterized by rather acceptable statistical test indices.

$$\text{LOG(IMP_G_LT)} = 0.871330 * \text{LOG(EXP_LT)} + 0.141030 * \text{LOG(CA_LT)}$$

4.3.8

	Coefficient	Std. Error	t-Statistic	Prob.
B(81)	0.871330	0.011711	73.40104	0.0000
B(82)	0.141030	0.012698	11.10613	0.0000
R-squared	0.961217	Mean dependent var		22.32912
Adjusted R-squared	0.959277	S.D. dependent var		0.135904
S.E. of regression	0.027425	Akaike info criterion		-3.268200
Sum squared resid	0.015043	Schwarz criterion		-3.169015
Log likelihood	48.95020	Durbin-Watson stat		2.063676

The statistical test of cointegration equation parameters for the import of goods is quite reasonable and R-squared characteristics – between those of the best in the model.

$$\text{DLOG(IMP_G_LT)} = -0.807339 * (\text{LOG(IMP_G_LT(-1))} - \%c81 * \text{LOG(EXP_LT(-1))} - \%c82 * \text{LOG(CA_LT(-1)))) + 0.826542 * \text{DLOG(EXP_LT)} + 0.160534 * \text{DLOG(CA_LT)}$$

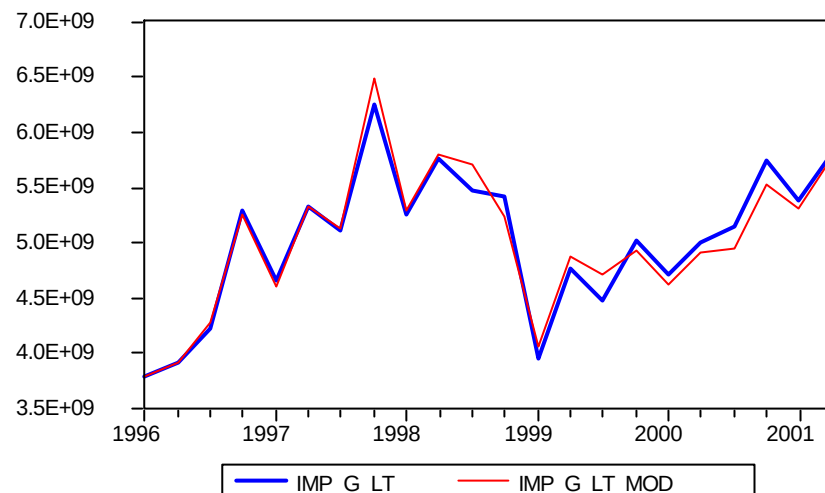
4.3.8a

	Coefficient	Std. Error	t-Statistic	Prob.
B(86)	-0.807339	0.251631	-3.208428	0.0049
B(87)	0.826542	0.067285	12.28413	0.0000
B(88)	0.160534	0.011296	13.21160	0.0000
R-squared	0.964471	Mean dependent var		0.020471
Adjusted R-squared	0.960524	S.D. dependent var		0.132583
S.E. of regression	0.026342	Akaike info criterion		-3.303704
Sum squared resid	0.012491	Schwarz criterion		-3.154487
Log likelihood	48.18889	Durbin-Watson stat		2.030849

The graphical view of comparative actual and imitated changes of import of goods for Lithuania is presented in the graph 4.3.8.

Graph 4.3.8

The Observed and Imitated Changes of the Import of Goods in Lithuania, 1996-2001



4.4. Getting the estimated model equations together within the system affirmed its suitable stability and adequacy for prognostic imitation purposes for the study of interactions between macroeconomic changes within small open economy and main monetary transmission flows.

5. PROGNOSTIC MODELING OF MONETARY TRANSMISSION WITHIN ECONOMICS: CASE OF LITHUANIA

5.1. The forecasting period was chosen with account of actual data prolongation and expected terms of formal integration of Baltic monetary systems into the European monetary area. The model presented in the section 4 of this report was tried for its suitability to follow expected differences between the significances of variables normalized in relation to EUR (future period) from those variables normalized in relation to USD (past period). This approximate approach can help to evaluate the suitability of some instrumental variables for modeling more effectively the monetary policy measures in the Baltic States in the transitional period to euroization.

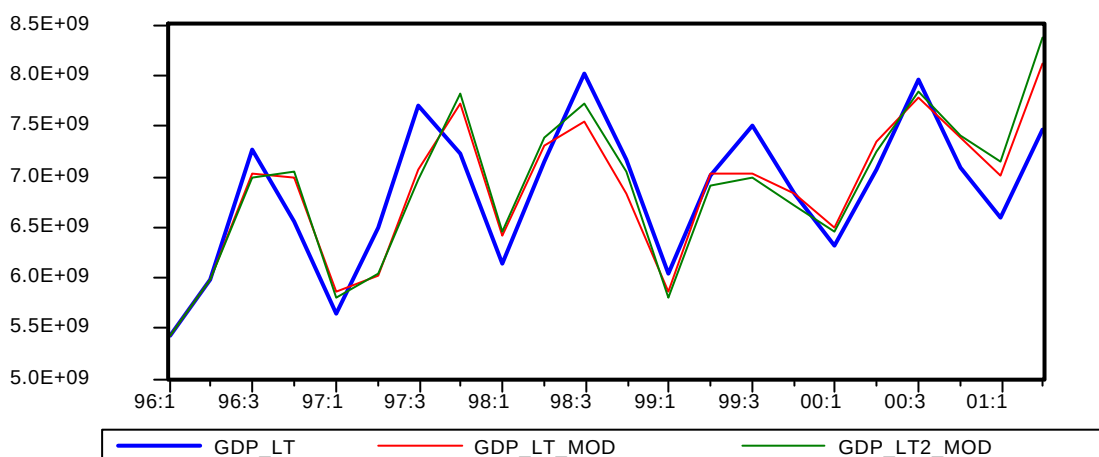
The logics of the equation system under review were based on some Balassa-Samuelson and Mundell-Fleming ideas interconnecting monetary flows and changes in GDP production sectors through foreign trade processes. With account of different reliability of statistics on main variables, the basic equations of GDP and M2 were divided into sectoral components, interconnected and linked also with effective exchange rates, personal incomes, loan, interest rates a/o monetary transmission variables. Their influence on main components of the balance of payments, such as current account, export and import, also was measured and statistically tested as independent variables within equation system. However, the interrelations of sectoral output changes with similar sectoral changes of productivities and esp. under influence of EER fluctuations determined by monetary transmission processes and foreign money markets were to

complicated and impossible to solve as a result of absence of sectoral productivity a/o important components for such econometric evaluations.

5.2. Different components of the REER were chosen as an instrument interconnecting the equations and variables within their system. The detailed econometric results received on statistics of Lithuania were presented above and some additional calculations for Estonia are in the annex. The influence of currency pegging as it is revealed through changes in the EER of national currency and esp. calculated on different foreign trade directions (to euro area, CIS or ruble area, also in USD) is not so cardinal but really noticeable esp. near turning points for some institualizing monetary and economic flows. Some evaluation results of elementary normalization of main variables buildied into the model (as before) are presented in the graphs below; and more detailed evaluations of statistical tests of the equations are presented in the annex).

Graph 5.2.1

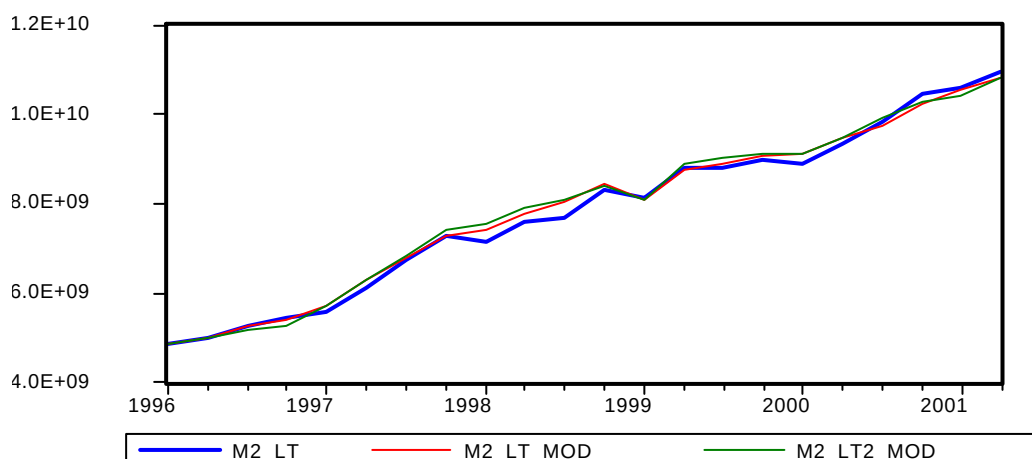
The Observed and Imitated Normalized Changes of the GDP in Lithuania, 1996-2001



Naturally, oversimplified imitation in some cases doesn't fix exactly the turning points (especially in 1999, 3 quarter; 2001, 1 quarter; also 1996 3 quarter; last case and in 1997, 3 quarter also with a slip at about quarter). One obviously seen explanation is that equation describes the process including not all main components and some of the last (as agriculture) determines more sharp seasonal fluctuations than follows from the model.

Graph 5.2.2

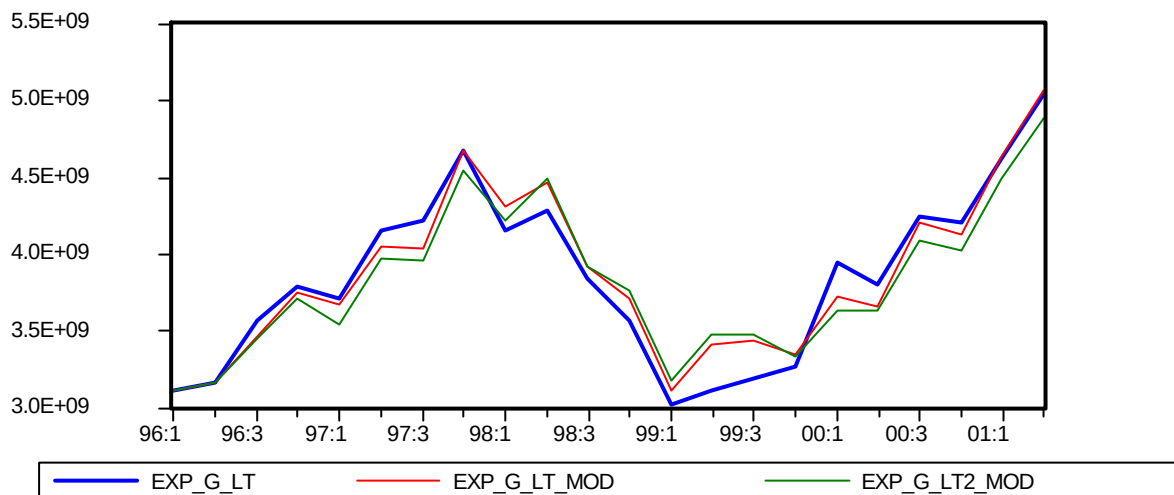
The Observed and Imitated Normalized Changes of the Money Supply in Lithuania, 1996-2001



The development of the money supply is simulated enough correctly and minor deviations may follow from external factors.

Graph 5.2.3

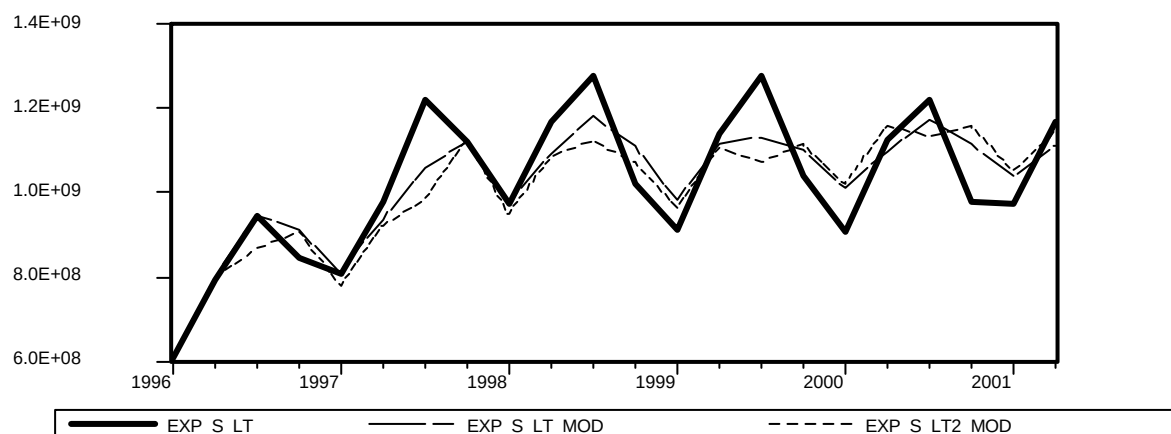
The Observed and Imitated Normalized Changes of the Export of Goods in Lithuania, 1996-2001



Are there clearly seen deviations of imitation from actual change resulting from short appreciation of EUR to USD in I-II quarters, 1999, and, after this short term, depreciation of EUR still Feb 2002? if so, then they may be used for identification of currency transmission influence according to Balassa and Mundell-Fleming effects even in cases when they are interconnected with other strong deviating effects noticed in the study before. As is well known, any appreciation of the currency usually decreases net export and the costs of production, and vice versa - in contrary case of currency depreciation. Their combined interactions determine the actual fluctuations of the EER and their impact on net export.

Graph 5.2.4.

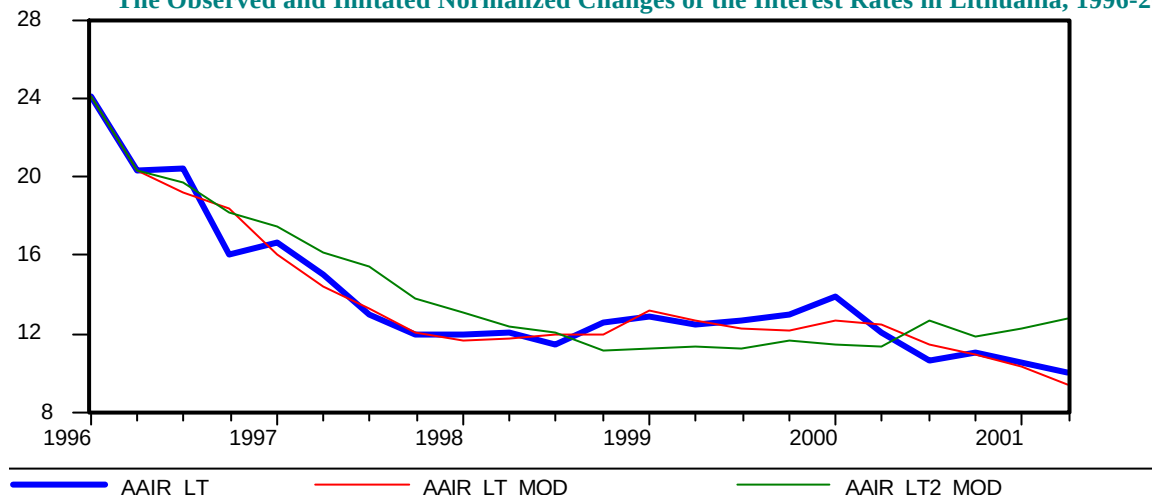
The Observed and Imitated Normalized Changes of the Export of Services in Lithuania, 1996-2001



The simulation of the export of goods requests more sophisticated econometric form so as turning points are fixed in the graph more smoothly than actually.

Graph 5.2.5

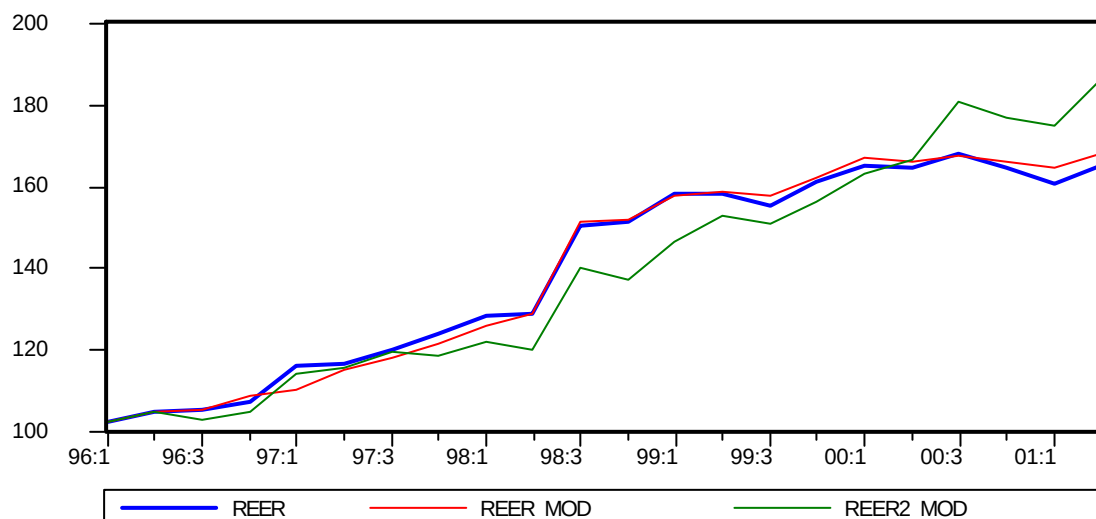
The Observed and Imitated Normalized Changes of the Interest Rates in Lithuania, 1996-2001



The interest rates as mentioned before are dependent of international money markets in some prolonged periods and preferable solution would be to include additional regressors reflecting their identified influence for better estimations.

Graph 5.2.6

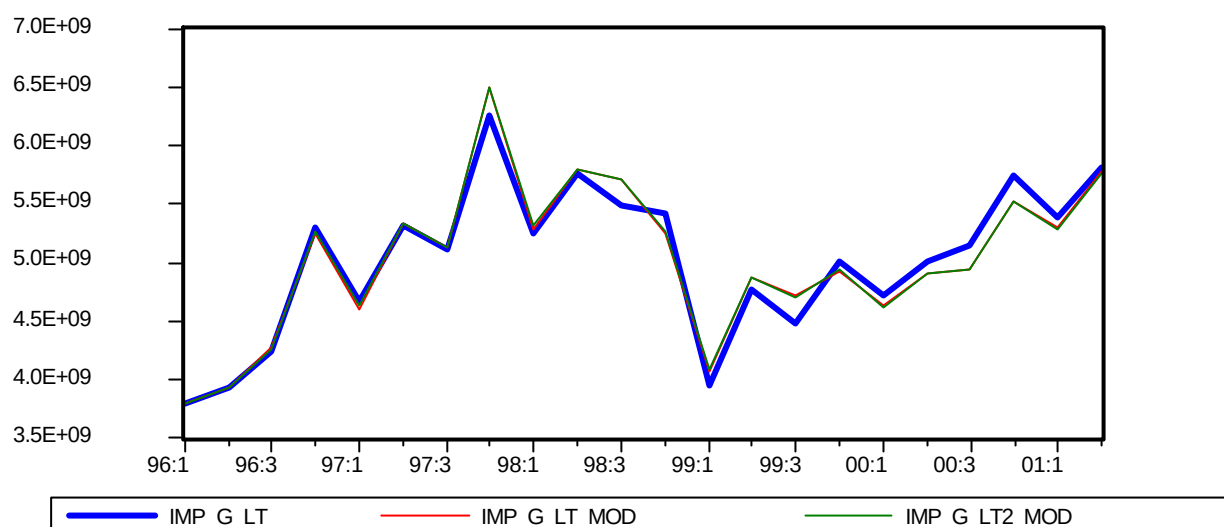
The Observed and Imitated Normalized Changes of the REER of LTL, 1996-2001



The form of the modeled normalized REER is mostly similar to actual its changes but there are some systemic deviations in prolonged period from about 1998: until 2000 it goes below the actual change and after – above its level.

Graph 5.2.7

The Observed and Imitated Normalized Changes of the Import of Goods in Lithuania, 1996-2001



It is important to mention that GDP production sectors and their derivatives as important regressors interconnecting real economy, monetary and foreign trade flows in the Baltics often can't be measured in detail in their back-end actions reflecting Balassa-Samuelson and Mundell-Fleming effects. Any such attempt based on theoretical considerations is strongly bordering with common sense as result of big deformations and deviations under unmeasured shadow economy exceeding tens of times the expected effect of currency re-pegging in small open economies.

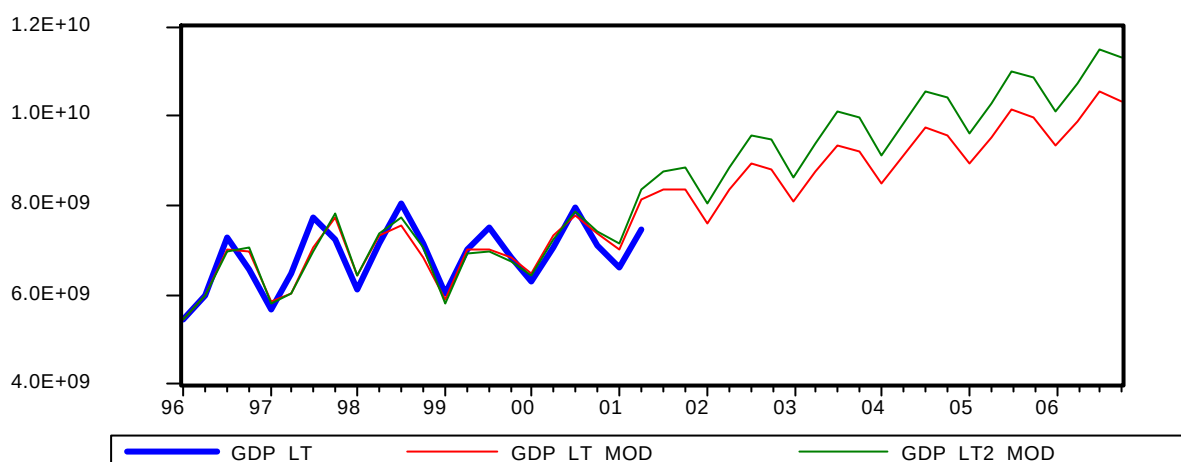
The preliminary calculations when econometrically evaluating the currency re-pegging effect onto monetary a/o economic transmission channels do not revealed enough significant identifiable results which would overpass the level of errors and omissions determined by evaluating various effects including those not measured by comparative official statistics incl. balance of payment and its main components.

The results of presented econometric approach were not so bright as those of leading partner in the ACE program, Dr. Lucio Vinhas de Souza and Elizabeth Ledrut (2001) but even in that case I would not risk doing conclusions concerning the priorities of the CBA or efficiency of float regime under influence of many important additional and still not modeled factors including political economic ones.

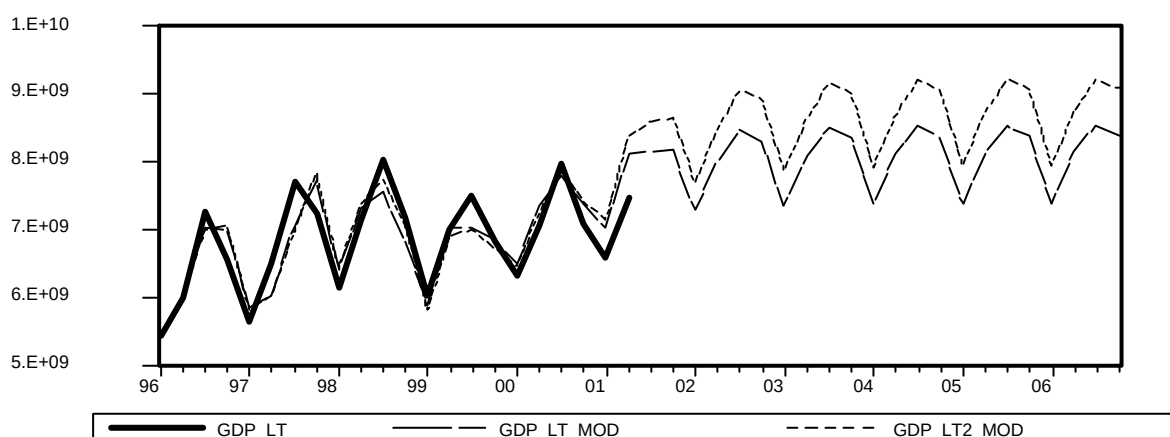
5.3. The forecasts with the model as above were done using exponential trend and in some cases smoothing the seasonal fluctuations (except imitation of the GDP). In some degree its results may be interpreted as optimistic ones so as exponential curve goes more sharply upward than linear one. Thick curve shows actual changes, dash curve – main development as before (under influence of pegging to USD as in previous period) and dotted curve – expected development with account of pegging to EUR (which approximate adequate is NEER_EUR). Some results can be seen from the graphs below.

Graph 5.3.1

**The Prognostic Simulation of Actual and Normalized Changes of the GDP in Lithuania, 1996-2007:
Optimistic and Pessimistic Variants**



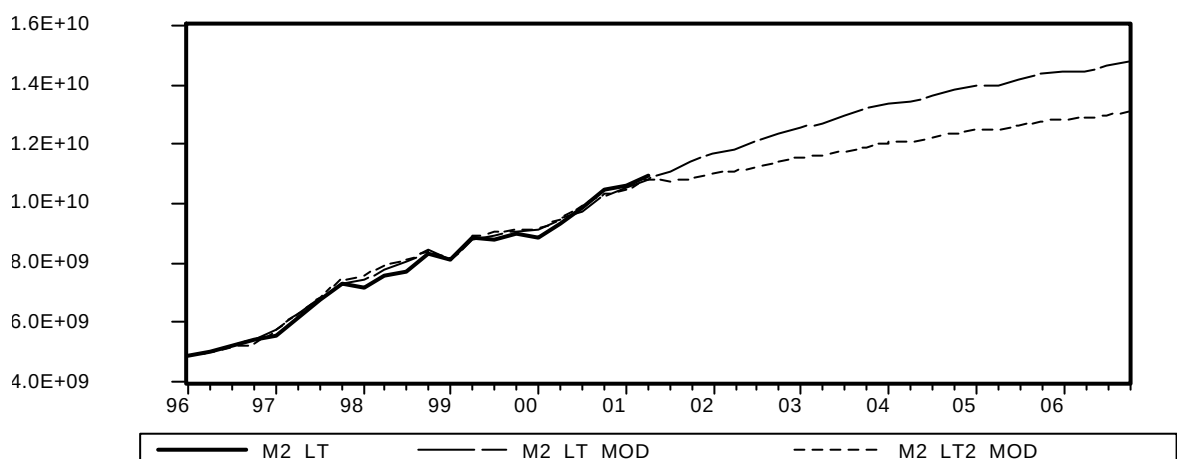
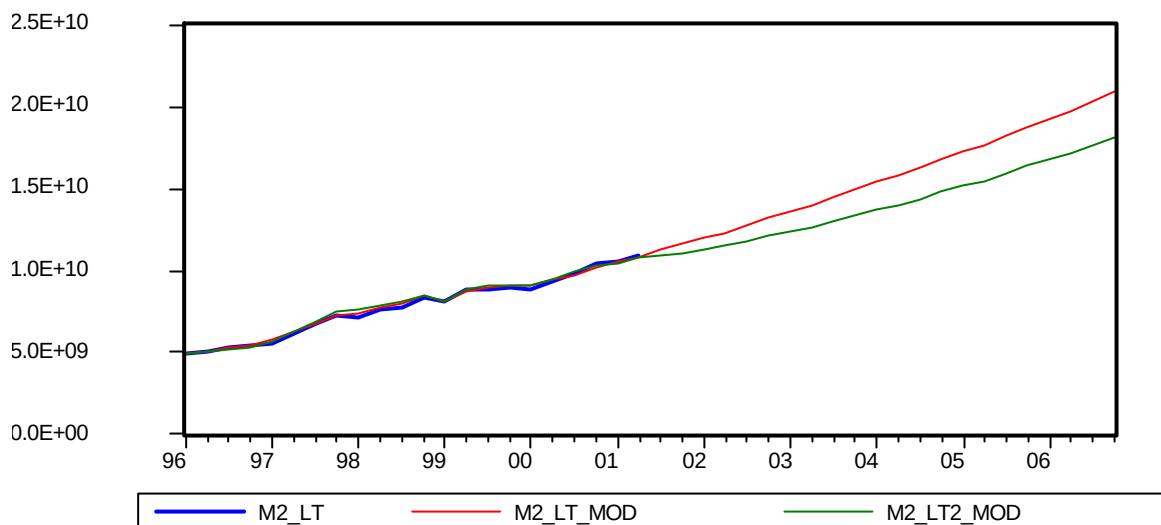
Both curves (actual and imitated changes of the GDP) goes rather similar way with repeating yearly cyclical fluctuations. The changes in the cross rates of the USD to EUR are not fixed in these movements enough significantly for 1999, when euro lost its weight to USD.



The increasing divergence between main and expected normalized trends in money supply of LTL may be influenced by expected changes in the market rates of EUR to USD which not influenced this supply in 1996-2001. With account of prevailing tendencies within more long previous period, the ratio of USD to EUR is expected to go slowly up from the level as was fixed in 2001.

Graph 5.3.2

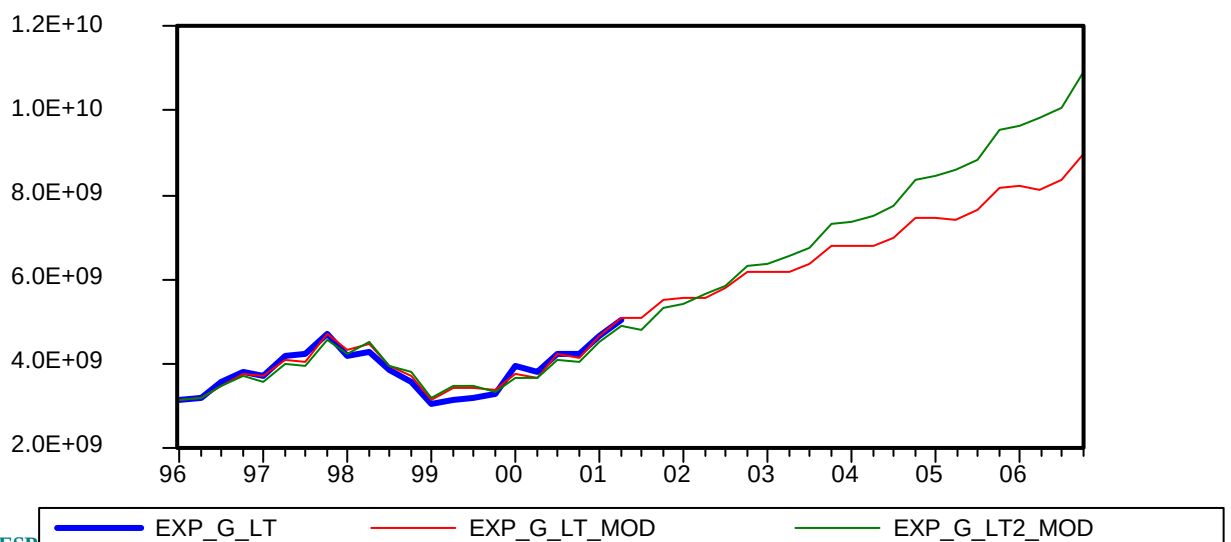
**The Prognostic Simulation of Actual and Normalized Changes of the Money Supply
in Lithuania, 1996-2007: Optimistic and Pessimistic Variants**

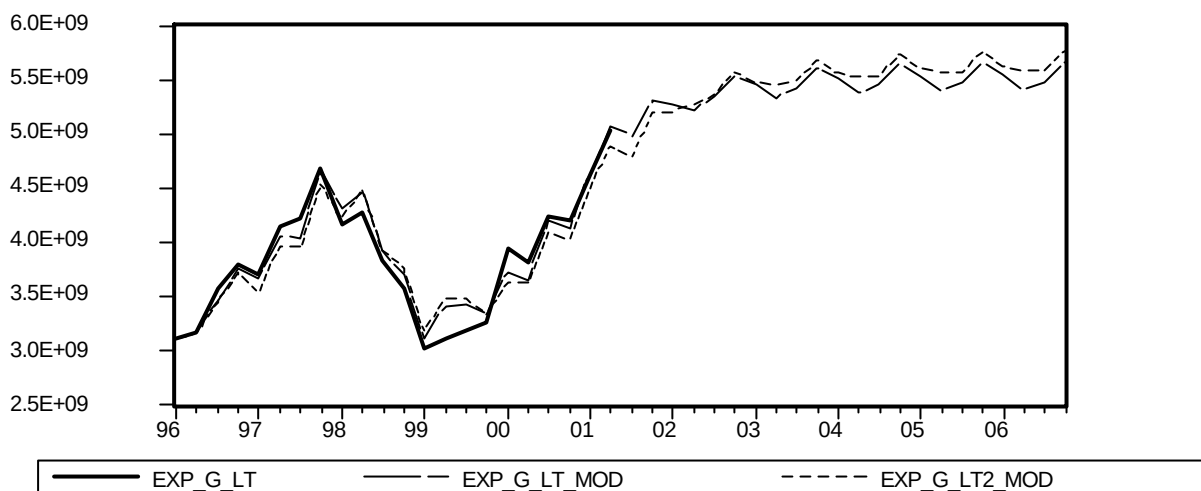


The imitation pattern of the perspective dynamics of the export of goods is rather routinous with insignificant yearly (seasonal /)waves both in inertive and normalized trends.

Graph 5.3.3

**The Prognostic Simulation of Actual and Normalized Changes of the Export of Goods
in Lithuania, 1996-2007: Optimistic and Pessimistic Variants**

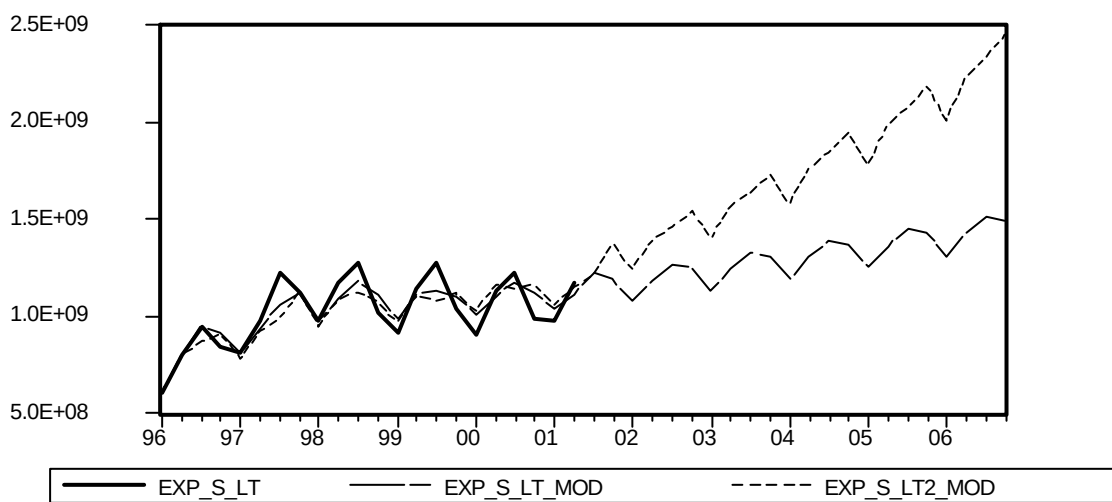


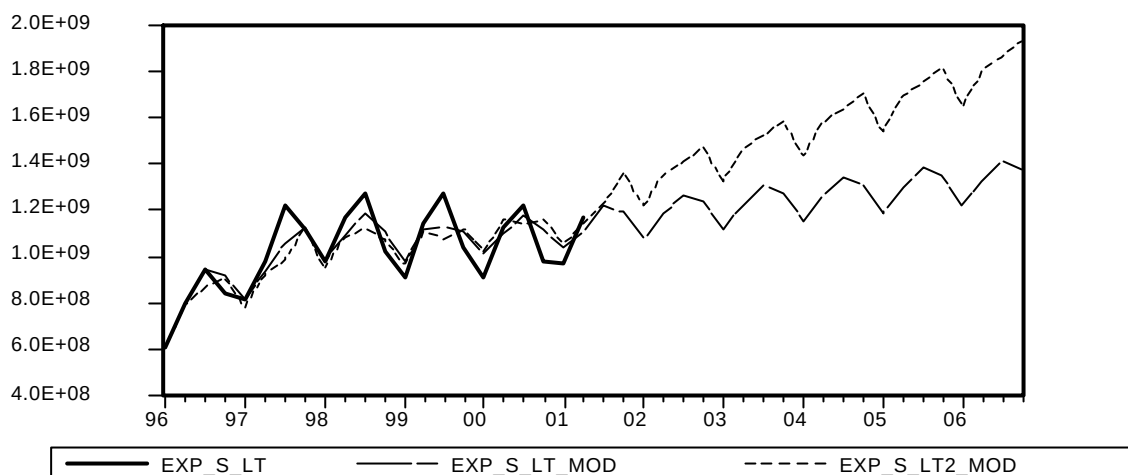


The statistical imitation of export of services is influenced by sharp seasonal fluctuations and not very successful. The increasing divergence between main and expected normalized trends in export of services in LTL may be influenced by expected changes in the market rates of EUR to USD fixed in the money markets in 1999-2001.

Graph 5.3.4

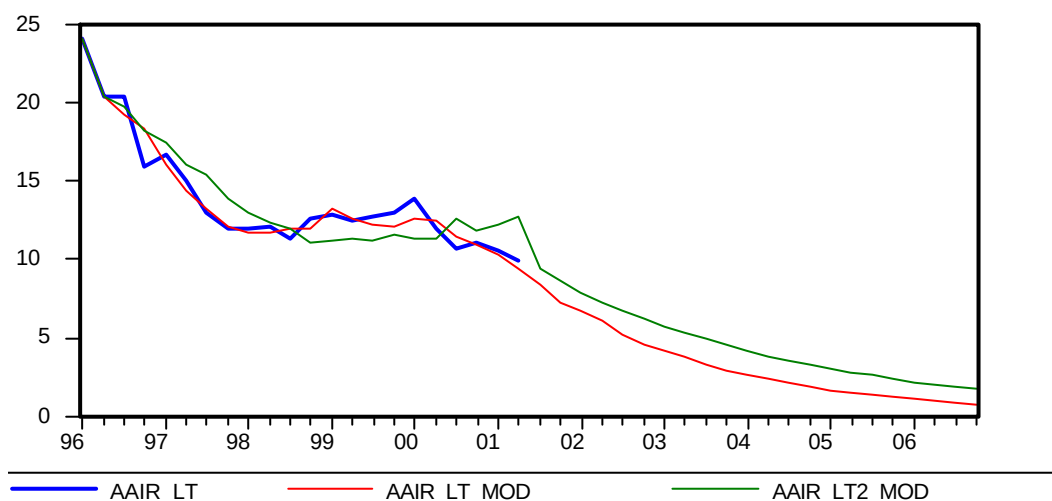
The Prognostic Simulation of Actual and Normalized Changes of the Export of Services in Lithuania, 1996-2007: Optimistic and Pessimistic Variants



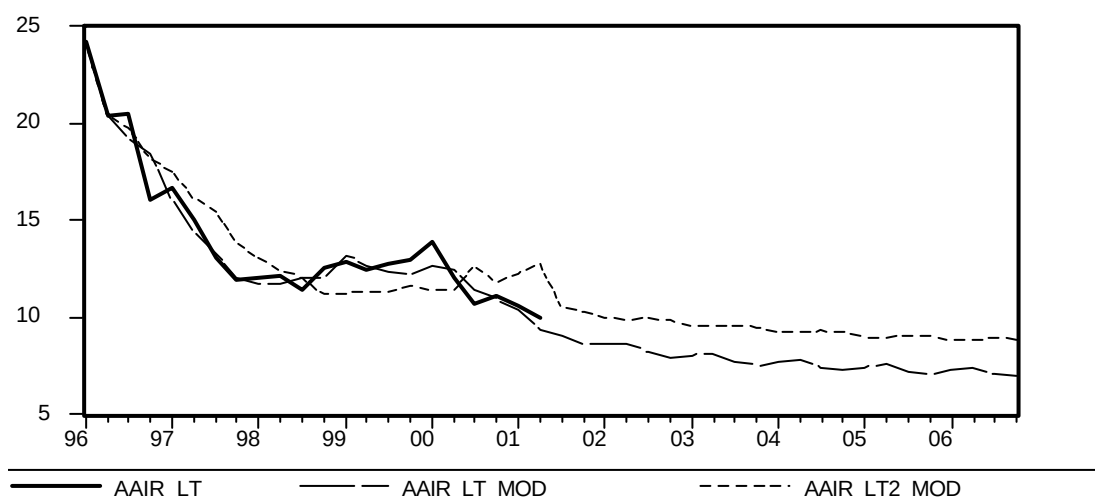


Graph 5.3.5

The Prognostic Simulation of Actual and Normalized Changes of the Interest Rates in Lithuania, 1996-2007: Optimistic and Pessimistic Variants



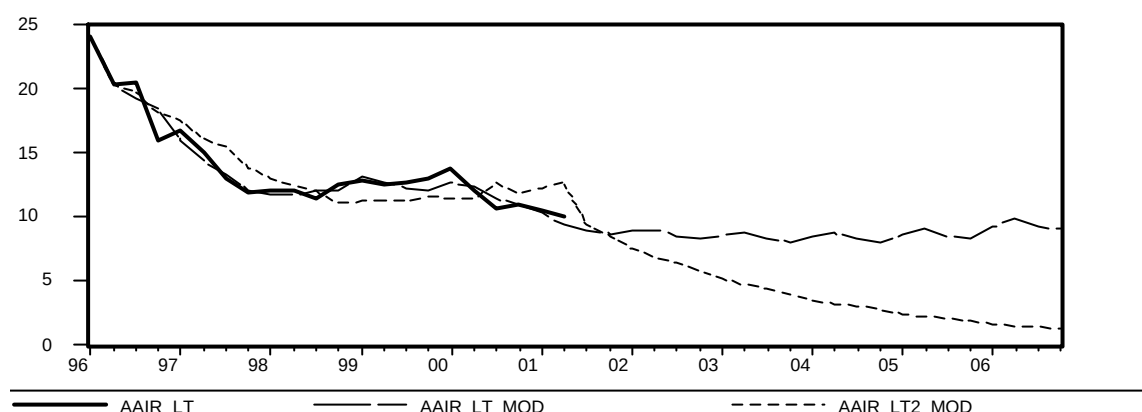
The pessimistic modification when expecting higher level of the IR is as below:



The same model used for imitation of expected perspective changes in the interest rates gives different trajectories of their normative and inertive development as seen from chart 5.3.5a below.

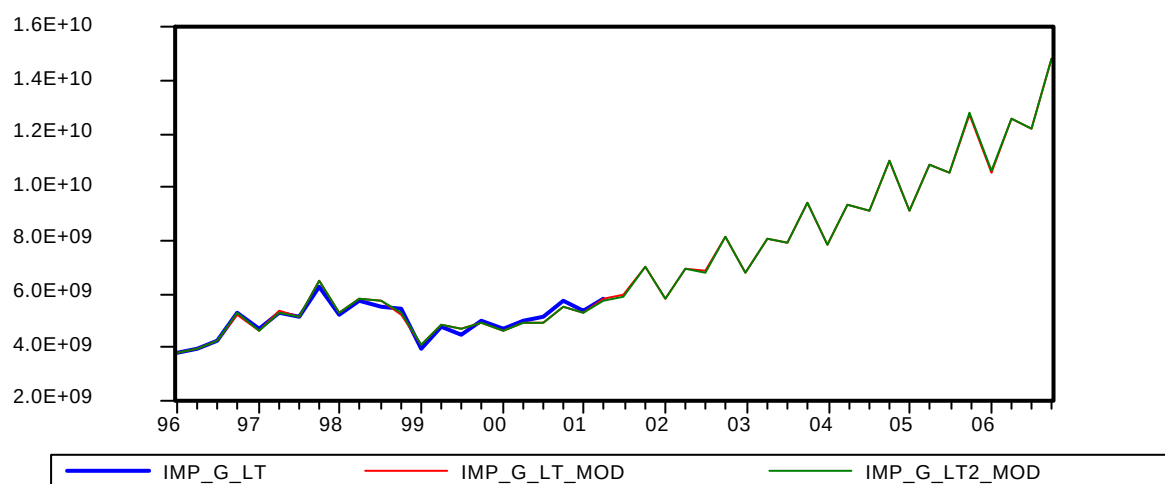
Graph 5.3.5a

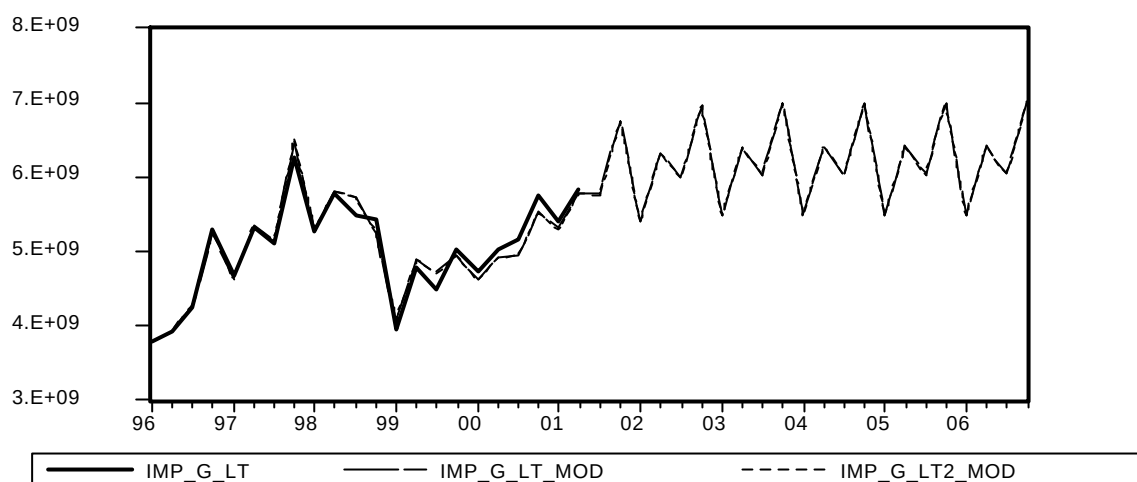
**The Prognostic Simulation of Actual and Normalized Changes of the Interest Rates
in Lithuania, 1996-2007: Linear Variant**



Graph 5.3.6

**The Prognostic Simulation of Actual and Normalized Changes of the Import of Goods
in Lithuania, 1996-2007: Optimistic and Pessimistic Variants**





No clearly seen difference between both forecasted lines may be interpreted as necessity to change main regressors including more strong interrelations with internal monetary transmission and foreign monetary market indicators when using this kind of approach for analysis of re-pegging influence.

6. PROGNOSTIC MODELING OF MONETARY TRANSMISSION: ESTONIA CASE

6.1. The adequacy of the model worked out in the section 5 of this study was experimented also on the base of Estonian macroeconomic statistics: main results of its applicability for the simplified evaluation of the currency pegging, exchange rate and/ or monetary transmission influence on economic development were mostly rather similar to the imitation of Lithuania's macroeconomic flows by their appearance.

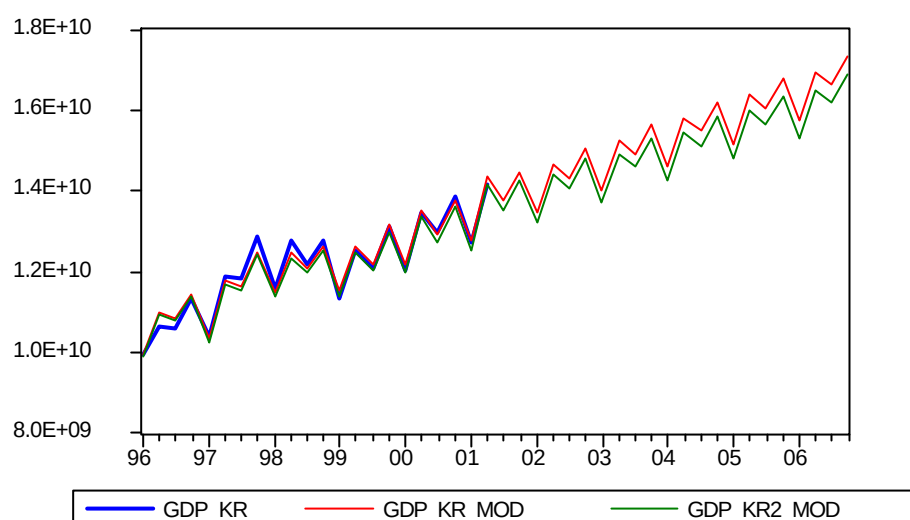
The calculations of parameters were conducted also in two versions: **a** – the imitation of actual changes and their trends and **b** – the imitation of normalized changes and their trends weighing main parameters with account of EEK rate to USD (as it is known, EEK was fixed to DEM and later to EUR) which approaches Estonian monetary system to actual and expected fluctuations of international currency markets through simplified changes within main exchange rate of EUR to USD. The estimation of model parameters in version **a** was based on 204 total system (unbalanced) observations for the period 1995:2–2001:4 using iterative least squares method. and forecasts of expected changes forwarded until 2007.

Some results of imitation for main determined variables can be seen from the graphs below based on the econometric calculations. Thick (blue) curve shows actual changes, dash (red) curve – trend of development (in fact, under influence of pegging to DEM as in previous period and later - pegging to EUR; its approximate adequate is NEER_EUR) and dotted (green) curve – expected development with account of changes in the market rate of EEK to USD.

Graph 6.1.1.

The Observed Changes of the GDP in Estonia,

Prognostic Simulation of Its Actual and Normalized Changes, 1996-2007

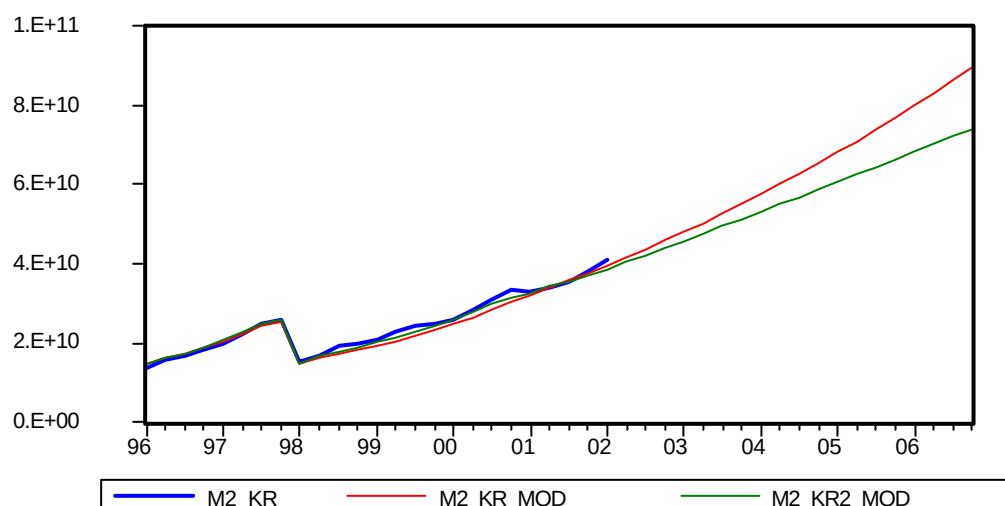


Both curves goes nearly repeating one another movements. The changes in the cross rates of the USD to EUR not fixed in these movements enough significantly, possibly as a result of low dependence from investments or payments in USD.

Graph 6.1.2.

The Observed Changes of the Money Supply in Estonia,

Prognostic Simulation of Its Actual and Normalized Changes, 1996-2007

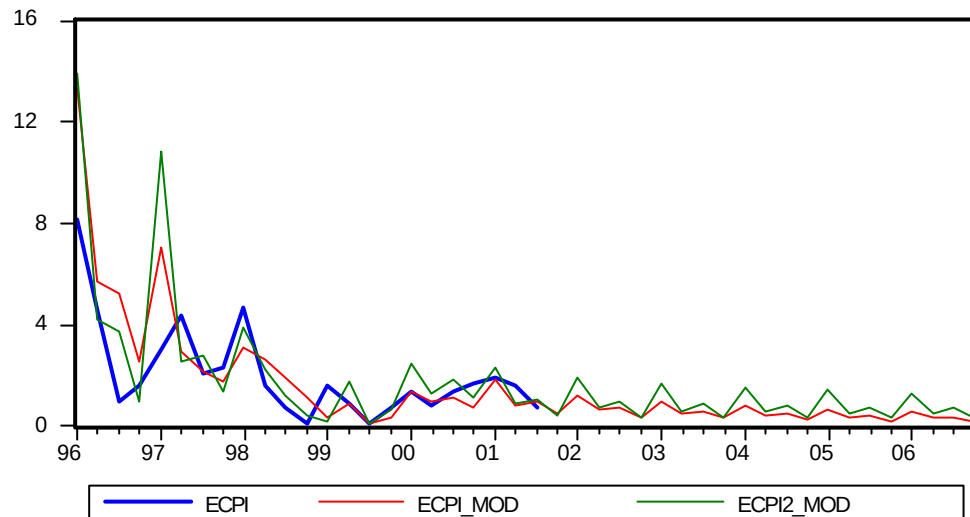


The increasing divergence between main and expected normalized trends in money supply of EEK may be influenced by expected changes in the market rates of EUR to USD. With account of prevailing tendencies within more long previous period, the ratio of USD to EUR went up.

Graph 6.1.3.

The Observed Changes of the CPI in Estonia,

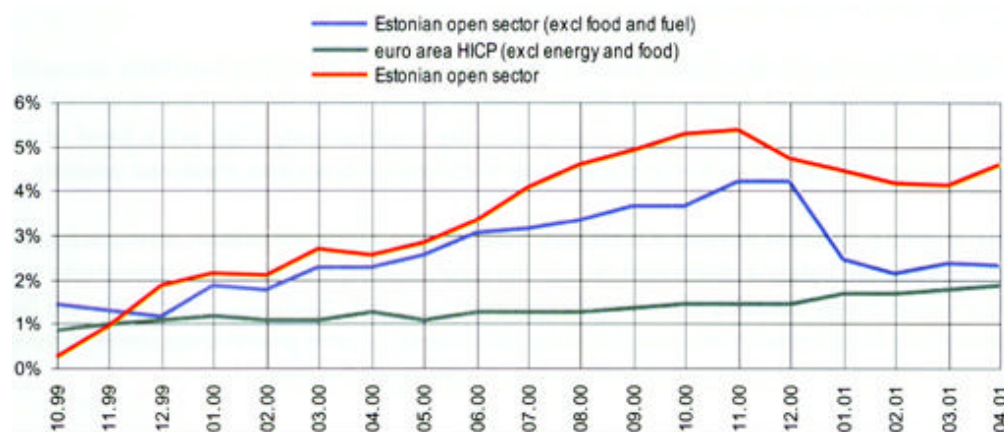
Prognostic Simulation of Its Actual and Normalized Changes, 1996-2007



The consumer prices are imitated as fluctuating seasonally more sharply in normalized trends than in regular trends EEK what may be influenced by different amplitude of fluctuations in the volume of GDP on market rates of EUR and USD. Euro zone consumer prices seasonally smoothed are much more stable as seen from graph 6.1.3a

Graph 6.1.3a.

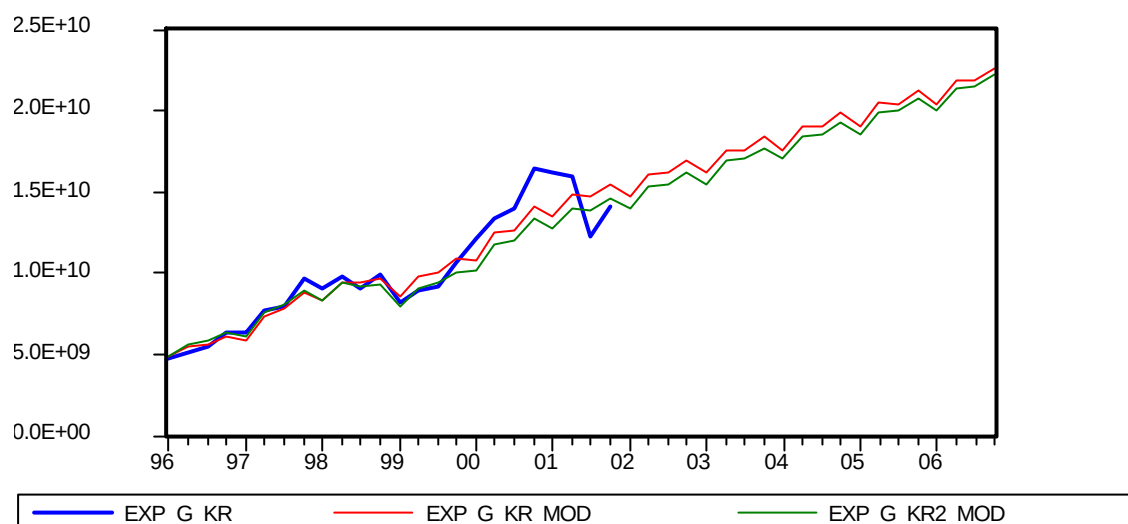
Comparative Consumer Prices in Estonia and Euro Area, 1997-2001



Source: Bank of Estonia, 2001.

Graph 6.1.4.

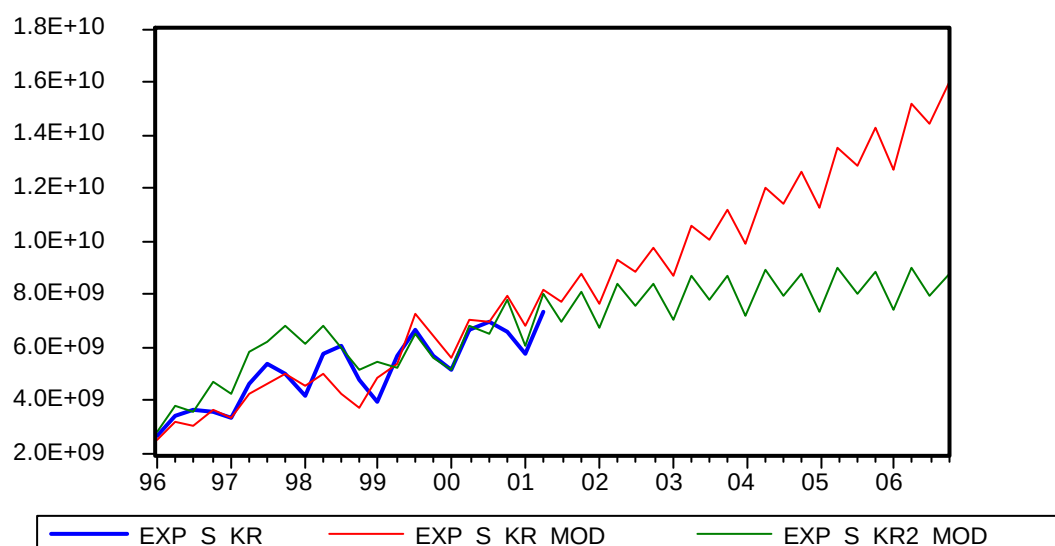
**The Observed Changes of the Export of Goods in Estonia,
Prognostic Simulation of Their Actual and Normalized Changes, 1996-2007**



The imitation pattern of the perspective dynamics of the export of goods was substantially influenced by 1999-2000 slump and 2000-2001 unprecedented upraise waves cyclically repeated in more smooth form both in inertive and normalized trends.

Graph 6.1.5.

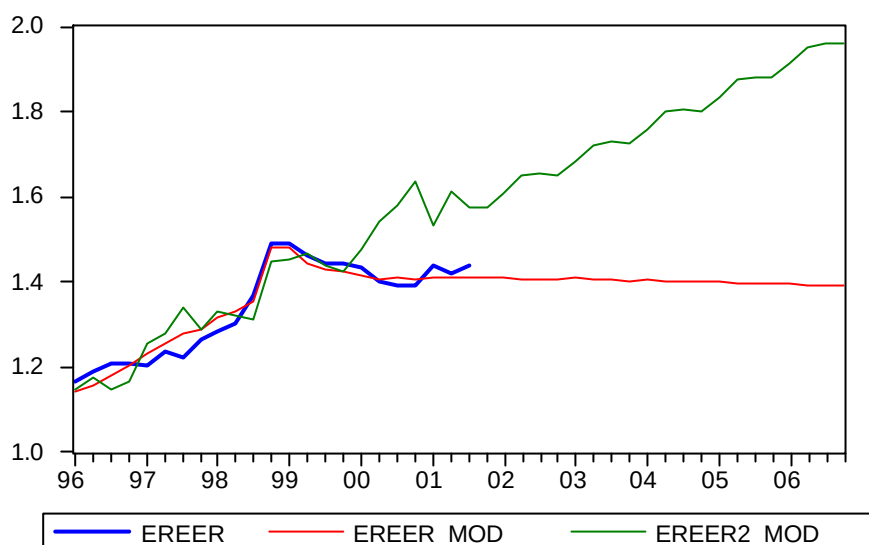
**The Observed Changes of the Export of Services in Estonia,
Prognostic Simulation of Their Actual and Normalized Changes, 1996-2007**



The statistical imitation of export of services is influenced by sharp seasonal fluctuations. The increasing divergence between main and expected normalized trends in export of services in EEK may be influenced by expected changes in the market rates of EUR to USD.

Graph 6.1.6.

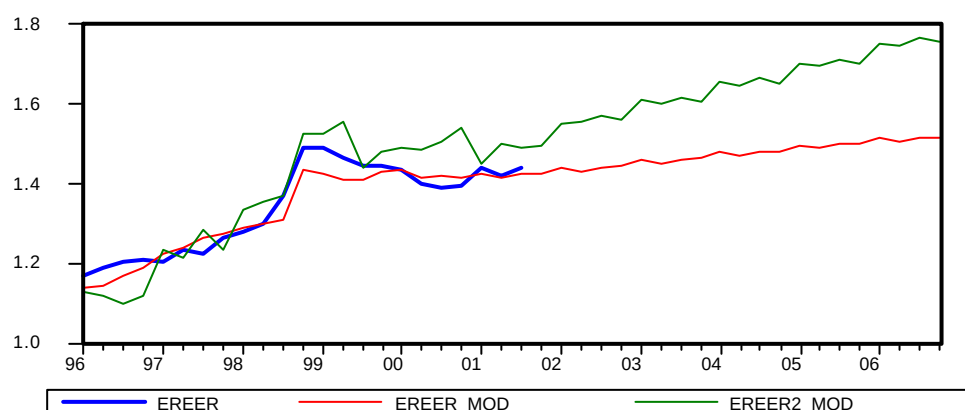
**The Observed Changes of the REER in Estonia,
Prognostic Simulation of Its Actual and Normalized Changes, 1996-2007: a variant**



The dynamics of the REER is rather complicated and presented in two modifications. Variant b is imitating the actual changes in 1996-2001 worse than variant a but it do not leads to such significant divergences between inertive and normalized trend forecasts; it is based on dependencies from loans, money supply and total export. Variant a is imitating about better the actual changes in 1996-2001 but it leads to not reliable forecast of normalized trend which sharply not coincidences with changes in fact for 2000-2001, accounts additional dependencies on changes in CPI, interest rates on loans and export of goods. The modification for variant b is presented below, graph 6.1.6a. In both cases the imitation of the REER still was not successful and needs to be ameliorated in the future.

Graph 6.1.6a

**The Observed Changes of the REER in Estonia,
Prognostic Simulation of Its Actual and Normalized Changes, 1996-2007: b variant**

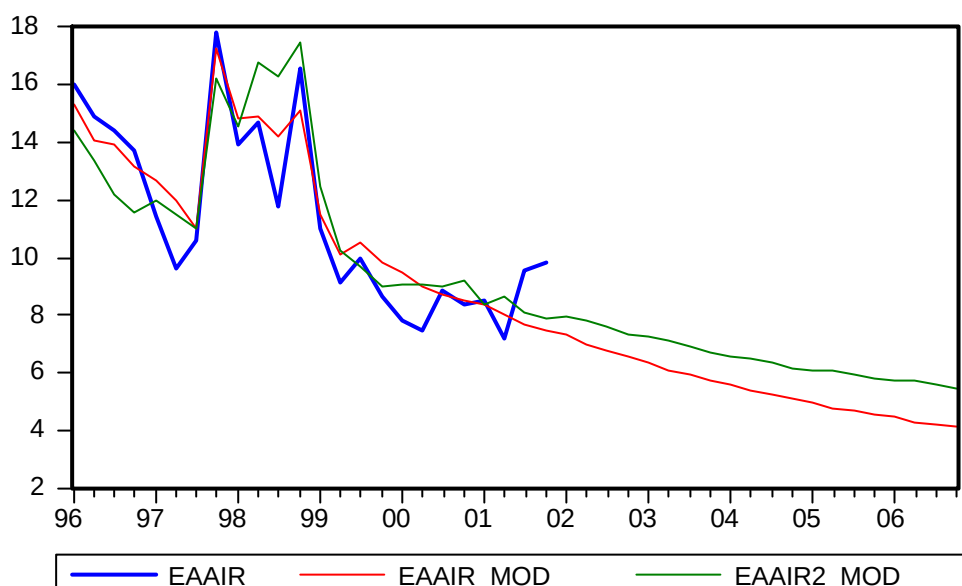


The similar problematic was attempt to find the simplified equation delineating correctly the complicated changes of the interest rates for loans depending not only from economic mechanism but also of external factors (cf. especially opposite changes of normalized curve to actual one in 1998-99 in the graph 6.1.7). The reasonable and admissible

was successful imitation of lowering interest rates in both inertive and normalized trend forecasts under intensifying influence of international markets when approaching to the EU.

Graph 6.1.7.

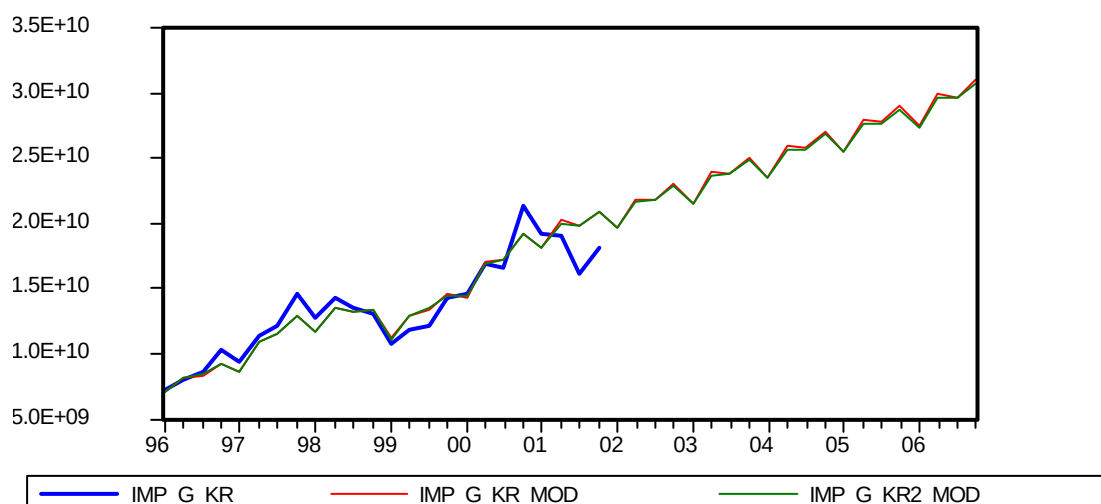
**The Observed Changes of the Interest Rates in Estonia,
Prognostic Simulation of Their Actual and Normalized Changes, 1996-2007**



The imitation of the import of goods in Estonia is mostly successful as in the case of Lithuania but not fruitful in aspect of its applicability for a study of influence of exchange pegging so as both curves – actual and normalized trend– goes about coinciding one another for all period (cf. below).

Graph 6.1.8.

**The Observed Changes of the Import of Goods in Estonia,
Prognostic Simulation of Its Actual and Normalized Changes, 1996-2007**



6.2. The significances of model parameters calculated as a residuals of cointegration for equations of variant **a** are presented in the table 6.2.1, for some independent variables determining REER – in both modifications: a and b (more detailed evaluation of parameters for variant b is presented below, in the table 6.3.1).

Table 6.2.1

The Results Of The Model Parameters Estimation by Iterative Least Squares: a¹⁷

	Coefficient	Std. Error	t-Statistic	Prob.
B(16)	-0.465066	0.179845	-2.585932	0.0105
B(17)	0.444893	0.027534	16.15820	0.0000
B(18)	0.527908	0.026439	19.96728	0.0000
B(26)	-0.092474	0.068624	-1.347547	0.1796
B(27)	0.049971	0.013113	3.810745	0.0002
B(28)	0.233820	0.128574	1.818555	0.0707
B(29)	-0.600989	0.038871	-15.46111	0.0000
B(36)	-0.671789	0.303335	-2.214677	0.0281
B(37)	7.336319	4.197019	1.747983	0.0822
B(38)	-11.78524	5.157288	-2.285163	0.0235
B(39)	-2.439035	0.874452	-2.789217	0.0059
B(46)	-0.378033	0.124890	-3.026918	0.0028
B(47)	0.833329	0.070350	11.84545	0.0000
B(56)	-0.746176	0.137942	-5.409345	0.0000
B(57)	0.062222	0.019914	3.124476	0.0021
B(58)	0.606616	0.197611	3.069746	0.0025
B(59)	-0.119668	0.019332	-6.190175	0.0000
B(60)	-2.307672	0.546263	-4.224474	0.0000
B(66a)	-0.238547	0.092298	-2.584532	0.0106
B(66b)	-0.324554	0.124807	-2.600447	0.0102
B(67a)	0.054915	0.018685	2.939018	0.0037
B(67b)	0.012183	0.048948	0.248892	0.8038
B(68a)	-0.053471	0.040249	-1.328499	0.1858
B(68b)	-0.043876	0.103289	-0.424789	0.6716
B(69)	0.075581	0.019915	3.795099	0.0002
B(76)	-0.954940	0.191901	-4.976207	0.0000
B(77)	-0.056693	0.026145	-2.168449	0.0315
B(78)	-0.541878	0.237530	-2.281308	0.0237
B(79)	1.945743	0.835567	2.328649	0.0210
B(80)	0.489485	0.116790	4.191171	0.0000
B(86)	-0.597696	0.197075	-3.032830	0.0028
B(87)	0.583518	0.175233	3.329953	0.0011
B(88)	0.561647	0.247009	2.273791	0.0242

Determinant residual covariance was equal 2.64E-21.

In more detail, the parameters calculated as a residuals of cointegration for the GDP of Estonia were as follows:

$$\begin{aligned} \text{DLOG(EGDP_KR)} = & \text{B(16)} * (\text{LOG(EGDP_KR(-1))} - 1.881119828 - 0.544380757 * \text{LOG(EGVA_SERV_KR(-1))} \\ & - 0.4332205705 * \text{LOG(EGVA_TRD_KR(-1))}) + \text{B(17)} * \text{DLOG(EGVA_TRD_KR)} + \text{B(18)} * \\ & \text{DLOG(EGVA_SERV_KR)} \end{aligned} \quad 6.2.1$$

where D – difference operator in logarithmic form expressing relative change in the variable against previous quarter,

E tter E when it added to variables denotes Estonia, p.ex.:

EGDP - gross domestic product in Estonia,

GVA_TRD – gross value added in wholesale and retail trade in Estonia,

EGVA_SERV – gross value added in services in Estonia.

The meanings of other variables in equations are similar to those used in equations for Lithuania (previous divisions 4 and 5 of this study).

The equation is expected to identify how much the gross domestic product is approximated not by its direct components but by gross value added within these sectors.

R-squared	0.977383	Mean dependent var	0.015565
Adjusted R-squared	0.975327	S.D. dependent var	0.078921
S.E. of regression	0.012397	Sum squared resid	0.003381
Durbin-Watson stat	1.805617		

The parameters of equation for money supply (M2) calculated as a residuals of cointegration for Estonia are presented in 6.2.2:

$$\begin{aligned} \text{DLOG(EM2_KR)} = & \text{B(26)} * (\text{LOG(EM2_KR(-1))} - 0.7168100729 * \text{LOG(EM2_KR(-2))} - 0.2877108236 * \\ & \text{LOG(EGDP_KR(-1))} - 0.4288193676 * \text{LOG(ENEER(-1))}) + \text{B(27)} + \text{B(28)} * \text{DLOG(EL_KR)} + \text{B(29)} * \text{D1} \end{aligned} \quad 6.2.2$$

Results of statistical testing of parameters for equation 6.2.2 are as below:

R-squared	0.927960	Mean dependent var	0.044935
Adjusted R-squared	0.917668	S.D. dependent var	0.129425
S.E. of regression	0.037137	Sum squared resid	0.028962
Durbin-Watson stat	1.966090		

The parameters of equation for consumer price index (ECPI) calculated as a residuals of cointegration for Estonia are presented in 6.2.3:

$$\begin{aligned} \text{DLOG(ECPI)} = & \text{B(36)} * (\text{LOG(ECPI(-1))} - 1.036468181 * \text{Q3(-1)} - 2.835754869 * \text{LOG(EW_KR(-1))} - \\ & 1.045098047 * \text{LOG(EGDP_KR(-1))} - 2.839679747 * \text{D4(-1)}) + \text{B(37)} * \text{DLOG(EW_KR)} + \text{B(38)} \end{aligned}$$

¹⁷ Sample: 1995:2 2001:4, included observations: 27, total system (unbalanced) observations 204.

$$*DLOG(EGDP_KR)+B(39)*D2$$

6.2.3

Results of statistical testing of parameters for equation 6.2.3 are as below:

R-squared	0.456164	Mean dependent var	-0.063874
Adjusted R-squared	0.378473	S.D. dependent var	1.080515
S.E. of regression	0.851845	Sum squared resid	15.23845
Durbin-Watson stat	2.284540		

The parameters of equation for export of goods (EEXP_G_KR) calculated as a residuals of cointegration for Estonia are presented in 6.2.4:

$$DLOG(EEXP_G_KR)=B(46)*(LOG(EEXP_G_KR(-1)) -0.9800631794*LOG(EIMP_G_KR(-1))-0.5422179155*LOG(EREER(-1)))+B(47)*DLOG(EIMP_G_KR)$$

6.2.4

Results of statistical testing of parameters for equation 6.2.4 are as below:

R-squared	0.828587	Mean dependent var	0.041983
Adjusted R-squared	0.821731	S.D. dependent var	0.111375
S.E. of regression	0.047025	Sum squared resid	0.055283
Durbin-Watson stat	2.156503		

The parameters of equation for export of services (EEXP_S_KR) calculated as a residuals of cointegration for Estonia are presented in 6.2.5:

$$DLOG(EEXP_S_KR)=B(56)*(LOG(EEXP_S_KR(-1)) -0.8356419142*LOG(EGVA_SERV_KR(-1)) -0.09474002352*LOG(ECPI(-1)) - 1.378016298*LOG(ER_KR_USD(-1)))+B(57)+B(58)*DLOG(EGVA_SERV_KR)+B(59)*DLOG(ECPI)+B(60)*DLOG(ENEER)$$

6.2.5

Results of statistical testing of parameters for equation 6.2.5 are as below:

R-squared	0.784159	Mean dependent var	0.058871
Adjusted R-squared	0.740991	S.D. dependent var	0.188926
S.E. of regression	0.096150	Sum squared resid	0.184897
Durbin-Watson stat	1.947879		

The parameters of equation for REER calculated as a residuals of cointegration for Estonia are presented in 6.2.6:

$$DLOG(EREER) = B(66) * (LOG(EREER(-1)) - 0.07172325665 * LOG(EM2_KR(-1)) - 0.08620497737 * LOG(EL_FX_KR(-1))) + B(67) + DLOG(EL_FX_KR) + B(68) * DLOG(EEXP_KR) + B(69) * D4$$

6.2.6

Convergence achieved after 2 iterations

Results of statistical testing of parameters for equation 6.2.6 are as below:

R-squared	0.547708	Mean dependent var	0.011827
Adjusted R-squared	0.486031	S.D. dependent var	0.027551
S.E. of regression	0.019751	Sum squared resid	0.008583
Durbin-Watson stat	2.279407		

The parameters of equation for the interest rate (EAAIR) calculated as a residuals of cointegration for Estonia are presented in 6.2.5:

$$\begin{aligned} \text{DLOG(EAAIR)} = & \text{B(76)} * (\text{LOG(EAAIR(-1))} - 14.20872195 - 0.7935796088 * \text{LOG(EM2_KR(-1))} - \\ & 0.3141388968 * \text{LOG(EL_EEK_KR(-))}) + \text{B(77)} + \text{B(78)} * \text{DLOG(EM2_KR)} + \text{B(79)} * \text{DLOG(EREER)} \\ & + \text{B(80)} * \text{D5} \end{aligned} \quad 6.2.7$$

Results of statistical testing of parameters for equation 6.2.7 are as below:

R-squared	0.719972	Mean dependent var	-0.026760
Adjusted R-squared	0.666633	S.D. dependent var	0.194977
S.E. of regression	0.112575	Sum squared resid	0.266138
Durbin-Watson stat	1.973154		

The parameters of equation for the import of goods (EIMP_G_KR) calculated as a residuals of cointegration for Estonia are presented in 6.2.8:

$$\begin{aligned} \text{DLOG(EIMP_G_KR)} = & \text{B(86)} * (\text{LOG(EIMP_G_KR(-1))} - 0.8596302064 * \text{LOG(EEXP_KR(-1))} - \\ & 0.1362961073 * \text{LOG(EGDP_KR(-1))}) + \text{B(87)} * \text{DLOG(EEXP_KR)} + \text{B(88)} * \text{DLOG(EGDP_KR)} \end{aligned} \quad 6.2.8$$

Results of statistical testing of parameters for equation 6.2.8 are as below:

R-squared	0.793431	Mean dependent var	0.045958
Adjusted R-squared	0.774652	S.D. dependent var	0.122581
S.E. of regression	0.058190	Sum squared resid	0.074495
Durbin-Watson stat	1.572568		

6.3. Below are presented the significances of model parameters calculations for equations of variant **b** (normalized imitation of the basic interrelations of variables for Estonia). It is important to mention that in about 1/3 cases the probabilistic characteristics of variant **b** were worse than those of variant **a** and exceeding the admissible level of probability. It is possible to expect similar misbalancing of the systemic imitation of the Lithuania's model equations in case of weighing their parameters to EUR especially for the period before repegging LTL to EUR.

Table 6.3.1

The Results Of The Model Parameters Estimation by Iterative Least Squares: b¹⁸

	Coefficient	Std. Error	t-Statistic	Prob.
B(16)	0.000163	0.000630	0.259389	0.7957
B(17)	0.460510	0.033668	13.67814	0.0000
B(18)	0.538171	0.033522	16.05432	0.0000
B(26)	-0.131881	0.060124	-2.193483	0.0298
B(27)	0.041892	0.010876	3.851647	0.0002
B(28)	0.192868	0.125019	1.542708	0.1250
B(29)	0.320931	0.111182	2.886528	0.0045
B(30)	-8.830211	0.538297	-16.40396	0.0000
B(36)	-1.108882	0.314470	-3.526193	0.0006
B(37)	6.975175	4.050224	1.722170	0.0871
B(38)	-12.34205	4.499201	-2.743164	0.0068
B(39)	-36.72522	11.10907	-3.305876	0.0012
B(46)	-0.196527	0.124051	-1.584242	0.1152
B(47)	0.834740	0.088148	9.469799	0.0000
B(56)	-0.011892	0.020000	-0.594623	0.5530
B(57)	0.042174	0.078117	0.539874	0.5901
B(58)	0.662957	0.280389	2.364423	0.0193
B(59)	-0.097442	0.032064	-3.038960	0.0028
B(60)	-1.336292	0.476462	-2.804613	0.0057
B(66)	-0.324554	0.124807	-2.600447	0.0102
B(67)	0.012183	0.048948	0.248892	0.8038
B(68)	-0.043876	0.103289	-0.424789	0.6716
B(69)	2.133114	0.646317	3.300416	0.0012
B(76)	-0.498874	0.164283	-3.036681	0.0028
B(77)	-0.062507	0.029775	-2.099341	0.0375
B(78)	-0.233798	0.271931	-0.859768	0.3913
B(79)	1.565061	0.516052	3.032759	0.0029
B(80)	6.914599	1.999179	3.458720	0.0007
B(86)	-0.514213	0.206418	-2.491119	0.0138
B(87)	0.702158	0.173968	4.036127	0.0001
B(88)	0.423352	0.207260	2.042618	0.0428
Determinant residual covariance	1.78E-20			

¹⁸ Sample: 1996:1 - 2001:4, included observations: 24, total system (unbalanced) observations 181.
Convergence achieved after 2 iterations

The parameters of equation for the GDP (EGDP) calculated as normalized residuals of cointegration for Estonia are presented in 6.3.1:

$$\begin{aligned} \text{DLOG(EGDP_KR/ER_KR_USD)} = & \text{B(16)*[LOG(EGDP_KR(-1)/ER_KR_USD(-1)) -} \\ & \text{9.049808904-0.6207957203 * LOG(EGVA_TRD_KR(-1)/ER_KR_USD(-1)) - 0.2379710928 *} \\ & \text{LOG(EGVA_SERV_KR(-1)/ER_KR_USD(-1))]+B(17) * G(EGVA_TRD_KR/ER_KR_USD) +} \\ & \text{B(18)*DLOG(EGVA_SERV_KR/ER_KR_USD)} \end{aligned} \quad 6.3.1$$

Results of statistical testing of parameters for equation 6.3.1 are about the same as for 6.2.1 what means that neutralization of USD influence to EGDP do not changed the applicability of the equation for the imitation of dynamics of EGDP:

R-squared	0.979604	Mean dependent var	-0.008225
Adjusted R-squared	0.977457	S.D. dependent var	0.091742
S.E. of regression	0.013774	Sum squared resid	0.003605
Durbin-Watson stat	2.298118		

The parameters of equation for the money supply (EM2) calculated as normalized residuals of cointegration for Estonia are presented in 6.3.2:

$$\begin{aligned} \text{DLOG(EM2_KR/ER_KR_USD)} = & \text{B(26)*[LOG(EM2_KR(-1)/ER_KR_USD(-1))-} \\ & \text{0.766871857*LOG(EM2_KR(-2)/ER_KR_USD(-2))-0.240597438*LOG(EGDP_KR(-1)/ER_KR_USD(-} \\ & \text{1))]+B(27)+B(28) *DLOG(EL_KR/ER_KR_USD)+B(29)* DLOG(ENEER/ER_KR_USD)+} \\ & \text{B(30)*(D1/ER_KR_USD)} \end{aligned} \quad 6.3.2$$

Results of statistical testing of parameters for equation 6.3.2 after neutralization of USD influence on EM2 also do not changed the applicability of the equation for the imitation of dynamics of money supply:

R-squared	0.947588	Mean dependent var	0.023350
Adjusted R-squared	0.935940	S.D. dependent var	0.139836
S.E. of regression	0.035392	Sum squared resid	0.022547
Durbin-Watson stat	2.076581		

The parameters of equation for the consumer prices (ECPI) calculated as normalized residuals of cointegration for Estonia are presented in 6.3.3:

$$\begin{aligned} \text{DLOG(ECPI/ER_KR_USD)} = & \text{B(36)*[LOG(ECPI(-1)/ER_KR_USD(-1))-15.20291204*(Q3(-} \\ & \text{1)/ER_KR_USD(-1)) - 5.731116329*LOG(EW_KR(-1)/ER_KR_USD(-1))-} \\ & \text{1.473115513*LOG(EGDP_KR(-1)/ER_KR_USD(-1)) - 28.21785389*(D4(-1)/ER_KR_USD(-} \\ & \text{1))]+B(37)*DLOG(EW_KR/ER_KR_USD)+B(38) +*DLOG(EGDP_KR/ER_KR_USD)} \\ & \text{+B(39)*(D2/ER_KR_USD)} \end{aligned} \quad 6.3.3$$

Results of statistical testing of parameters for equation 6.3.3 after neutralization of USD influence on CPI shows much better determination (normal and adjusted R-squared) coefficients than basic equation (6.2.3) when imitating the dynamics of CPI:

R-squared	0.652312	Mean dependent var	-0.083635
Adjusted R-squared	0.594364	S.D. dependent var	1.139024
S.E. of regression	0.725439	Sum squared resid	9.472708
Durbin-Watson stat	2.441799		

The parameters of equation for the export of goods (EEXP_G_KR) calculated as normalized residuals of cointegration for Estonia are presented in 6.3.4:

$$\begin{aligned} \text{DLOG(EEXP_G_KR/ER_KR_USD)} = & B(46) * (\text{LOG(EEXP_G_KR(-1)/ER_KR_USD(-1))} - \\ & 0.9387829206 * \text{LOG(EIMP_G_KR(-1)/ER_KR_USD(-1))} - 0.3928430772 * \text{LOG(EREER(-1)/ER_KR_USD} \\ & (-1))) + B(47) * \text{DLOG(EIMP_G_KR/ER_KR_USD)} \end{aligned} \quad 6.3.4$$

The statistical testing of parameters for equation 6.3.4 after neutralization of USD influence on export of goods shows also better determination (normal and adjusted R-squared) coefficients than basic equation (6.2.4):

R-squared	0.811567	Mean dependent var	0.022016
Adjusted R-squared	0.803002	S.D. dependent var	0.121568
S.E. of regression	0.053957	Sum squared resid	0.064051
Durbin-Watson stat	2.301382		

The parameters of equation for the export of services (EEXP_S_KR) calculated as normalized residuals of cointegration for Estonia are presented in 6.3.5:

$$\begin{aligned} \text{DLOG(EEXP_S_KR/ER_KR_USD)} = & B(56) * (\text{LOG(EEXP_S_KR(-1)/ER_KR_USD(-1))} - 30.23759683 - \\ & -0.5584756686 * \text{LOG(EGVA_SERV_KR(-1)/ER_KR_USD(-1))} - 1.378016298 * \text{LOG(ECPI(-} \\ & 1)/ER_KR_USD(-1))) + B(57) + B(58) * \text{DLOG(EGVA_SERV_KR/ER_KR_USD)} + B(59) \\ & * \text{DLOG(ECPI/ER_KR_USD)} + B(60) + \text{DLOG(ENEER/ER_KR_USD)} \end{aligned} \quad 6.3.5$$

The statistical testing of parameters for equation 6.3.5 shows that normalizing the influence of USD market rates on export of goods is substantially reducing the determination (normal and adjusted R-squared) coefficients comparing to their significances in the basic equation (6.2.5):

R-squared	0.600192	Mean dependent var	0.020413
Adjusted R-squared	0.506119	S.D. dependent var	0.179102
S.E. of regression	0.125867	Sum squared resid	0.269323
Durbin-Watson stat	2.413383		

The parameters of equation for the real EER (EEREER) calculated as normalized residuals of cointegration for Estonia are presented in 6.3.6:

$$\begin{aligned} \text{DLOG(EREER/ER_KR_USD)} = & \text{B(66)} * (\text{LOG(EREER(-1)/ER_KR_USD(-1))} - 0.105084281 * \text{LOG(EM2_KR(-1)/ER_KR_USD(-1))} - 0.008298005365 * \text{LOG(EL_FX_KR(-1)/ER_KR_USD(-1))}) + \text{B(67)} + \\ & \text{DLOG(EL_FX_KR/ER_KR_USD)} + \text{B(68)} + \text{DLOG(EEXP_KR/ER_KR_USD)} + \text{B(69)} * (\text{D4/ER_KR_USD}) \end{aligned}$$

6.3.6

The statistical testing of parameters for equation 6.3.6 shows that normalizing the influence of USD market rates on REER insignificantly ameliorates the determination (normal and adjusted R-squared) coefficients comparing to their significances in the basic equation (6.2.6) but in both cases their level is insufficient:

R-squared	0.448478	Mean dependent var	-0.007466
Adjusted R-squared	0.361396	S.D. dependent var	0.059692
S.E. of regression	0.047701	Sum squared resid	0.043233
Durbin-Watson stat	1.944510		

The parameters of equation for the interest rate on loans (EEREER) calculated as normalized residuals of cointegration for Estonia are presented in 6.3.7:

$$\begin{aligned} \text{DLOG(EAAIR/ER_KR_USD)} = & \text{B(76)} * (\text{LOG(EAAIR(-1)/ER_KR_USD(-1))} - 15.80099985 - 1.521843677 * \\ & \text{LOG(EM2_KR(-1)/ER_KR_USD(-1))} - 0.807301649 * \text{LOG(EL_EEK_KR(-1)/ER_KR_USD(-1))}) + \\ & \text{B(77)} + \text{B(78)} * \text{DLOG(EM2_KR/ER_KR_USD)} + \text{B(79)} * \text{DLOG(EREER/ER_KR_USD)} + \\ & \text{B(80)} * (\text{D5/ER_KR_USD}) \end{aligned}$$

6.3.7

The statistical testing of their parameters (for equation 6.3.7) shows that normalizing the influence of USD market rates on the interest rate on loans is slightly ameliorating the determination (normal and adjusted R-squared) coefficients comparing to their significances in the basic equation (6.2.7):

R-squared	0.723853	Mean dependent var	-0.041285
Adjusted R-squared	0.662487	S.D. dependent var	0.231183
S.E. of regression	0.134308	Sum squared resid	0.324693
Durbin-Watson stat	2.215063		

The parameters of equation for the import of goods (EIMP_G_KR) calculated as normalized residuals of cointegration for Estonia are presented in 6.3.8:

$$\begin{aligned} \text{DLOG(EIMP_G_KR/ER_KR_USD)} = & \text{B(86)} * (\text{LOG(EIMP_G_KR(-1)/ER_KR_USD(-1))} - 0.8575398294 * \\ & \text{LOG(EEXP_KR(-1)/ER_KR_USD(-1))} - 0.1378730423 * \text{LOG(EGDP_KR(-1)/ER_KR_USD(-1))}) + \text{B(87)} \\ & * \text{DLOG(EEXP_KR/ER_KR_USD)} + \text{B(88)} * \text{DLOG(EGDP_KR/ER_KR_USD)} \end{aligned}$$

6.3.8

The statistical testing of their parameters (for equation 6.3.8) shows that normalizing the influence of USD market rates on the import of goods is also ameliorating the determination (normal and adjusted R-squared) coefficients comparing to their significances in the basic equation (6.2.8):

R-squared	0.835845	Mean dependent var	0.018450
Adjusted R-squared	0.818565	S.D. dependent var	0.127468
S.E. of regression	0.054295	Sum squared resid	0.056012
Durbin-Watson stat	1.618800		

6.4. The imitation not included the cases of financial shocks pressing the monetary transmission strongly enough in the euro area who would threat the Baltics in less content under stabilizing buffers of the CBA. The currency regime is expected to determine the exchange mechanism all period until final date of their entry to the ERM but it can not be absolute panacea in cases of strong disturbances in world monetary markets or, what would be even worse, in situation of disintegration of these markets. The different options in real and nominal convergence, also the anticipated trade-offs have to be discussed from the aspect of possibility to diminish the current account deficit in the Baltic States to a sustainable level.

CONCLUSIONS

As concerns the evaluations on the side of actual monetary policy interrelated with the re-pegging of national currencies, the conclusions from this study are following:

1. Lithuania and Estonia are determined to support the currency board regime as institutional base for pre-accession period after re-pegging their national currencies directly to the euro (02.2002 and 01.1999 adequately).
2. Previous anchoring substantially helped to stop three-digit inflation down to one digit price growth. Some approximate imitations and arguments permit to expect that a float regime would open some additional chances for small open economies but mostly at the account of greater risk.
3. Both Estonia and Lithuania intensively reoriented their external monetary flows and trade operations to Eurozone and Euro markets – Estonia from 1994-5, Lithuania - from 1999. Latvia oriented its currency to euro going smoothly through pegging it to the SDR basket before. All three Baltic States also mostly adequately restructured their state reserves, state debt, other financial indicators and flows.
4. The traditional Mundell-Fleming models interconnecting exchange rate changes with structural changes in finance and economic system present perspective analytical solution for the determining impact of currencies in the transition economies pegging and/or repegging now to Euro. However, their adaptation for the analysis of monetary transmission channels in the Baltic countries is complicated as a result of errors and omissions in evaluating shadow sector within official statistics over passing the values of econometrical evaluations measuring various effects.
5. The modeling of monetary policy alternatives and its instrumentation in Lithuania, differently from Estonia, was not oriented directly to the determination of its effect in the pre-accession to the EU period. It requests more

substantial support and much more professional attention, inspiring the more perspective solutions in the national monetary regimes and the simulation of effects of various monetary policy instruments in Lithuania.

6. The pressure of domestic money appreciation under CBA in the Baltics and esp. in Lithuania which just changed currency anchor is discouraging export to the EU as a result of lowering rate of EUR to USD and induced anxiety about shock risk in future BoP following fluctuating rates of energy resources.

On the side of the econometric imitations aiming to identify the possible impact of repegging of national currencies to macroeconomic indicators, the approach to measure the impact of the EER of these currencies concerning different zones was attempted as an instrumental for the comparative purposes:

1. The identification of simplified interconnections between changes of the EER of the Baltic currencies to EUR, on the one side, and main structural variables of the GNP, esp. gross value added, also balances of the main BoP components, on the other side, can help to evaluate in principle the pegging influence on the macroeconomic development. The effect on national macroeconomics of the re-pegging of LTL and EEK from one anchor to another under CBA was influenced also by the conjuncture of their anchor currencies and their interrelations, in some cases more than by internal and sectoral changes (as may be seen from the structural reorientation of their balances of payment and esp. comparing actual and normalized curves of foreign trade of goods).
2. The econometric simulation of the reinforcing or loosening the criteria for convergence of fundamental differences and adjusting the decisive structures through flows of incomes and changes in prices helps as a proxy when evaluating the regional monetary transmission channels.

The prognostic simulation of actual and normalized changes of the monetary transmission channels for 1996 - 2007 gives many additional impulses for understanding of mechanisms and interaction of variables after national currencies re-pegging processes in the Baltics.

I, the undersigned, certify that the information submitted with this report is new, complete and correct to the best of my knowledge.

Name: Antanas Buracas

Signature: **AntBuracas**

Date : April 15, 2002.....

References

1. Abazorius Darius (2000). Strategy of the Bank of Lithuania during the integration into the EU //Report: *Currency Boards - Experience and Prospects* Seminar organized by Eesti Pank, 5-6 May, Tallinn.
2. *Balance of Payments, Exchange Rates, and Competitiveness in Transition Economies* (1999) / ed. by Mario I. Blejer, IMF, Marko Skreb. Croatian National Bank. Kluwer Academic Publishers, Boston.
3. Balino Tomas J.T., V. Sundararajan (1997). Monetary Policy Instruments: Design of Instrument Mix and Co-ordination of Instrument Design // *Instruments of Monetary Management. Issues and Country Experiences*. IMF, Wash.

4. Beblavý M. (2001). Exchange rate and exchange rate regime in Slovakia. Recent developments // Preprint, Vienna workshop: *Alternatives for Exchange Rate Regime in Advanced Transition Economies*, 21 Sept. / www.icegec.org.
5. Budrytė Algė, Kvedaras Virmantas (2000). Prospects of the Lithuanian Macroeconometric Model. // *Pinigø studijos (Monetary Studies)*, March.
6. Buraėas Antanas (2000). Aims of Monetary Policy and Options to Instrumentate it in Lithuania // *Management of Organizations: Systematic Research*, Nr. 15.
7. Buraėas Antanas, Sakalauskas Virgilijus (2001) Mechanisms of Currency Pegging Impact on National Economy // *Economics of Engineering*, Kaunas, 2001.
8. Buraėas Antanas (2002) The Interactions Between Effective Exchange Rate and Macroeconomic Structure: Case Of Lithuania // *Management of Organizations: Systematic Research*, Nr.24.
9. Buraėas Antanas, Zuokas Danas (2002) The Interactions Between Effective Exchange Rate and Macroeconomic Structure: Attempts of Econometric Modeling // *Management of Organizations: Systematic Research*, Nr.25.
10. Calvo G. (1985). Currency Substitution and Real Exchange Rate // *Journal of International Money and Finance*, N 4, pp. 175-188.
11. Croce Enzo, Mohsin S. Khan (2000) Monetary Regimes and Inflation Targeting // *Finance and Development*, Sept., IMF Survey Vol. 37 Nr. 3.
12. Deutsche Bundesbank (June 2000). Macro-Econometric Multi-Country Model: MEMMOD. // <http://www.bundesbank.de/en/monatsbericht/sonderpub/memmod.pdf>, p.22-24, 62-73.
13. Economic Policy in the Framework of Accession to the EU and EMU (2001) // The international conference hold under auspices of the EU delegation in Vilnius, 8-9 Feb. / Preprints of the reports.
14. Filipozzi Fabio (2000). *Equilibrium Exchange Rate Of The Estonian Kroon, Its Dynamics And Impacts of Deviations* // Tallinn.
15. Fischer Stanley (2001). Exchange Rate Regimes: Is the Bipolar View Correct? // IMF, Jan. 06.
16. Fisher S., Sahay R., Végh C. (1998). How Far is Eastern Europe from Brussels? // *IMF Working Papers*, WP/98/53.
17. Foreest Pieter W. van, Vries Casper G. de (2001) Endogenous Response to Monetary Integration
18. Ghosh, Gulde and Wolf. (1999). Currency Boards: More than a Quick Fix? // Paper prepared for the 30th Panel Meeting of Economic Policy, Helsinki, Oct. 15-16.
19. Goldberg M. D., Frydman R., Macroeconomic Fundamentals and the DM/\$ Exchange Rate: Imperfect Knowledge and the Monetary Model // Paper presented at a workshop on *Exchange Rate and Monetary Policy Issues*, Institute for Advanced Studies, Vienna, Austria, April 19-20, 2001.
20. Guidelines for the Application of the BoL's Monetary Policy Instruments (1999). // BoL, Vilnius.
21. Gulde A. -M., Kahkonen J, Keler P. Pros and Cons of Currency Board Arrangements in the Lead-up to EU Accession and Participation in the Euro Zone // *IMF PDP* 2000 N 1.
22. Halpern László, Judit Neményi (2001). Fiscal Foundation of Convergence to the European Union. Hungarian Economy Towards EU Accession. // Preprint to be published by MIT Press in *'Transition: the First Decade* ed. by M. I. Blejer and M. Skreb.
23. IMF Completes First Lithuania Review (2001). // *International Monetary Fund News Brief*, Jan. 10.

24. Isard Peter, Hamid Faruquee, G. Russell Kincaid, and Martin Fetherston (2001). *Methodology for Current Account and Exchange Rate Assessments* // International Monetary Fund Occasional Paper Nr.209.
25. Kalinauskas Žilvinas (2000). Comparative Study of the Baltic States' Payment Balances in 1st Half of 2000 // *Pinigø studijos (Monetary Studies)*, nr 4.
26. Kalinauskas Žilvinas, Tamošiūnas Raimundas (2000) An Applied Equilibrium Model For Lithuanian Economy // *Pinigø studijos (Monetary Studies)*, n.1.
27. Kandil Magda (Nov. 2000). The Asymmetric Effects of Exchange Rate Fluctuations: Theory and Evidence from Developing Countries // *IMF Working Paper, WP/00/184*.
28. Keller Peter M. (2000). Recent Experience with Currency Boards and Fixed Exchange Rates in the Baltic Countries and Bulgaria and Some Lessons for the Future // *Currency Boards -Experience and Prospects*. Seminar organized by Eesti Pank, 5-6 May, Tallinn.
29. Klyuev Vladimir (2001). A Model of Exchange Rate Regime Choice in the Transitional Economies of Central and Eastern Europe / *IMF WP, 01/140*, September.
30. Kuodis Raimondas (2000). Lithuania: Recent Macroeconomic Developments and Future Prospects // *Currency Boards - Experience and Prospects*. Seminar organized by Eesti Pank, 5-6 May, Tallinn.
31. Kuttner Kenneth N., Posen Adam S. (2001). What Makes the Best Nominal Anchor? // *Federal Reserve Bank of New York*, Institute for International Economics. April 16 (Preprint).
32. Kuttner K. N., Posen A S. (1999). Does Talk Matter After All? Inflation Targeting And Central Bank Behavior // *Federal Reserve Bank of New York*, Institute for International Economics.
33. Levy-Yeyati Eduardo (2000), Sturzenegger Federico. Exchange Rate Regimes And Economic Performance // *IMF WP, October*.
34. Lithuanian Macroeconomic Review (2001). Vilniaus bankas..
35. Masson Paul R. (1999). Monetary and Exchange Rate Policy of Transition Economies of Central and Eastern Europe after the Launch of EMU // *IMF Policy Discussion Paper*.
36. Menzie David Chinn (1997). The Usual Suspects? Productivity And Demand Shocks And Asia-Pacific Real Exchange Rates // University of California, Santa Cruz, June 24.
37. *The Monetary Policy of the ECB*. Executive Board of the ECB, 2001.
38. Orlowsky Lucjan T. (2000) A Dynamic Approach to Inflation Targeting in Transition Economies // *ZEI (Zentrum für Europäische Integrationsforschung) Working Paper*, B.11.
39. Orlowski Lucjan T. (2001) Monetary Convergence of the EU Candidates to the Euro: A Theoretical Framework and Policy Implications. // Paper to the conference: *How to Pave the Road to E(M)U: The Monetary Side of the Enlargement Process and Its Fiscal Support* organized by the Deutsche Bundesbank, the National Bank of Hungary and the Center for Financial Studies, Eltville, Germany, October 26-27, 2001 // <http://www.bundesbank.de/de/vo/pdf/orlowski.pdf>
40. Pikkani Rasmus (2000) *The Monetary Sector under a Currency Board Arrangement: Specification and Estimation of a Model with Estonian Data* // Tallinn.
41. Price, Lionel (1996). Economic Analysis in a Central Bank - Models Versus Judgments // *Handbooks in Central Banking*, Centre for Central Banking Studies, Bank of England, L., No 3.
42. Purchasing Power Parities. Background (2001). OECD // <http://www.oecd.org/std/ppp/pppbackground.htm>.

43. Rajasalu Teet a/o (2000). Implications of the Introduction of the Euro for the Estonian Financial Markets // *Eesti Pank Bulletin*, No 4.
44. Repše Einars (2000). Latvia: Focus on Country Development // *Finance and Development*, Sept., IMF, Vol. 37 Nr. 3.
45. Republic of Lithuania: 2000. Article IV Consultation and First Review Under the Stand-By Arrangement-- Staff Report; Public Information Notice and News Brief on the Executive Board Discussion; and Statement by the Authorities on the Republic of Lithuania // *IMF Country Report* No. 01/19 // Jan 2001.
46. Republic of Lithuania: Statistical Appendix and Tax Summary (2001) // *IMF Country Report* No. 01/40 // February 21.
47. Souza Lucio Vinhas de (2001). EMU and Enlargement: Current Strategies of Exchange Rate Linkages // Paper for the EP, Project No IV/99/13 *Economic and Monetary Union and Enlargement*, Output 4/6.
48. Thimann Christian (2001). Monetary Policy Challenges in Transition and Towards Accession // The international conference *Economic Policy in the Framework of Accession to the EU and EMU* held under auspices of the EU delegation in Vilnius, 8-9 Feb.

I, the undersigned, certify that the information submitted with this report is new, complete and correct to the best of my knowledge.

Name: Antanas Buracas

Signature: **AntBuracas**

Date : April 15, 2002.....